Contribution of Certification Bodies and Sustainability Standards to Sustainable Development Goals: An Integrated Grey Systems Approach

Muhammad Ikram[a], Qingyu Zhang[a*], Robert Sroufe[b], Marcos Ferraso[c]

[a] *College of Management, Research Institute of Business Analytics and Supply Chain Management, Shenzhen University, Shenzhen, 518060, China*

[b] *Donahue Graduate School of Business, Duquesne University, 820 Rockwell Hall, 600 Forbes Avenue, Pittsburg, PA, USA*

[c] *Institute of Scientific Research and graduate school, Universidad de Lima, Lima, Peru Av. Javier Prado Ests, 4600 Santiago de Surco 15023 Lima-Peru*

mikram@szu.edu.cn, q.yu.zhang@gmail.com, sroufer@duq.edu, admmarcosferasso@gmail.com

Abstract

The International Organization for Standardization and the Global Reporting Initiative promote their alignment with the United Nations' Sustainable Development Goals. This convergence of standards and information is causing the selection of a certification body to register these sustainability standards to become more complicated. Therefore, this study fills a gap in the literature by developing a framework to prioritize the most critical attributes for selecting a certification body. Initially, we focus on seven attributes using the Grey Delphi method. Further, Grey Analytical Hierarchy Process is employed to find primary and sub-indicators relative importance. Finally, we used a novel Grey Absolute Decision Analysis model to rank the widely adapted top six certification bodies. Results reveal that Reputation, Payment Method & Cost, and Quality of Auditors are significant indicators with weighted scores of 0.2120, 0.1704, and 0.1406. In contrast, the auditor's knowledge about the sustainability standards, and review of the meetings obtained the highest weight score 0.5009 and 0.4918 and are seen as essential among 35 sub-criteria. Within this study, we identify which ISO and GRI sustainability standards contribute to a given SDG. We also find Bureau Veritas and Société Générale de Surveillance are the best-suited certification bodies help to achieve SDGs as they obtained the highest weights 0.8731, and 0.7340. This study is among the first of its kind to address selecting the right certification body for aligning with the SDGs by using the integration of three novel grey models. Outcomes of this study can help assist scholars, managers, government agencies, and decision-makers in selecting certification bodies to achieve SDGs while simultaneously improving sustainability practices.

Keywords: Sustainable Development Goals; Certification bodies; Sustainability Standards; ISO; GRI; Grey systems approach

Nomenclature

$[a_j^1, a_j^s]$	Range values of grey classes
a_j^k	Lower bound of *kth* grey classes
b_j^k	Upper bound of *kth* grey classes
$f_j^k(x)$	Weight function of k grey class
η_j^k	Wights of *j* indicator in synthetic clustering

* Corresponding Author (Qingyu Zhang; q.yu.zhang@gmail.com)

$\{\sigma_j^k\} = \{\sigma_j^{k^*}\}$	Evaluated decision vectors
$\otimes G_{ij}$	Grey number
$\otimes \overline{G_{ij}}$	The Upper limit of the grey number
$\otimes \underline{G_{ij}}$	The Lower limit of the grey number
λ	Grey coefficient
(a_j)	Decision Object
$[a_{ij}]$	Response decision measurement
$C(k)$	Decision criteria
$A_{(j)}$	Alternative
ε	Absolute grey relational grade
$[\varepsilon]$	Pairwise comparison matrix
$\alpha_i = \frac{1}{N}(\varepsilon_{i1} + \varepsilon_{i2} .... + \varepsilon_{iN})$	Javed alpha
$\delta(k)$	Simulated weight of criteria
$(\beta_1(k), \beta_2(k), ..., \beta_N(k))$	Geometric mean
ȑ	Grey Absolute Decision Analysis Index
(R_j', R_j'')	Set of relative weights

1. Introduction

The Sustainable Development Goals (SDGs) started at the United Nations Conference on Sustainable Development in Rio de Janeiro in 2012. The objective was to produce a set of universal goals that meet our world's wicked environmental, political, and economic challenges. These challenges are rooted in an understanding that the greater the industrial development in a country, the greater the potential for detrimental impacts to the very resources capital development depends on, the earth, people, and social systems (Skirbekk, 1994). The SDGs replaced the Millennium Development Goals (MDGs), which started a global effort in 2000 with measurable, globally-agreed objectives for addressing poverty, hunger, diseases, expanding education to all children, and other priorities. The SDGs are a call to action for the world to take a more sustainable path into the future. All 17 Goals interconnect, meaning success in one affects success for others. Dealing with the threat of climate change impacts how we manage our fragile natural resources, achieving gender equality or better health helps eradicate poverty, and fostering peace and inclusive societies will reduce inequalities and help economies prosper. The SDGs coincided with another historic agreement reached in 2015 at the COP21 Paris Climate Conference. Together with the Sendai Framework for Disaster Risk Reduction, signed in Japan in March 2015, these agreements provide a set of common standards and achievable targets to reduce carbon emissions, manage the risks of climate change and natural disasters, and rebuild more resiliently after a crisis (UNEP, 2020)

In 2015 the United Nations presented the 2030 Agenda with accompanying 17 Sustainable Development Goals (SDGs). These goals have gotten the attention of academics, business leaders, activists, and policy decision-makers to stimulate a transformation in world affairs (Huan et al., 2021; Zhao et al., 2020). In 2019, United Nations (UN) reported that financial support is a big hurdle to implement the SDGs agenda in developing countries (UN World, 2019). At the same time, the political leaders worldwide are not fully aware of Agenda of 2030. Local governments, international agencies, various businesses, and stakeholders also have no clear distinctions of responsibilities to fulfill these global goals in developed and developing countries (Zhao et al., 2020). It probably comes as no surprise then that Lamichane et al. (2020) found that achieving the SDGs has been happening at a minimal level in developing countries.

Climate change, increasing carbon dioxide emission, continued natural disasters (Ofosu-Adarkwa et al., 2020), and event the COVID-19 pandemic have disrupted the SDGs' plans and have made it more challenging for companies and governments to achieve. Therefore, it is increasingly important for the SDGs and Agenda 2030 to find new ways to progress. Hence, International Organizations for Standardization (ISO), Global Reporting Initiative (GRI), and other international bodies have already provided a tangible mechanism to help achieve SGDs by implementing sustainability tools.

According to Det Norske Veritas Germanischer Lloyd's (DNVGL) forecasting report on SDGs' global progress, the world will not meet even half of SDGs till 2030 (DNVGL, 2015). The report presents the world's division into five regions, i.e., USA, China, OECD, BRISE (Brazil, Russia, India, South Africa and other emerging economies), and ROW (rest of other world) as characterized by their size and economic development status. The details of the regions and progress of SDGs are in Figure 1. The report indicates that the USA and OECD countries (excl. the USA) have likely to be reached (Goal 2, goal 3, goal 4, goal 6, goal 9, and goal 11) a target fulfillment of more than 95%. At the same time, China is also on the right track and has achieved more than 95% (goal2, goal3, goal4, goal7, and goal9). Furthermore, BRISE has not achieved any goal of more than 95% and is at the transition phase. Finally, the ROW has achieved only one goal (goal 12) at more than 95%. The rest of the other goals are not likely to be reached by more than 50%. The report also indicates that collective efforts are required to reach the SDGs. The green indicator shows the goal likely to be reached (i.e., target fulfillment of 95% achieved). In contrast, orange indicates the goal reached more than 50% between today's status, and the goal is likely closed but not fully reached. Meanwhile, the red indicator shows that the goal reached less than 50% reached. Additionally, SDGs 2, 3, 4, 6, and 9 have a positive outcome and move in the right direction. Conversely, the pace is too slow to reach SDG climate action goal 13, affordable and clean energy goal 7 due to energy production through fossil fuels and non-renewable energy consumption, which accumulate predicted carbon emissions of 2,900 Giga tones by the year 2037.

Figure 1. Likelihood of meeting the 17 SDGs in five world regions. (*Source (DNVGL, 2015)*)

ISO is one of the largest standard developers, with more than 22,000 sustainability management standards used in 170 countries to fulfill the SDGs requirements (ISO, 2018a). ISO standards cover all the aspects of social, economic, and environmental concerns and help enable more sustainable practices. ISO standards ensure consistency, transparency, and transfer of sustainability practices across the world. Moreover, ISO standards play a vital role in international trade partnerships. ISO introduced the SDGs action plan regarding Agenda 2030, in which ISO standards provide essential tools to assist governments, industrial sectors, and consumers in achieving each of the SDGs.

Across many of the SDGs, ISO has developed significant standards that contribute to sustainability. For example, the Environmental Management System (EMS) ISO 14001 standard covers SDG 13 on climate action and 15 regarding life on land. Further, ISO 26000, the Social Responsibility standards, provides guidelines and supports SDGs 2 Zero Hunger, 5 Gender Equality, and 10 Reduce Inequalities. Add to this, ISO 24000 addressing Sustainable Procurement, and this standard helps to achieve SDGs 1 No Poverty, 2 Zero Hunger, and 12 Sustainable Production and Consumption. ISO 15001 helps develop an Energy Management System (EnMS) to enable SDG 7 Affordable and Clean Energy. The complete details of how the ISO technical committee developed each standard to help achieve SDGs are within the ISO website (ISO, 2018b).

GRI is the second-largest sustainability reporting non-profit and multi-stakeholder international organization. It was established in 1997 and launched in partnership with the United Nations Environmental Program (UNEP) in 2000 (Sethi et al., 2017). GRI provides assistance to ensure economic, social, and environmental sustainability through sustainability guidelines and standards. GRI is a catalyst for a more sustainable future that provides a consistent and transparent sustainability disclosure and standards to fulfill stakeholder needs (Hu et al., 2019; Gallego-Álvarez et al., 2018). The GRI is a defacto global sustainability reporting standard used in more than 90 countries. It provides services for multinational companies, government agencies, Small Medium Enterprises (SMEs), NGOs, industrial groups, and United Nations Global Compact (Gómez Martín et al., 2020; Tsalis et al., 2020). Since 2016, GRI has been included in the UN SDGs and developed standards and disclosure, contributing to these goals. GRI standard 207: 1-4, 203-2 and 207-2 align with SDG 1; standard 403 contributes to SDG 3 Good Health and well-being; GRI standard 404 enables guidelines for SDG 4 Quality Education; GRI standard 201 covers the SDG 8 Decent Work and Economic Growth; GRI standard 302 helps climate vulnerability SDG 13 Climate Action; GRI 403 promote peaceful societies for sustainable development SDG 16; GRI 303 contributes to the availability of clean water and sanitation SDG 6. The complete details of GRI standards that cover each SDG are available on the GRI Website (GRI, 2020).

When implementing sustainability standards such as certified and assured tools, non-certified tools, international guidelines, management programs, and disclosures, companies need to decide if an external certification body should align with SDGs. To make the business more sustainable, the companies need to become ISO, GRI, and other standards registered companies. These external sustainability certifications can positively affect organizational performance (Ali and Yusuf, 2019). With registration by highly reputable certification bodies, sustainability certification can provide more recognition and competitive advantage to companies (Chkanikova and Sroufe, 2020).

There are various certification bodies (CB) such as DNVGL, Bureau Veritas (BV), SGS, Lloyd's Register (LR), BSI, TUV, Intertek, and others. They provide their services to companies to register ISO, GRI, and other sustainability standards with competitive prices, quality auditors, and assurance to develop more sustainable business practices aligned with SDGs through effective implementation. The selection of an appropriate CB can be a difficult decision for companies. Despite this difficulty, the certification literature has paid little attention to this decision, and few studies have investigated this issue. This lack of attention is evident when comparing this management problem with other management standards, such as integrating management standards (IMS) and barriers to effective IMS implementation.

Further investigation is required to analyze how companies choose their certification bodies to align with and help achieve SDGs. Selecting the inappropriate certification body can have high financial and operational risks. Choosing the appropriate certification bodies to help companies implement sustainability standards can provide various benefits such as cost reduction, competitive advantage, customer complaints reduction, productivity, and customer satisfaction.

Thus, it is important to understand better the range of criteria considered in these decisions and our primary research questions:

RQ1. What sustainability tools are required to align with and help achieve Sustainable Development Goals?

RQ2. What are the main criteria involved in selecting certification bodies, and how to implement sustainability tools?

RQ3. What is the relative importance of these criteria, and how companies prioritize them?

To help answer these research questions, we have identified essential criteria from the literature that include (i) Reputation (ii) Operational Outcomes (iii) Quality of Auditor (iv)Contract Condition (v) Quality of Implementation (vi) Payment Method & Cost (vii) Knowledge of Product and Business. These main certification body criteria are then further classified into sub-criteria. One contribution to this research field is to assess the criteria and sub-criteria while filling a gap in the literature. This study's information can help decision-makers select the right certification body to align practices and help achieve SDGs. The second contribution of this study is to, for the first time, compare the top six certification bodies, providing the outcomes to managers, governments, and policymakers with new insights for how the selection of a certification body can help align practices with the SDGs. Third, we develop a framework for which sustainability tools/standards are required to achieve SDGs. Finally, this study is the first of its kind to use the integration of three novel grey models such as Grey Delphi (to finalize the classification of certification body selection criteria); Grey analytic hierarchy process approach GAHP (compute the weights for criteria and sub-criteria); and we use a novel Grey Absolute Decision Analysis (GADA) method based on GAHP to identify and prioritize the categories of the top six certification bodies.

The paper's structure is as follows: Section 2 presents the conceptual background of SDGs, sustainability standards, study criteria, and sub-criteria for selecting the certification bodies to align with and help achieve SDGs. Section 3 discusses the steps involved in developing Grey Delphi, GAHP, and GADA methodologies. Section 4 presents the results and discussion. The study's conclusion and policy implication, along with options for future research, are in the last section.

2. Literature Review

2.1 The United Nations Sustainable Development Goals (SDGs)

The basis of the SDGs was introduced by United Nations (UN), through the 'Our Common Future' report produced by the Commission on Environment and Development, also known as the 'Brundtland report' (Brundtland, 1987). Within this report, the definition of sustainable development is expressed as meeting the needs of a current generation without compromising the needs of future generations. Sustainable development's primary focus is a better life for human beings while staying within environmental limits. Later, a three-dimensional model of sustainability was proposed by Elkington. Widely known as the Triple Bottom Line (TBL), representing the economic development (profit), environmental progress (planet), and the social equity (people) dimensions (Elkington, 1997). Next, the UN Agenda for Development updated the sustainable development definition by integrating the previous Brundtland definition and the TBL dimensions. Sustainable development is considered in a multidimensional approach promoting a higher quality of life for human beings. It is noted that the interdependent basis of sustainable development is dependent upon economic development, social progress, and environmental protection (Abid et al., 2021; Barbosa-Póvoa et al., 2018).

According to Govindan et al. (2013), although there are significant contributions from the Brundtland Report definition, the challenge for sustainability lies in its operationalization. It also needs to be defined in a way so that it can be measured. Add to this the idea that Corporate Social Responsibility (CSR) is considered essential to organizations' long-term success. It integrates the TBL model when stating that profitability, social responsibility, and environmental preservation are crucial CSR elements (Pan et al., 2020). The literature shows no universal definition, although, for sustainability practices, the available definitions include economic, social, and environmental dimensions. The central idea is to create a balanced development successfully grounded in social progress and preserve the natural environment, with the generation of value for stakeholders (such as partners, customers, employees, and society).

Aligned with the planned Agenda for Sustainable Development 2030, created by the UN and launched in 2015, the 17 SDGs divided into the economy, society, and biosphere (in Figure 2), represent a historical demarcation of the sustainable development agenda. In this approach, the SDGs integrate the three pillars of sustainability: environmental sustainability, social development, and economic development (Rehman et al., 2020; White and Lee, 2009). Underlying this is the importance of sustainability goals for creating global policies and governance. The SDGs include core features of development concerning issues and opportunities involving the COVID-19 pandemic, energy, water, oceans, transportation, urbanization, technology, science, and climate. The SDGs were conceived to be addressed as individual goals and indicators for monitoring progress and reporting in the UN's reports (United Nations, 2020).

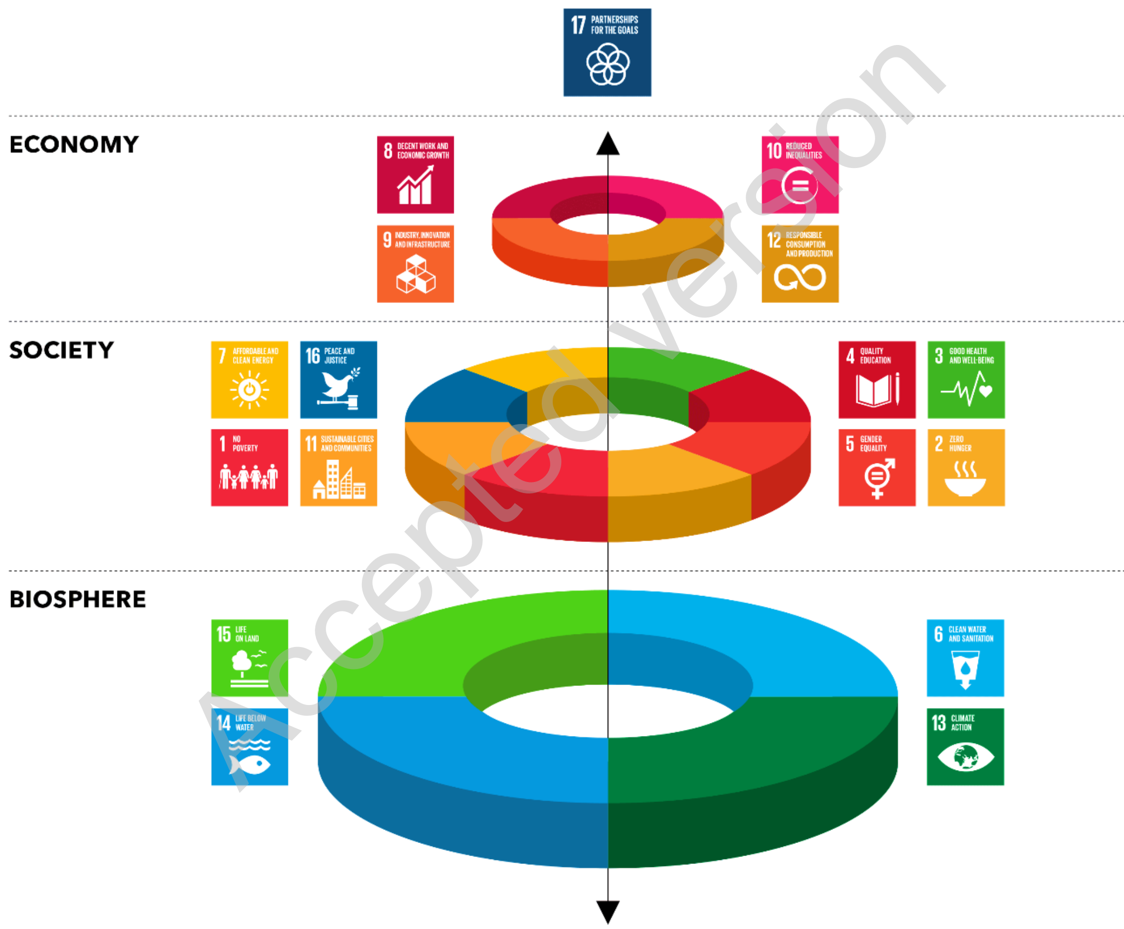


Figure 2. The United Nations Sustainable Development Goals (SDGs); *source (United Nations, 2020)*

2.2 Sustainability Tools to Achieve Sustainable Development Goals

The International Organization of Standardization – ISO is formed by members of more than 160 countries that provide services and a supportive platform for developing standardization to guide entire industries. The standards are presented and discussed by experts' committees appointed by relevant organizations or even state members (ISO, 2017). The standards defined by ISO members form the basis for the definition of global standards and then incorporate this into quality management infrastructure that is the combination of initiatives, institutions, organizations, activities, and people that help ensure products and services meet the requirements of customers (Ikram et al., 2020b). Even though quality management can be operationalized in other measurement or guiding procedures, the ISO standards are considered core patterns and provide relevant mechanisms for codifying and transforming knowledge into organizational knowledge. Table 1 summarizes sustainability tools that foster the achievement of SDGs.

The voluntary sustainability standards joined forces with globalization, turning sustainability into a regulatory mechanism to achieve stakeholders' requirements regarding companies' internationalization and their value chains (Diaz-Balteiro et al., 2017; Chowdhury et al., 2020). The standards that stand out were the management systems established by the ISO related to quality management (ISO 9001:2015), the environmental management system (ISO 14001:2015). The British Standards Institution published the Occupational Health and Safety Standards (OHSAS) (OHSAS 18001; ISO 45001:2018). One primary feature of these management systems is the possibility of being certified by Certification Bodies that perform independent third-party auditing services. They also assess whether the adherence of a given management system according to the determined international standards (e.g., ISOs 9001, 14001, 45001, and the OHSAS 18001) and recommendations for achieving intended results (Fonseca and Carvalho, 2019).

Table 1: Summary of Sustainability Standards Aligned with and Helping to Achieve Sustainable Development Goals (SDGs)

Goal	Goal Description according to SDGs	Required Sustainability Standard	Standards Descriptions
Goal 1: No Poverty	"End poverty in all its forms everywhere." This goal targets the eradication of extreme poverty everywhere and implementing solutions and protection measures. To achieve such a goal, it needs a shared platform containing best practices in economic systems, providing sustainable food production, and balancing resource use and consumption towards SD.	ISO 20400, ISO 37001, GRI 207	ISO 20400:2017: guides organizations on integrating sustainability within procurement (per ISO 26000). ISO 37001: refers to the anti-bribery management system that targets transparency and trust related to organizations' credibility and reputation. GRI 207: relates to economic topic-specific standards. This standard refers to the economic dimension of organizations' sustainability, focusing on their financial conditions.
Goal 2: Zero Hunger	"End hunger achieve food security and improved nutrition and promote sustainable agriculture." This goal is devoted to increasing rural infrastructure investments, agricultural research, technology development, and plant and livestock gene banks to enhance agricultural productive capacity in developing countries.	ISO 22000 family, ISO 26000, ISO 20400, ISO 34101 series, GRI 411	ISO 22000: relates the requirements for food safety management system by mapping the organizations' needs and controlling food safety hazards. ISO 26000: relates the social responsibility, guiding organizations in assessing their commitments towards the environment and society. ISO 20400 (described in goal 1). ISO 34101: refers to sustainable cocoa. This standard provides management systems for cocoa sustainability production and traceability. GRI 411: relates the impacts on indigenous peoples' rights, proposing remediation plans that organizations can follow.
Goal 3: Good Health and Well-Being	"Ensure healthy lives and promote well-being for all at all ages." This goal relates to health and population, sustainable transport, and National Sustainable Development Strategies (NSDS).	ISO 11137 series, ISO 7153, ISO 37101, GRI 403	ISO 11137-1:2006: specifies the sterilization of health care products on radiation and establishes the requirements for development, validation, and routine control of sterilization processes for medical devices. ISO 7153-1:2016: relates to metal materials to manufacture surgical instruments (general surgery, orthopedics, and dentistry). ISO 37101:2016: specifies the SD in communities, guiding cities and communities to establish SD policies based on a holistic approach.

			GRI 403: relates to occupational health and safety.
Goal 4: Quality Education	"Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all." Education for Sustainable Development (ESD) is an integral element of quality education and an enabler of SD.	ISO 21001, ISO 29993, GRI 404	ISO 21001:2018: offers the requirements and guidance for management systems for educational organizations. ISO 29993: relates to learning services outside formal education, including all types of life-long learning. GRI 404: refers to training and education and organizations' approach for training and upgrading employees' skills, performance, and career development.
Goal 5: Gender Equality	"Achieve gender equality and empower all women and girls". This goal is related to gender equality and women's empowerment.	ISO 26000, SA8000, GRI 404	ISO 26000: (described in goal 2). SA8000: guides organizations to develop, maintain and implement socially acceptable practices in workplaces. GRI 404: (described in goal 4).
Goal 6: Clean Water and Sanitation	"Ensure availability and sustainable management of water and sanitation for all." Water resources suffered from an increase in pollution and severe water stress around the world. Water crises are causing implications for public health, environmental sustainability, food, energy security, and economic development.	ISO 24518, ISO 24521, ISO 30500, GRI 303	ISO 24518:2015: specifies the activities relating to drinking water and wastewater services and water utilities crisis management. ISO 24521:2016: guides the management of essential on-site domestic wastewater services. ISO 30500:2018: refers to non-sewer sanitation systems, establishing the general safety and performance requirements for designing and testing prefabricated integrated treatment units. GRI 303: refers to water and effluents and how organizations manage their uses.
Goal 7: Affordable and Clean Energy	"Ensure access to affordable, reliable, sustainable, and modern energy for all." Energy is a core element for achieving SDGs due to climate change. Then, ensuring affordable, reliable, and sustainable sources of energy will allow better opportunities for future generations.	ISO 50001, EU Ecolabel, GRI 302	ISO 50001: relates to energy management, establishes basic guidelines for organizations committed to addressing its impacts, conserving resources, and improving energy management. EU Ecolabel: a label of environmental excellence, Ecolabel promotes the circular economy, less waste, and CO2 generation and manufacturing processes. GRI 302: presents the general requirements about energy use.
Goal 8: Decent Work and Economic Growth	"Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all."	ISO 45001, SA8000, GRI 201	ISO 45001: relates to occupational health and safety, this standard guides organizations in improving employee safety, reducing workplace risks, and creating safer working conditions. SA8000: (described in goal 5).

	This goal is related to the green economy, sustainable tourism, employment, decent work for all, and social protection.		GRI 201: guides the general requirements on economic performance.
Goal 9: Industry, Innovation, and Infrastructure	"Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation." Related to industry and sustainable transportation, this goal intends to reinforce social and industrial development to promote social objectives and employment creation, gender equality, labor standards, and access to health and education.	ISO 56002, ISO 44001, GRI 203	ISO 56002: provides general guidance for an innovation management system to organizations. ISO 44001:2017: establishes the requirements and framework for collaborative business relationship management systems. GRI 203: refers to the requirements on indirect economic impacts of organizations.
Goal 10: Reduce Inequalities	"Reduce inequality within and among countries." This goal aims to advance societies on complex quality and compliance systems, ensure proper market functioning, protect people's health and safety, and preserve the environment.	ISO 26000, SA8000, SMETA, GRI 405	ISO 26000: (described in goal 2). SA8000: (described in goal 5). SMETA: guides organizations on proving they are working in an ethical manner (ethical audit). GRI 405: refers to general guidance for diversity and equal opportunity requirements for organizations.
Goal 11: Sustainable Cities and Communities	"Make cities and human settlements inclusive, safe, resilient, and sustainable." It is related to disaster risk reduction, sustainable transportation, voluntary local reviews, sustainable cities and human settlements, and NSDS.	ISO 37101, GRI 203	ISO 37101: (described in goal 3). GRI 203: (described in goal 9).
Goal 12: Responsible Consumption and Production	"Ensure sustainable consumption and production patterns." It encompasses chemicals and waste, sustainable consumption and production, and sustainable tourism. The way societies are producing and consuming is fundamental when addressing SD.	ISO 20400, ISO 15392, ISO 20245, GRI 302	ISO 20400: (described in goal 1). ISO 15392: provides general guidance for organizations towards sustainability in buildings and civil engineering works. ISO 20245: refers to the cross-border trade of second-hand goods. GRI 302: (described in goal 7).
Goal 13: Climate Action	"Take urgent action to combat climate change and its impacts". This goal comprises the atmosphere, climate change, small island developing States, and NSDS.	ISO 14001, EMAS, EU Ecolabel, GRI 302	ISO 14001:2015: offers requirements and guidance for environmental management systems. EMAS: The Eco-Management and Audit Scheme was developed by European Commission to assess and report their environmental performance. EU Ecolabel: (described in goal 7). GRI 302: (described in goal 7).
Goal 14:			ISO/TC 234: presents the requirements for standardization in the field of fisheries and aquaculture.

Life Below Water	"Conserve and sustainably use the oceans, seas and marine resources for SD". Within this goal, oceans, seas, small island developing States, and sustainable tourism are addressed.	ISO/TC 234, ISO/TC 8, GRI 305	ISO/TC 8: comprises the ships and marine technology requirements. GRI 305: related to emissions covering greenhouse gases (GHGs).
Goal 15: Life on Land	"Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss." It involves the biodiversity and ecosystems, forests, mountains, NSDS, desertification, land degradation, and drought.	ISO 14055, ISO 14001, EMAS, EU Ecolabel, ISO 38200, GRI 306	ISO 14055-1:2017: presents the good practices framework and guidelines for establishing good practices for combatting land degradation and desertification. ISO 14001: (described in goal 13). EMAS: (described in goal 13). EU Ecolabel: (described in goal 7). ISO 38200: comprises the requirements for chain of custody of wood and wood-based products. GRI 306: refers to effluents and waste management.
Goal 16: Peace, Justice, and strong institutions	Promote peaceful and inclusive societies for SD, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels. This goal includes the information for integrated decision-making and participation, the institutional frameworks and international cooperation for SD, violence against children, and NSDS.	ISO 19600, ISO/DIS 37000, ISO/TC 309, GRI 403	ISO 19600:2014: offers guidelines for compliance management systems. ISO/DIS 37000: presents guidance for governance and ethics in organizations. ISO/TC 309: specifies the standardization for direction, control, and accountability for organizations' governance. GRI 403: (described in goal 3).
Goal 17: Partnership for the Goals	"Strengthen the means of implementation and revitalize the global partnership for SD." It involves capacity development, finance, financial inclusion, multi-stakeholder partnerships and voluntary commitments, science, technology, trade, and NSDS.	GRI 207, ISO	GRI 207: (described in goal 1). ISO: independent, a non-governmental international organization for standardization.

Source: International Organization for Standardization, (ISO, 2019); Global Reporting Initiative (GRI, 2018); Social Accountability (SA8000, 2016); UN, 2020); Sedex Members Ethical Trade Audit (SMETA, 2018); Eco-Management and Audit Scheme (EMAS, 2019); European Union Ecolabel (EU Ecolabel, 2018)

2.3 Establishing criteria for selecting certification bodies

Since certifications are essential for several organizations to achieve their trade goals, there are several motivations to obtain certifications from certification bodies. These enterprises and their auditors suffer pressure from these organizations to get more certifications (Alvarenga et al., 2018). They also audit organizations to see whether they are compliant or not. The assurance of these procedures is complex. Certifications require the certification bodies' auditors' professionalism in being impartial, checking whether an organization's compliance is following the predetermined standard. In assessing quality standards, the audits have a broader task in checking if what the company says it is doing is what they are doing in practice (Hoy and Foley, 2015). In ISO 9000, and the principles for continuous improvements, external auditors are expected to look at evidence and facts rather than only look at what is written in reports when comparing financial audit practices. From the side of organizations, they expect auditors to act more as consultants than auditors. It is noteworthy that auditors underestimate the needed balance related to the continuous improvements and compliance with the standard within the auditing process (Ikram et al., 2020c).

Sustainability certification is granted to an organization after an accredited external auditing service (Tavares de Aquino and Maciel de Melo, 2016). Along the certification process, an organization must present the proofs that it meets certification requirements. The organization already selected the certification/registration body that it is targeting to achieve at this stage. The organization's goal related to this external auditing is to confirm whether the organization meets SD requirements (Castka et al., 2015). The external audit can contribute to the company's opportunities to improve its sustainability practices towards the SDGs (Bian et al., 2020). An organization decides which standards/certification process it will be enrolled in by selecting a specific certification body.

In this study, we selected the top six reputable certification bodies such as BV, SGS, DNVGL, LR, BSI, and TÜV based on their sizes, support of SDGs, competitive prices, a team of quality auditors, financial position, and factors presented in Table 2. We posit the organization's decision is based on many criteria, such as getting industry knowledge, recognition concern, costs reduction, gaining experience, improving quality of communication, among others. These criteria come from both the literature and support from elements of the certification standards. The knowledge of the organization's product and business is grounded in ensuring the auditor understands the client's goals while possessing knowledge of the industry. A CB's reputation is vital as this third parties' involvement helps build confidence in the certification process and its outcomes. There is always a concern with payments and operational outcomes, as this reflects the importance of providing a cost/benefit for the certification. As was noted in the introduction of this study, organizations in developing countries often struggle with available resources for certifications as a lack of resources prohibits certification and therefore progress on meeting the UN SDGs and improvements in corporate social responsibility (CSR) Lamichane et al. (2020). We would like to note the importance of high-quality auditors. Organizations seeking CBs need help solving ambiguous problems, judgment based on prior experience, and good communication. To this end, selecting the correct CB can quickly help get an organization up to speed with technical and management decisions. Additionally, the vast range of auditing services prices and quality levels led are also considered critical factors for an organization to decide which service to hire (USTUN and DEMIRTAS, 2008).

Due to the variety of certification bodies, an organization may face difficulties choosing the best certification body according to its goal. If the organization makes a poor decision in selecting its suppliers and certification body, direct and indirect consequences arise (Chan and Kumar, 2007). In this way, Prajogo et al. (2020) reported the appearance of inconsistencies related to the certification body's quality of service, especially those offered in developing countries. In summary, selecting the wrong supplier can lead the organization to incur operational and financial difficulties.

The establishment of successful partnerships depends on selecting suitable suppliers and strategic choices when managing suppliers (Wang et al., 2020). Selecting the right certification body operationalizes strategic choices, and these decisions involve tangible and intangible criteria. Previous researchers have stated that a general model for decision-making comprising preestablished criteria sets is almost impracticable (Huang and Keskar, 2007). We see a practical approach to identifying criteria based on our review and summary of the literature. Each of these selection criteria has additional sub-criteria. The sub-criteria help to provide a deeper level of insight. We add to this additional expert feedback. The resulting criteria and sub-criteria are in Table 2 and Table 3.

Table 2: Comparative Analysis of Top Certification Bodies

Certification Body	Location	Support to Sustainable Development Goals (SDGs)	Number of employees	Providing Services to Countries	Years of Business	Customers	Assessment, Rating and Awards	Revenue in 2019 (USD)	Source
DNVGL	Norway	√	12000	+100	156	100,000	NA	2386.89 million USD	(DNVGL, 2019)
BV	France	√	+78000	+140	192	400,000	MSCI (Rated A), Dow Jones Sustainability Index (B Rated), Sustainanalytics (Low Risk, Ranked 3)	6041.77 million USD	(Bureau Veritas, 2019)
SGS	Switzerland	√	+94000	+140	140	200,000	Member of Dow Jones Sustainability Index, ASM sustainability award Gold class, Received the new platinum rating from Ecovadis	7243.99 million USD	(SGS, 2019)
LR	United Kingdom	√	8000	80	260	60,000	NA	945.1 million USD	(Lloyd's Register, 2019)
BSI	United Kingdom	√	5089	195	118	84,000	NA	725.53 million USD	(BSI, 2019)
TÜV	Germany	√	25000	50	150	90000	NA	2469.86 million USD	(TÜV SÜD, 2019)

DNVGL=*Det Norske Veritas Germanischer Lloyd*, **BV**= *Bureau Veritas*, **SGS**= *Société Générale de Surveillance*

LR= *Lloyds Register*, **BSI**= *British Standards Institution*, **TUV**= *Technischer Überwachungs-Verein*

Table 3: Decision Sub-criteria of Certification Bodies Selection and Supporting Information

Criteria	Sub-Criteria	Supporting Information	Authors
Reputation (REP)	The financial position of certification bodies gives confidence to customers. (RC1) The level of acceptance of the certification bodies in the market. (RC2) The global recognition of certification body. (RC3)	At ISO 17021, topic 4.5.1, the CB provides public access, or disclosure of financial situation, CB's hierarchical structure, the license status, i.e. (the granting, maintenance of certification). These procedures are adopted to gain the confidence and credibility of the CB.	(Chan & Chan, 2010; Haeri & Rezaei, 2019)

	The total number of auditors working in the certification body. (RC4) The total number of customers the certification body has. (RC5)		
Operational outcomes (OP)	Improve effectiveness and consistency of the operating procedures (OP1) Improved conformance to specification (OP2) Reduction in customer complaints (OP3) The effective tracking of documents and records (OP4) The reduction in managerial losses (OP5) Creating a transparent organizational culture (OP6)	Collective action theory: Companies may find incentives in private benefits for engaging a sustainability certification. Among the benefits are customer loyalty, corporate reputation, regulatory relief, improved product prices, and increased market shares. Moreover, the benefits from sustainability certification include improvements in economic efficiency by integrating sustainability into products' supply. Another benefit is reducing the coordination costs that arise in inter-firm relations, harmonizing sourcing requirements, and reaching production scales.	(D. I. Prajogo, 2011; Cai & Jun, 2018; Kannan et al., 2014)
Quality of Auditor (QA)	The auditors have much knowledge about the management standards (QA1) The auditors are responsive to the needs of the organization (QA2) The auditors have a valid auditing license (QA3) The auditors conduct the auditing work in an appropriate way (QA4) The auditors have high ethical standards (QA5) The auditors have practical communication skills (QA6)	At ISO 10019, topic 4.2.6, it is finding the auditor's requirements related to work experience (such as technical, management, and professional features). Some of the experience's features recommended for the auditor are related to solving problems, judgment ability, and communication skills with parties. The auditor's profile may consider one or more specified requirements.	(Poksinska et al., 2006; Power & Terziovski, 2007)
Contract Condition (CC)	The certification body's requirement on the maturity of the standard (CC1) The clarification of term and conditions (CC2) To comply with certification body rules and policies (CC3) The cancellation/termination of contract (CC4) Forbidden to disclose confidential information without prior written approval (CC5) Limitation of Liability and Indemnity of certification body (CC6)	In CB25, recommendations about consultancy selection procedures are found by analyzing the strategic adaptation concerning an organization's target goals, organizational culture, and contract conditions.	(Heras-Saizarbitoria & Boiral, 2013; Hernandez-Vivanco et al., 2019)

Quality of Implementation (QI)	Documented the quality policy and standard operating procedures (QI1) Review and evaluation of management system documents (QI2) Conducting audits on a semiannual basis to check the performance (QI3) Conducting the awareness session about the importance of management standards (QI4) Properly communicate the audit result to the organization's top management team (QI5) To check management review meeting minutes (QI6) Follow up the corrective actions on non-conformances (QI7) Corruption free and transparency in implementation (QI8)	The CB25 states that auditors need to prove their technical competency and professional profiles implemented in previous projects. These proofs ensure that the auditor can manage the project under his/her responsibility and ensure the management standard.	(Ivanova et al., 2014; Ikram, Sroufe, et al., 2020)
Payment Method & Cost (PC)	The price for the overall three-year program of certification audit and surveillance (PC1) Payment methods and conditions that fit the internal planning of an organization (PC2) Location influences on cost, for example, auditors transport costs (PC3)	CB25 stated the requirements for selecting a consultant, such as defining projects' deadlines and determining prices based on cost/benefits, previously formalizing the contract. At ISO 10019 (annex A, topic A.1.2.e), it is expressed that the costs for supporting all activities must be written in the contract.	(Castka, et al., 2020; Jamal & Sunder, 2011)
Knowledge about the client's products and business (KN)	Ability to determine the specific cultural and social customs related to the company's production process (KN1) Knowledge of the sector of activity (by type of process) and the associated risks concerning the place of the company processes (KN2) Technical knowledge is needed to address constraints that may emerge from the partnership (KN3)	In ISO 10019 (topic 4.2.5), recommendations about the CB should possess reasonable knowledge on an organization's processes and products and know customers' expectations previously of offering the consultancy service. Moreover, consultants must understand and capture core points from the industry in which the company operates.	(Power & Terziovski, 2005; Prajogo et al., 2016; Azapagic, 2015)

Note: CB= Certification body

3. Research Methods

Within this study, we have applied Grey Delphi, Grey AHP, and GADA, respectively, to sort out the most relevant indicators from the literature and assess the influence of shortlisting the main criteria and sub-criteria. The schematic of the methodology is in Figure 3. The main advantage of grey system theory is that it ensures accuracy with a small set of data and partial information (Javed et al., 2020). The initial step is to assess the main criteria involved in selecting certification bodies that help align with and achieve SDGs through a thorough literature review. To shortlist only the relevant and most important indicators, Grey Delphi is employed. In fuzzy Delphi, experts were asked to score each indicator for screening out fewer essential barriers. The next step is to apply GAHP for obtaining the relative importance of finalized CB indicators and sub-indicators. Finally, the overall ranking of sub-criteria is achieved by summing each alternative's respective weights under individual criteria involved in selecting the right certification body. Then, GADA obtains comparative weights of the top six most adoptable CB's.

To the best of the author's knowledge, there is no standard available for how many experts should be involved in this analysis. Prior studies used a different number of experts for data analysis to obtain reliable results. For example, Shete et al. (2020) selected six experts for data analysis for sustainable supply chain innovation to achieve sustainability in supply chain management. Whereas, Torkabadi et al. (2018) selected five experts using the FAHP to identify the supply chain barriers to improve sustainable production and consumption.

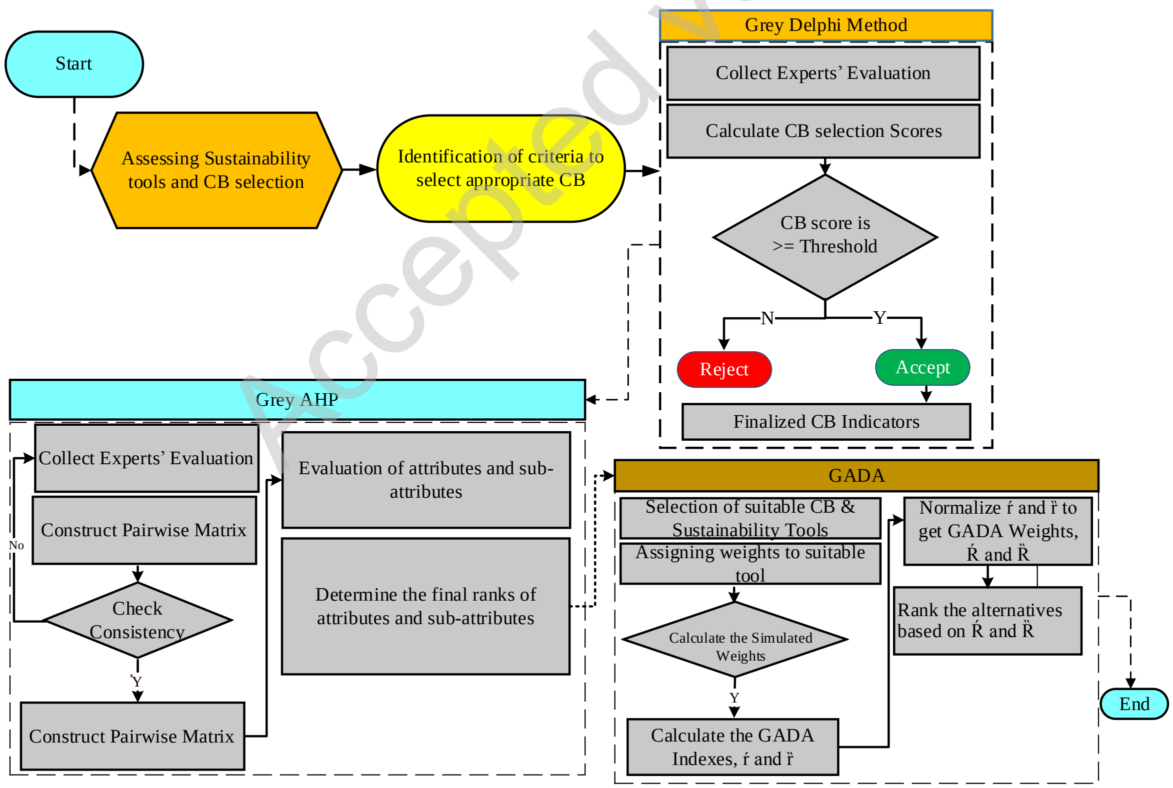


Figure 3: Grey Integrated Research Methodology

Note: CB=Certification body, GAHP= Grey Analytical Hierarchy Process, GADA=Grey Absolute Decision Analysis, R' =Relative Weight

Therefore, in this study, eight academic and industrial experts were involved. We asked participants through an online survey: academic professors, research analysts, policymakers, stakeholders, and managers to give a score based on TFN and grey linguistic variables to prioritize the certification body's selection indicators. The experts have more than 15 years of work experience, and they are very familiar with the industry, sustainability standards, and certification research context. Their details are in the supplementary information (S1), whereas the questionnaire used for experts' feedback is in the supplementary information (S2).

3.1 Grey Delphi Method

We used the Grey Delphi method by combining Deng's Grey system Theory and the Delphi method (Ma et al., 2011) to select the evaluation indicator using the grey weight function based on Delphi questionnaires. This study employs grey numbers, shown in Table 4, to construct the membership degree function. The development process of the Grey Delphi Method comprises four steps, which are as follows:

Step 1: This step involves the development and defining the range values of *s* grey classes for evaluation indicator *j* $[a_j^1, a_j^s]$ and split into *k* grey classes.

Step 2: As per the evaluation criterion, half trapezoidal whitening weight function is used for *k* and *s* grey classes and *k*=1. The formulation is as follows:

$$f_j^1(x)=\begin{cases} 1, & x \le a_j^1 \\ \frac{b_j^1-x}{b_j^1-b_j^1}, & a_j^1 < x \le b_j^1, \\ 0 & x > b_j^1, \end{cases} \tag{1}$$

$$f_j^s(x)=\begin{cases} 0, & x < a_{j,}^s \\ \frac{x-b_j^s}{b_j^s-b_j^s}, & a_j^s \le x < b_j^s, \\ 1 & x \ge b_j^s, \end{cases} \tag{2}$$

The triangular whitening weight function is used for k=*l* (*l*=2, 3…*s*-1). The formulation is as follows:

$$f_j^k(x)=\begin{cases} 0, & x \notin [a_{j,}^k\ b_j^k], \\ \frac{x-a_j^k}{\lambda_j^k-a_j^k}, & x \in [a_{j,}^k\ \lambda_j^k], \\ \frac{b_j^k-x}{b_j^k-\lambda_j^k} & x \in [\lambda_j^k, a_j^k], \end{cases} \tag{3}$$

Where, $\lambda_j^k=[a_{j+}^k\ b_j^k]/2$. a_j^k represent the initial point is called the lower bound of *kth* grey classes, whereas the upper bound of *kth* grey classes is denoted by b_j^k respectively.

Step 3: The calculation of synthetic coefficient denoted by σ_j^k is done in this step. The formulation equation is:

$$\sigma_j^k=\sum_{p=1}^{l} f_j^k(x).\ \eta_j^p \tag{4}$$

Where $f_j^k(x)$ represents the whitening weight function of k grey class for indicator *j*, whereas p presents the experts' feedback. Here, η_j^k indicates the wight of *j* indicator in synthetic, clustering, the name and number of experts' feedback in category *p*.

Step 4: The final step involves the identification of evaluated decision vectors to check whether the indicator as per set judge criterion $max_{1\leq k\leq s}\left\{\sigma_j^k\right\}=\left\{\sigma_j^{k^*}\right\}$ belongs to class k^*.

Table 4: Grey Delphi Scale

Grey class	Linguistics	Range
1	Not Very Important	(1, 2.5)
2	Not Important	(0.5, 3.5)
3	Undecided	(1.5, 4.5)
4	Important	(2.5, 5.5)
5	Very Important	(3.5, 5)

3.2 Grey Analytical Hierarchy Process

We employed Grey Analytical Hierarchy Process (G-AHP) by integrating Grey System theory with Saaty's AHP to determine the weights. In Saaty's AHP, a 1-9 scale is used to construct pairwise judgment matrices. Whereas, in GAHP five grade-level scales such as (1, 3, 5, 7, 9) are used to evaluate the experts' feedback. The details of GAHP linguistic values are in Table 5. The main advantage of using GAHP, over AHP is that GAHP provides a more accurate result with a partial and uncertain set of expert feedback (Zareinejad et al., 2014). Moreover, we select GAHP over Fuzzy AHP because the former already includes fuzzy conditions and, therefore, expertly handles fuzziness in the decision-making.

The development process of G-AHP comprises four following steps:

Step 1: Draw the hierarchal structure by defining the study goal, main criteria, sub-criteria, and Alternatives for the problem to be solved

Step 2: Construct matrices of pairwise comparisons with the help of the grey scale presented in Table 5. Let E indicates the G-AHP pairwise comparison matrix as follows:

$$E=\begin{bmatrix}\otimes G_{11} & \cdots & \otimes G_{1n}\\ \vdots & \ddots & \vdots\\ \otimes G_{m1} & \cdots & \otimes G_{mn}\end{bmatrix}=\begin{bmatrix}\left(\underline{G}_{11,}\overline{G}_{11}\right) & \cdots & \left(\underline{G}_{1n,}\overline{G}_{1n}\right)\\ \vdots & \ddots & \vdots\\ \left(\underline{G}_{m1,}\overline{G}_{m1}\right) & \cdots & \left(\underline{G}_{mn,}\overline{G}_{mn}\right)\end{bmatrix} \quad (5)$$

Where grey number is denoted by $\otimes G_{ij}$, $\otimes G_{ij}$ =1 if i=j, and $\otimes \overline{G_{ij}}$ and $\otimes \underline{G_{ij}}$ are the upper and lower limit of $\otimes G_{ij}$ respectively.

Step 3: The normalization of the G-AHP pairwise matrix as in presented as follows:

$$E^*=\begin{bmatrix}\otimes G_{11}{}^* & \cdots & \otimes G_{1n}{}^*\\ \vdots & \ddots & \vdots\\ \otimes G_{m1}{}^* & \cdots & \otimes G_{mn}{}^*\end{bmatrix}=\begin{bmatrix}\left(\underline{G}_{11,}{}^*\overline{G}_{11}{}^*\right) & \cdots & \left(\underline{G}_{1n,}{}^*\overline{G}_{1n}{}^*\right)\\ \vdots & \ddots & \vdots\\ \left(\underline{G}_{m1,}{}^*\overline{G}_{m1}{}^*\right) & \cdots & \left(\underline{G}_{mn,}{}^*\overline{G}_{mn}{}^*\right)\end{bmatrix} \quad (6)$$

The upper and lower limit $\otimes \overline{G_{11}}^*$ and $\otimes \underline{G_{11}}^*$ are the normalized limits obtained as follows:

$$\underline{G_{11}}^*=\left[\frac{2\underline{G}_{ij}}{\sum_{i=1}^{m}\underline{G}_{ij}+\sum_{i=1}^{m}\overline{G_{11}}}\right] \quad (7)$$

$$\overline{G_{11}}^*=\left[\frac{2\overline{G_{ij}}}{\sum_{i=1}^{m}\underline{G}_{ij}+\sum_{i=1}^{m}\overline{G_{11}}}\right] \quad (8)$$

Step 4: Finally, the calculation of relative grey weights for the normalized lower and upper limit as presented as:

$$\otimes \underline{W_i} = \frac{\Sigma\, \underline{G_{ij}^{*}}}{n} \tag{9}$$

$$\otimes \overline{W_i} = \frac{\Sigma\, \overline{G_{ij}^{*}}}{n} \tag{10}$$

By converting the grey weights $\otimes W_i$ to definite weights W_i as:

$$W_i = \left[(1 - \lambda) * \otimes \underline{W_i}\right] + \left[\lambda * \otimes \overline{W_i}\right] \tag{11}$$

λ is the grey coefficient whose value is set as 0.5. The value of the grey coefficient can be varied based on the decision-makers.

The consistency ratio (CR) is an important indicator to check the reliability of each GAHP pairwise matrices' results was computed. The acceptability of result by using GAHP, when the value of CR records less than 0.1, otherwise the obtained results consider unreliable and process of obtaining experts' feedback must be revised (Ikram et al., 2020)

Table 5. Grey Analytical Hierarchy Process Scale

Scale	Linguistics	⊗*G*
1	Not Very Important	(1, 2)
3	Not Important	(2, 4)
5	Undecided	(4, 6)
7	Important	(6, 8)
9	Very Important	(8, 10)

3.3 Grey Absolute Decision Analysis (GADA)

Deng introduced the concept of grey system theory in 1981. It has been widely used in these applications in various areas, e.g., healthcare, environmental sustainability, supply chain, publications growth analysis, project management engineering, and energy. In the grey systems paradigm, perfect data does not exist (Javed et al., 2020b). Data that may seem perfectly useful may appear partially useful tomorrow when new information is made available. There are many sources of uncertainty in primary data (Sharma and Seal, 2019). No quantitative information can ideally represent a qualitative concept. Thus, the data's uncertainty or impreciseness corresponds to subjectivity or qualitative concepts, and this can represent the practices looked at in this study.

The Grey Absolute Decision Analysis (GADA) model, proposed by Javed et al. (2020), is a new development in the theory of Multi-Attribute Decision Making (MADM). It is a breakthrough as, unlike conventional methods, it does not require the normalization of data. Thus, the model is practically dimensionless and convenient to use. The key advantage of this method lies in its simplicity and independence over the criteria dimensions. The construction of GADA comprises six steps:

Step 1. Preparing data reporting

This step involves the recording of responses from the expert and constructing a matrix of response decision measurement represented by $[a(k)]$ for "higher the better" criteria, C(k) for each alternative, $A_{(j)}$ for an optimal solution.

$$A_{(1)} \quad \cdots \quad A_{(S)}$$

$$a(k)=\begin{array}{c}E_1\\ \ldots\\ E_N\end{array}\begin{bmatrix}a_{1(1)} & \cdots & a_{1(S)}\\ \vdots & \ddots & \vdots\\ a_{N(1)} & \cdots & a_{N(S)}\end{bmatrix} \tag{12}$$

The reciprocal operation is needed to separate better from lower criteria. The above matrix can be written as:

$$\begin{array}{cc} & \begin{array}{ccc}A_{(1)} & \ldots & A_{(S)}\end{array}\\ r(k)=\begin{array}{c}E_1\\ \ldots\\ E_N\end{array} & \begin{bmatrix}r_{1(1)} & \cdots & r_{S(1)}\\ \vdots & \ddots & \vdots\\ r_{1(N)} & \cdots & r_{S(N)}\end{bmatrix}\end{array} \tag{13}$$

While it should be observed that in the case of lower the better criteria, we need to Reciprocate the Operation (RO) for data analysis, thus, we can say $r_j = RO(a_i)$. Whereas, when in the 'higher the better case, the feedback from experts can be used for data analysis thus $[r_{ij}]=[a_{ij}]$. The GADA approach's main advantage is that there is no need for data normalization due to its dimensionless property.

Step 2. **Calculating alpha values and AGRG Pairwise Comparison Matrix**

Absolute Grey Relational Grade (ε) is calculated among E_i, and can be written as: $E_s; i=1,2,3....N$, which is the AGR pairwise comparison matrix denoted by $[\varepsilon]$ and used to obtain the pairwise comparison matrix. Further, Javed alpha (α) is measured by the average of each row yields (Javed and Liu, 2019). When the value of ε is equal to 1, which shows a strong association, and the ε value at 0.5, presents a poor and weak relationship, the overall value of ε within E_i would always be equal to 1. If the value of ε is recorded less than or near to 0.5, which lies in the unacceptable region, which turns into recollecting the expert's feedback. For one criterion, the equation can be developed as:

$$\begin{array}{cc} & \begin{array}{ccc}E_1 & \ldots & E_N\end{array}\\ \begin{array}{c}E_1\\ \ldots\\ E_N\end{array} & \begin{bmatrix}\varepsilon_{11} & \cdots & \varepsilon_{1N}\\ \vdots & \ddots & \vdots\\ \varepsilon_{N1} & \cdots & \varepsilon_{NN}\end{bmatrix}\begin{bmatrix}\alpha_1\\ \ldots\\ \alpha_N\end{bmatrix}\begin{bmatrix}\sqrt{\alpha_1}\\ \ldots\\ \sqrt{\alpha_N}\end{bmatrix}\end{array} \tag{14}$$

Where $\alpha_i = \frac{1}{N}(\varepsilon_{i1}+\varepsilon_{i2}....+\varepsilon_{iN})$ is called *Javed's alpha*. The equation can be developed for *M* criteria $C(k); K=1,2,....M$.

$$\begin{array}{cc} & \begin{array}{ccc}C(1) & \ldots & C(M)\end{array}\\ \begin{array}{c}E_1\\ \ldots\\ E_N\end{array} & \begin{bmatrix}\sqrt{\alpha_1(1)} & \cdots & \sqrt{\alpha_1(M)}\\ \vdots & \ddots & \vdots\\ \sqrt{\alpha_N}(1) & \cdots & \sqrt{\alpha_N(M)}\end{bmatrix}\end{array} \tag{15}$$

Step 3. Calculating perceived criteria weights

Now collect perceived weights of criteria from the experts, whereas, *k*=1,2,3,…, *M* (*M*) is total number of criteria

$$\begin{array}{cc} & \begin{array}{ccc}E_1 & \ldots & E_N\end{array}\\ \begin{array}{c}C(1)\\ \ldots\\ C(M)\end{array} & \begin{bmatrix}\beta_1(1) & \cdots & \beta_N(1)\\ \vdots & \ddots & \vdots\\ \beta_1(M) & \cdots & \beta_N(M)\end{bmatrix}\end{array} \tag{16}$$

$$\begin{array}{c} C(1) \\ \dots \\ C(M) \end{array} \begin{bmatrix} \beta(1) \\ \dots \\ \beta(M) \end{bmatrix}$$

Where, $\beta(k)$ =geometric mean $\beta_1(k), \beta_2(k), \beta_3(k), \dots.. \beta_{N(k)}$ and $\sum \beta(k) = 1$.

Step 4. Calculating simulated criteria's weights

Calculating Simulated weight criteria's $\delta(k)$ by multiplying the $\theta(k)\,\beta_i(k)$ and $\sqrt{\alpha_i(k)}$ to get $\Theta(\mathrm{k})$. This approach of obtaining $\Theta(\mathrm{k})$ was proposed by Javed et al. (2020).

$$\begin{array}{cc} & \begin{array}{ccc} E_1 & \dots & E_N \end{array} \\ \begin{array}{c} C(1) \\ \dots \\ C(M) \end{array} & \begin{bmatrix} \Theta_1(1) & \cdots & \Theta_N(1) \\ \vdots & \ddots & \vdots \\ \Theta_1(M) & \cdots & \Theta_N(M) \end{bmatrix} \end{array} \tag{17}$$

$$\begin{array}{c} C(1) \\ \dots \\ C(M) \end{array} \begin{bmatrix} \delta(1) \\ \dots \\ \delta(M) \end{bmatrix}$$

Where, $\delta(k)$ = geometric mean $\Theta_1(k), \Theta_2(k) \dots \Theta_N(k)$ $\theta_i(k) = \omega_i(k) \times \sqrt{\alpha_i(k)}\,.\,\beta_i(k)$

Step 5. Calculating the overall weight of criterion against each alternative

In the MCDA approach, each criterion's weight is obtained by transforming the individual's weight to group weight with the help of a geometric mean (Stević et al., 2018). Hence, in GADA, the formula of overall weight calculation is as follows:

$$\begin{array}{cc} & \begin{array}{ccc} A_1 & \dots & A_S \end{array} \\ \begin{array}{c} C(1) \\ \dots \\ C(M) \end{array} & \begin{bmatrix} \grave{\mathrm{r}}_1(1) & \cdots & \grave{\mathrm{r}}_N(1) \\ \vdots & \ddots & \vdots \\ \grave{\mathrm{r}}_1(M) & \cdots & \grave{\mathrm{r}}_N(M) \end{bmatrix} \end{array} \tag{18}$$

Where for each criterion $C(k)$ and for $j = 1, 2, 3, 4...., S$.

$$\grave{\grave{\mathrm{r}}}_j = \left(\prod_{j=1}^{S} \grave{\mathrm{r}}_j^{\delta_k}\right)^{1/\sum_{k=1}^{M} \delta_k} \tag{19}$$

The relative weights $\grave{\grave{\mathrm{R}}}_j$ would be yielded by normalization of weights of alternatives within each criterion, can be written as:

$$\grave{\grave{\mathrm{R}}}_j = \frac{\grave{\grave{\mathrm{r}}}_j}{\sum_{j=1}^{S} \grave{\grave{\mathrm{r}}}_j} \tag{20}$$

The weights mentioned above can be arranged in a definite sequence to obtain the local rankings of alternatives.

Step 6. Calculating the relative weight of each criterion against each alternative

In the final step of the GADA construction method, the calculation of global weights is developed by aggregating the simulated weights of criteria such as:

$$\begin{array}{cc} & \begin{array}{ccc} A_1 & \dots & A_S \end{array} \\ \begin{bmatrix} \ddot{\mathrm{r}} \\ \ddot{\mathrm{R}} \\ Rank \end{bmatrix} = & \begin{bmatrix} \ddot{\mathrm{r}}_j & \cdots & \ddot{\mathrm{r}}_j \\ \ddot{\mathrm{R}}_j & \ddots & \ddot{\mathrm{R}}_j \\ - & \cdots & - \end{bmatrix} \end{array} \tag{21}$$

Where,

$$r_j'' = \left(\prod_{j=1}^{S} r_j'^{\delta(k)} \right)^{1/\sum_{K=1}^{M} \delta(k)} \tag{22}$$

With the normalized vector,

$$\ddot{\mathrm{R}}_j = \frac{\ddot{\mathrm{r}}_j}{\sum_{j=1}^{S} \ddot{\mathrm{r}}_j} \tag{23}$$

Later, $\ddot{\mathrm{R}}_j$ can be utilized to prioritize suitable S alternatives in order to provide the optimal solution.

4. Results and Discussion

The results of this study comprised three phases. In the first phase, the result of Grey Fuzzy is developed in which the main criteria and sub-criteria of certification bodies are finalized. The second phase presents the GAHP results in which the main and sub-indicators of certification bodies are ranked as per the experts' relative importance. Finally, the ranking of alternatives to show which certification body is the best option for managers to select to achieve the SGDs is in the last phase of the study.

4.1 Grey Delphi results

We used the five grey classes from 1 to 5 with range [1, 2.5], [0.5, 3.5], [1.5, 4.5], [2.5, 5.5] and [3.5,5] which represents not very important, no important, undecided, important, and very important, respectively. Further, this study calculated the synthetic clustering coefficient σ_j^k to obtain the values of decision vectors of certification bodies indicators. To check the validity of Grey Delphi result, there are two selection parameters: (a) If the important or very important (4 and 5) classes have maximum value in the vector decision that belongs to class k*, this certification body indicator is accepted. Whereas (b) If the class's value, namely important and very important, which is the ratio of class k* is more than 1.0 or over 50% degree except for undecided class, then this certification body indicator is accepted. Table 6 presents the eleven certification bodies' indicators. In contrast, seven indicators are selected, and the rest of the four certification bodies' indicators are not included as they do not meet the acceptability criteria. According to the expert judgmental and evaluation approach, the four indicators, such as political instability, location, ownership, institutional corruption, and nepotism, are rejected because they do not fit within the study's scope.

Table 6: Result of the Grey Delphi Method

Indicators	Grey decision vector	Selected √/ Rejected ×	Indicator code

Reputation	(0.67, 4.33, 8.67, 5.67, 2)	√	REP
Operational outcomes	(1.67, 3.67, 6.33, 6.67, 2.67)	√	OP
Location	(2.33, 6, 7.33, 4.67, 1)	×	
Quality of Auditor	(3, 8, 7.33, 2.67, 0.33)	√	QA
Contract Condition	(0, 0.67, 3.67, 7.67, 7.67)	√	CC
Political instability	(0, 3.33, 10.67, 5.67, 1.67)	×	
Quality of Implementation	(6.67, 4.33, 8.67, 8.67, 2)	√	QI
Ownership	(5.8, 5.33, 2, 0.33)	×	
Payment Method & Cost	(3, 4, 3.67, 6, 3.67)	√	PC
Institutional corruption and nepotism	(0, 2, 7.33, 7, 4.33	×	
Knowledge of products and business	(1.33, 2.33, 5.67, 8.33, 3.33)	√	KN

4.2 Grey analytical hierarchy process results

In this study, GAHP is used to compute the weights of main and sub-indicators to select certification bodies to align with SDGs. Firstly, we draw the hierarchical structure of the problem presented in Figure 4. There are three steps involved in GAHP. The first step involves the study's goal to be achieved; second, a ranking of main and sub-criteria of top five certification bodies worldwide, which ensure the achievement of SD; finally, the overall ranking by using GAHP is presented in the third section respectively.

4.2.1 Ranking of main indicators

The relative weights of criteria and sub-criteria are obtained by solving the pairwise matrix. The results are in Table 7. It can be seen that the Reputation (REP) received the highest weight of 21.20 % followed by Payment Method & Cost (PC) 17.04%, Operational outcomes (OP) 16.10%, Quality of Auditor (QA) 14.06%, Quality of Implementation (QI) 13.87%, Contract Condition (CC) 10.20% and Knowledge of products and business (KN) 7.53%. It must be noted that the difference between the weights of all the main criteria is not more than 3%. Hence, we can say that all of these barriers are important. Nonetheless, REP is relatively more significant, and KN is comparatively less significant.

We can see why reputation is important as certification bodies help with transparency, disclosure of information, and build confidence in the CB, certification process, and its outcomes. This visibility is helpful to all parties and promotes confidence in the certification process of reputable CBs. There is good face validity with payments and operational outcomes also ranking high. These support the importance of providing a cost/benefit for the certification process and strong value proposition. As was noted in the introduction of this study, organizations in developing countries often struggle with available resources for certifications as a lack of resources prohibits certification and therefore progress on meeting the UN SDGs and improvements in corporate social responsibility (CSR) Lamichane et al. (2020). We would like to note the importance of high-quality auditors. Organizations seeking CB need help quickly solving problems, good judgment, and communication with these third parties, so resources are not wasted. The CB can quickly help with technical and management options based on prior experience. The quality of implementation, contract, and knowledge of the organization's product and business is grounded in ensuring the auditor can manage the project, understand the goals and organizational culture of the client while possessing practical knowledge of the industry at a macro-level and the client's products and processes at a micro-level.

Table 7: Ranking of selecting Certification Body criteria

Main Criteria	Code	Weight	Rank
Reputation	REP	0.2120	1st
Paymen Method & Cost	PC	0.1704	2nd
Operational Outcomes	OP	0.1610	3rd
Quality of Auditor	QA	0.1406	4th
Quality of Implementation	QI	0.1387	5th
Contract Condition	CC	0.1020	6th
Knowledg of Products & Business	KN	0.0753	7th

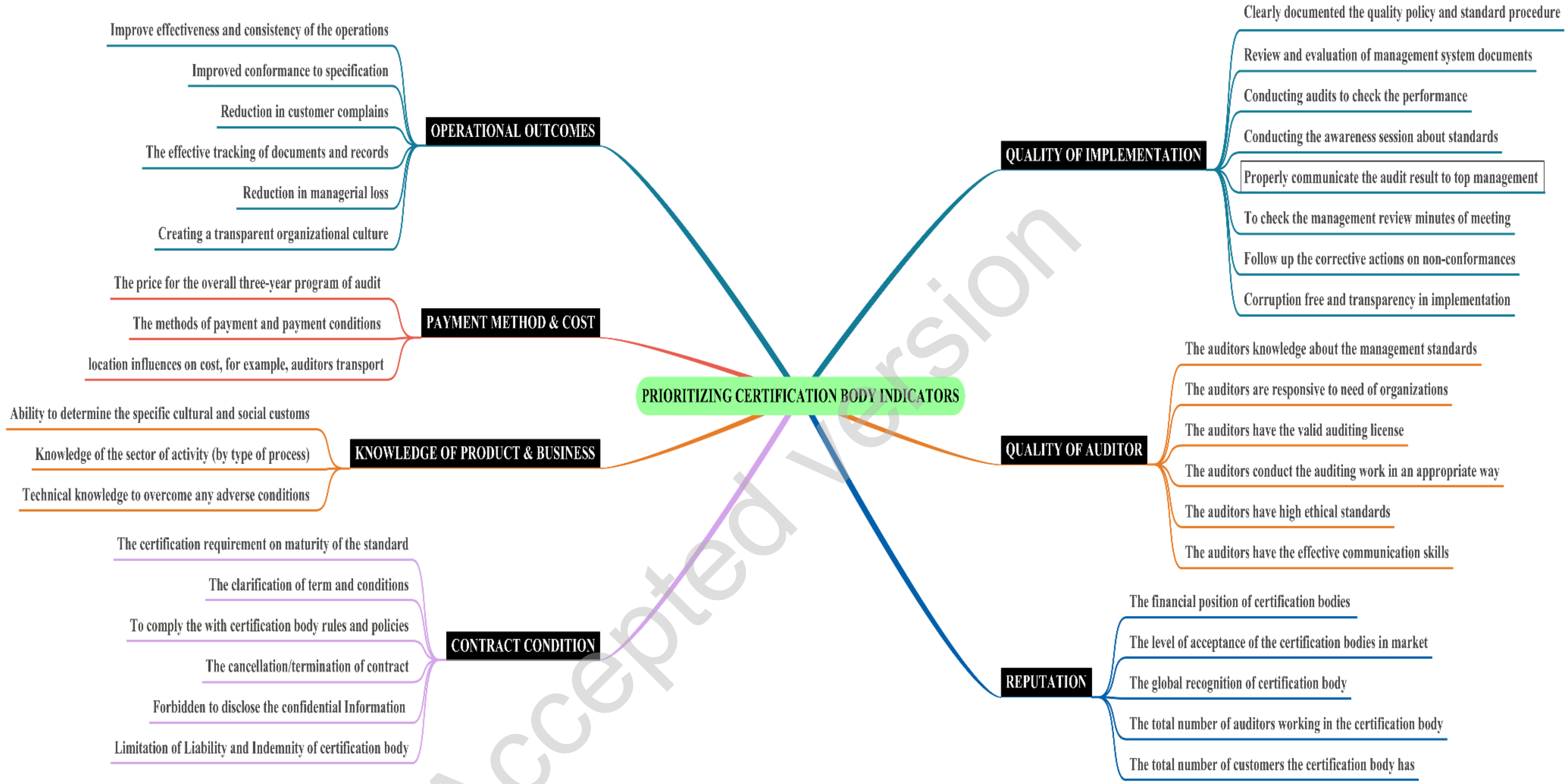


Figure 4: Hierarchal Structure Operationalized in this Study; *source (self-developed)*

4.2.2 Ranking of sub-Indicators

Once we obtained the main criteria weights, we now quantify the degree to which each indicator affects each certification body alternative. To do so, we have constructed 37 pairwise comparison matrices, one for each category of indicator. Next, experts were asked to compare sub-criteria under each category. The weights of sub-criteria under each category are calculated by solving these matrices, as shown in Fig 5-10. It should be noted that any sub-criteria receiving higher weight shall be considered more affected for the selection of certification body to align with and help achieve SD. Hence, the lower the weights, the least effective the rankings of the sub-indicators.

Under REP, weights received by sub-criteria (from higher to lower) are as follows: The level of acceptance of the certification bodies in the market (REP2) 30.84%> the global recognition of certification body (REP3) 23.13%> can be seen in Figure 5. The total number of auditors working in the certification body (REP4) 16.20% > The total number of customers the certification body has (REP5) 15.10% > The financial position of certification bodies to give more confidence to customers (REP1) 14.73%. Other studies that investigated the service sector (Ristono et al., 2018) observed that a third party's reputation plays a vital role in supplier selection. Since the process of certifying the management standard is a bit complex and expensive, organizations rely on reputable certification firms and qualified auditors to effectively implement sustainability certifications. Moreover, certification providers transfer value and credibility to accreditation companies. Therefore, selecting reputable certification bodies ensures internal and external stakeholders that the accreditation body effectively follows management standards (Alvarenga et al., 2018).

Under OP, the sequence is: Reduction in customer complaints (OP3) 25.02%> Improve effectiveness and consistency of the operating procedures (OP1) 23.94%> The reduction in managerial losses (OP5) 18.02%> The effective tracking of documents and records (OP4) 12.51%> Improved conformance to specification (OP2) 10.96%> Creating a transparent organizational culture (OP6) 9.52% presented in Figure 6. According to other studies, Ikram et al. (2020), Chkanikova & Sroufe (2020), our result indicates that the third-party sustainability certification enables dynamic capabilities, improves the production process, reduction in customer complaints, customer satisfaction, and opportunity for the organization to create competitive advantage.

The following sequence was obtained for a pairwise comparison of the Quality of Auditor category (QA): Figure 7 represents the ranking of Quality of Auditor sub-criteria category in which the auditors know the management standards (QA1) 32.30%> The auditors conduct the auditing work in an appropriate way (QA4) 20.21%> The auditors have high ethical standards (QA5) 14.64%> The auditors are responsive to need of organizations (QA2) 13.24%> The auditors have the valid auditing license (QA3) 11.01%> The auditors have practical communication skills (QA6) 8.61%. Previous studies Wiengarten et al. (2017), Singh et al. (2011) endorse our result in which effective implementation of ISO 9001 management standard needs to be handled by the quality auditor, which has an immediate positive effect on organization production outcome. External auditors' expertise, such as practical communication skills, ethical behavior, and enough knowledge about management standards, may boost the company's confidence and willingness to adopt sustainability standards (Hernandez, 2010).

The pairwise comparison of Contract of Condition (CC) presents the following sequence in Figure 8: The clarification of term and conditions (CC2) 28.06%> The cancellation/termination of the contract (CC4) 25.93%> Forbidden to disclose this confidential information to any person or entity without the prior written approval (CC5)

14.90> Limitation of Liability and Indemnity of certification body (CC6) 10.59%> To comply the with certification body rules and policies (CC3) 11%> The certification body's requirement on the maturity of the standard (CC1) 9.940%. Cândido et al. (2019) conducted a study to investigate if the certified companies may lose/withdraw their ISO certifications due to compliance issue with CB rule and regulation, cancellation of the contract because of less economic benefits, and facing difficulties to manage on-time cost and expenditures of the audit fee.

Figure 9 shows the following sequences of sub-criteria regarding the Under Quality of Implementation (QI): Clearly documented the quality policy and standard operating procedures (QI1) come first in the row and obtained the higher weights of 21.88% within the category followed by> Conducting audits at semiannual basis to check the performance (QI3) 16.38%> Conducting the awareness session about the importance of management standards (QI4) 13.66%> Follow up the corrective actions on non-conformances (QI7) 12.02%> Review and evaluation of management system documents (QI2) 10.44%, corruption-free and transparency in implementation (QI8) 9.671%> Properly communicate the audit result to organization top management team (QI5) 9.360%> and To check management's review of the minutes from meeting (Q16) received the lowest weight 6.6000 % within the category respectively. The certification firms adopt a systematic management approach through identification and evaluate the indicators that assist the effective implementation of management standards. Therefore, the sufficient quality of management standards implementation needs to overcome the repetition/duplication of documents through integration techniques to create synergy between systems (Ikram et al., 2020).

Under the Payment Method and Cost (PC) category, the weights received by sub-criteria (from higher to lower) are in Figure 10: The methods of payment and payment conditions that fit the internal planning of the organization (PC2) received the highest weight of 39.10% followed by> The price for the overall three-year program of certification audit and surveillance (PC1) 36.23%> location influences on cost, for example, auditors transport costs (PC3) 24.67%. Our result is similar to other studies in which payment method and cost are essential criteria when selecting a CB (Jespersen & Hasle, 2017; Poksinska et al., 2006). The cost of certification is considered more as an investment than an expense when quality or technical departments maintain strong connections with their customer through timely fulfilling orders (Chiarini, 2019).

Pairwise comparison of Knowledge about the client's products and business (KN) presents the following sequence: Knowledge of the sector of activity (by type of process) and the associated risks concerning the place of the company processes (KN2) 36.04%> Ability to determine the specific cultural and social customs related to the

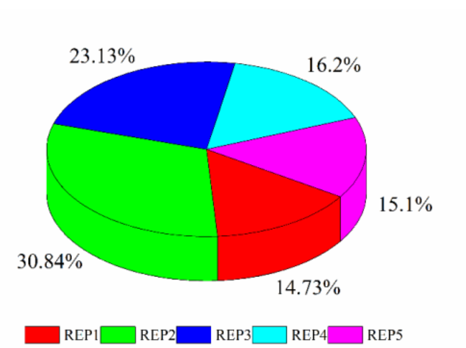


Fig. 5: Reputation Sub-criteria

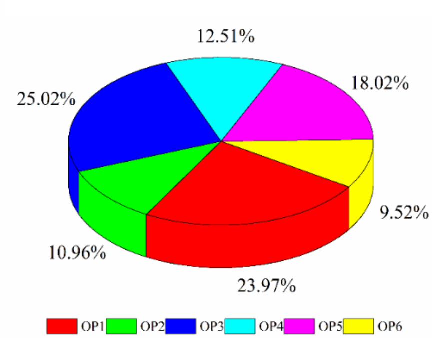
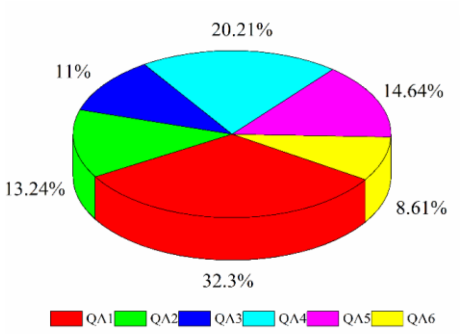


Fig. 6: Operational Outcome Sub-criteria



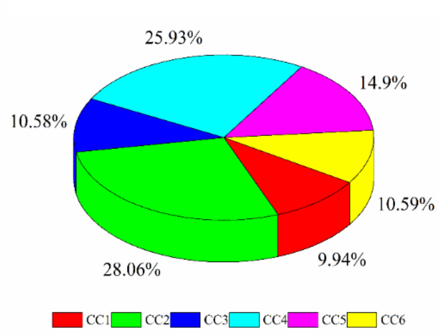
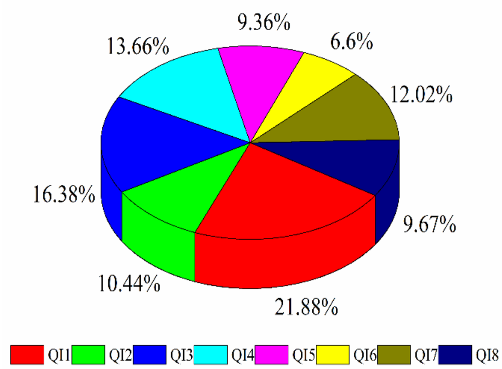


Fig. 7: Quality of Auditor Sub-criteria

Fig. 8: Contract Conditions Sub-criteria



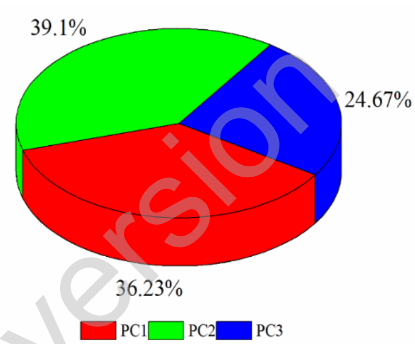


Fig. 9: Quality of Implementation Sub-criteria

Fig. 10: Payment Method & Cost Sub-criteria

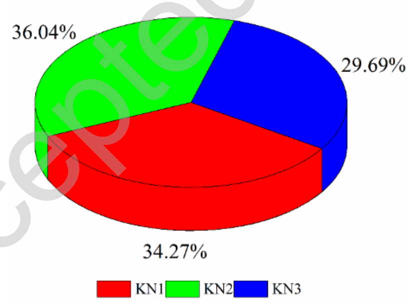


Fig. 10: Knowledge of Product and Business Sub-criteria

Note= The complete form of sub-criteria codes is in table 3.

company's production process (KN1) 34.27%> Technical knowledge to overcome any adverse conditions that are part of the partnership scope (KN3) 29.69%.

4.3 The final ranking of sub-indicators

After calculating the weights of sub-indicators, it is important to determine the overall ranking of sustainability tools' sub-indicators. Weights of sub-criteria under each category of main criteria are multiplied with that criteria weight to obtain the final ranking.

To check the reliability of each GAHP pairwise matrices, the CR was computed. When the value of CR records less than 0.1, the results using GAHP are acceptable; otherwise the obtained results are considered unreliable

and the process of obtaining experts' feedback must be revised (Ikram et al., 2020). The CR of all the attributes in this study was recorded less than 0.1 and is presented in Table 8. Finally, each sub-criteria's weights are summed to obtain cumulative weight, which is the base of the final ranking provided in Table 8. The final ranking of sub-attributes for the selection of appropriate certification bodies to develop SD is as follows:
QA1>QI6>QI1>KN2>QA5>KN1>PC1>QI4>>QA3>OP6>REP4>CC5>QI3>REP1>REP3>KN3>OP5>PC2>QI2>CC4>QI7>CC6>CC1>OP2>OP3>QI8>QI5>QA6>QA2>REP5>REP2>QA4>OP4> PC3>CC3>OP1>CC2.

To select the appropriate CB which ensure to achieve SD in the organization, the auditors have much knowledge about the management standards (QA1) receive the highest weight of 0.5009, followed by, To check the management review minutes of the meeting (Q16) 0.4918 and Clearly documented the quality policy and standard operating procedures (QI1) 0.4789 are ranked at the second and third position of the essential sub-criteria respectively (see Figure 11).

Table 8: Final Ranking

Main Attributes	Main criteria codes	Main Attribute Weights	Sub-attributes codes	Consistency Ratio (CR)	Normalized weight
Reputation	REP	0.2120	REP1	0.0083	0.3496
			REP2		0.1488
			REP3		0.3359
			REP4		0.3719
			REP5		0.1546
Operational outcomes	OP	0.1610	OP1	0.0086	0.0980
			OP2		0.2117
			OP3		0.1958
			OP4		0.1408
			OP5		0.3139
			OP6		0.3889
Quality of Auditor	QA	0.1406	QA1	0.0129	0.5009
			QA2		0.1589
			QA3		0.3889
			QA4		0.1471
			QA5		0.4439
			QA6		0.1705
Contract Condition	CC	0.1020	CC1	0.0003	0.2145
			CC2		0.0857
			CC3		0.1190
			CC4		0.2967
			CC5		0.3692
			CC6		0.2318
Quality of Implementation	QI	0.1387	QI1	0.0178	0.4789
			QI2		0.3044
			QI3		0.3543
			QI4		0.4189
			QI5		0.1891
			QI6		0.4918
			QI7		0.2917
			QI8		0.1932
Payment Method & Cost	PC	0.1704	PC1	0.0007	0.4339
			PC2		0.3069
			PC3		0.1192
	KN		KN1		0.4367

Knowledge of products and business	0.0753	KN2 KN3	0.0112	0.4717 0.3223

Note= The complete form of sub-criteria codes is presented in table 3.

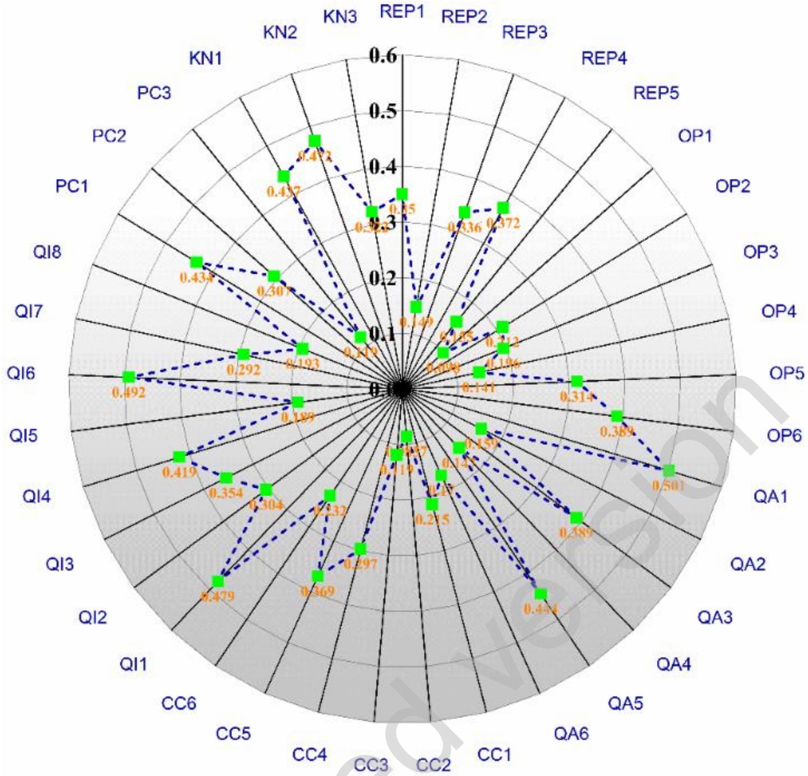


Fig. 11: Relative Weights of Sub-criteria**;** *source (self-developed)*

Note= The complete form of sub-criteria codes is presented in table 3.

4.4 Grey absolute decision analysis results

Aiming to expand insights from analyses, we want to know which sustainability tools effectively align with and manage sustainability practices. The ranking of six main categories of certification bodies, namely BVC (CB1), LR (CB2), DNVGL (CB-3), BSI (CB-4), SGS (CB-5), TÜV (CB-6), and systems, were analyzed by a novel GADA approach. The authors benefited from skilled experts who provided valuable inputs and were considered in a 'group decision-making approach' to determine results. The GADA method is an MCDA technique by which it is possible to obtain optimum relative weights of the criteria. This is achieved through the 'simulated weights of the criteria' $\delta(k)$, normalized to get more precise relative weights, which, in turn, is used for improving the ranking of other methods. We have considered all criteria, including, in this manuscript, as a benefit for developing corporate sustainability towards six types of tools involving systems, standards reporting, and programs.

In implementing the GADA method, we have six alternatives/possibilities and eight independent decision-makers experts, who evaluated them against two unequally weighted criteria C(1): benefit criteria (Support to SDGs, assessment rating and awards, providing services to countries); C(2) cost-type criteria. (number of employees, year of business, revenue, number of customers). These experts attributed weights to the set of two criteria, as follows (0.2, 0.8), (0.3, 0.7), (0.4, 0.6), (0.2, 0.8), (0.4, 0.6), (0.3, 0.7), (0.4, 0.6) and (0.2, 0.8). Table 9 scores were recorded on 5-

point Likert-scale, where 1 represents the lowest score, and 5 represents the highest score. For benefit, 5 implies the highest benefit, and cost at 1 implies the least expensive. Here, criterion 1 is the worst concerning the benefit, and concerning the cost criterion, 5 is the worst. Tables 10 and 11 presents the execution of the GADA method.

The certification bodies selection overall ranking for the development of SDGs are based on final scores presented in Table 11 and is as follows:

CB-1>CB-5>CB-3>CB-6>CB-4>CB-2

Based on global weight recording, our results recommend the BVC (CB-1) and SGS (CB-5) the best suited CB’s help to achieve SDGs by obtaining the highest weight 0.8731, and 0.7340 respectively, followed by DNVGL (CB-3) with 0.6782 weight, TUV (CB-6) scored at 0.5344, BSI (CB-4) with 0.5212 weight, and LR (CB-2) with a weight of 0.4213. The ordered ranking of different certification bodies and packages is significant due it provided the overall values for SDGs. While the weight comparison helps to prioritize certification bodies, all the identified CB’s can achieve SDGs. Additionally, these certification bodies are commonly adopted, ensuring business improvements related to quality and sustainable business practices (Souza and Alves, 2018).

Companies can benefit from our findings in looking at certification bodies in a new light, including our study's outcomes in decision-making and utilizing the approach taken in this study to develop a customized approach for their organization and certification decision-making needs. Organizations, the United Nations, and other authorities can benefit from this study because certification bodies enable tangible cost-benefit results toward global goals. Local governments, international agencies, businesses, and stakeholders all have opportunities to help fulfill these global goals. In 2019, United Nations (UN) reported that financial support is a big hurdle to implement the SDGs agenda in developing countries (UN World, 2019). This study's results and additional benefits come from understanding a more complex value proposition involved in certification bodies. We hope the results will help with a more intelligent allocation of scarce resources to help align investments with faster progress toward global goals for sustainable development.

Table 9: Experts’ Evaluation Input Data of Certification Bodies

Experts	**Benefit *C(1)***						**Cost *C(2)***					
	a_1	a_2	a_3	a_4	a_5	a_6	a_1	a_2	a_3	a_4	a_5	a_6
E_1	4	3	3	5	4	3	2	1	2	3	1	4
E_2	5	3	4	3	4	5	3	2	1	2	4	2
E_3	4	5	3	5	4	3	3	2	2	2	3	1
E_4	4	3	4	5	4	5	2	3	4	2	3	5
E_5	5	5	4	4	3	4	1	2	4	2	3	1
E_6	4	3	4	4	4	5	2	5	2	1	5	1
E_7	3	5	3	4	3	4	2	3	4	4	1	3
E_8	5	5	3	4	4	3	4	2	3	2	3	1

Note: E=Expert, a=alternative

Table 10. The Cost and Benefits Result Based on Expert Feedback

Experts	Benefit *C(1)*						Cost *C(2)*					
	a_1	a_2	a_3	a_4	a_5	a_6	a_1	a_2	a_3	a_4	a_5	a_6
E_1	4	3	3	5	4	3	0.5	1	0.5	0.3333	1	0.25
E_2	5	3	4	3	4	5	0.3333	0.5	1	0.5	0.25	0.5
E_3	4	5	3	5	4	3	0.3333	0.5	0.5	0.5	0.3333	1
E_4	4	3	4	5	4	5	0.5	0.3333	0.25	0.5	0.3333	0.2
E_5	5	5	4	4	3	4	1	0.5	0.25	0.5	0.3333	1
E_6	4	3	4	4	4	5	0.5	0.2	0.5	1	0.2	1
E_7	3	5	3	4	3	4	0.5	0.3333	0.25	0.25	1	0.3333
E_8	5	5	3	4	4	3	0.25	0.5	0.3333	0.5	0.3333	1
GM	4.6953	3.8730	3.4642	4.1952	3.7221	3.9040	0.4524	0.4395	0.4001	0.4753	0.3976	0.5491
Weighted GM using α (r̀)	4.8311	3.9005	3.1248	4.2143	3.9215	3.9110	0.4902	0.4777	0.3974	0.4502	0.4051	0.5880

Note: E=Expert, a=alternative, GM=Geometric mean, :ȑ=GADA

Table 11: Final Evaluation and Ranking of Certification Bodies

	(r̀)	a_1	a_2	a_3	a_4	a_5	a_6
Inter-criteria ranking	*C(1)*	4.8317	3.9001	3.1244	4.2146	3.9210	3.9112
	Ranking	1	5	6	2	3	4
	C(2)	0.4902	0.4777	0.3974	0.4502	0.4051	0.5880
	Ranking	2	3	6	5	5	1
Global ranking	ȑ	0.8733	0.5342	0.7344	0.6781	0.4210	0.5210
	Ȑ	0.3544	0.2901	0.2505	0.3112	0.2311	0.2090
	Ranking	1	4	3	2	6	5

Note:ȑ=GADA index, Ȑ =GADA weight

5. Conclusions

While we hope companies, governments and decision makers can benefit from this study, yet also need to acknowledge some limitations. First, we enlisted the help of eight experts from different companies and sectors. Although this study aligns with other studies using MCDA models in quality management problems, it would be interesting, for future work, to increase the number of experts participating in the study to increase the validation of results. We also recommend the inclusion of experts from other economies, i.e., from the well-established to emergent, to gather a diversified set of options related to different experts' profiles. In doing so, we recommend the Delphi method panels to compare and enrich the criteria and sub-criteria considered in this research. Finally, the selection criteria can change according to specific conditions and vary according to external disturbances (such as the current COVID-19 pandemic). Future work should focus on companies from different countries. We recommend building a comparative scorecard to identify global insights related to certification body selection processes worldwide.

In meeting the objectives and primary research questions of this study, we have multiple contributions to the field and new insights regarding relationships and the SDGs. First, this study contributes to academic research that attempts to explain the right certification bodies' selection to align with and achieve SDGs. The outcomes of his study helps to establish seven important criteria and thirty-seven sub-criteria. These criteria are developed by starting with an extensive review of the literature, and then we build on this with the help of experts. Simultaneously, we used a novel Grey Delphi method to meet this objective. Initially, we identified eleven main criteria from the literature. We classified them into seven of the most influential and vital criteria, such as Reputation, Operational Outcomes, Quality of Auditor, Contract Condition, Quality of Implementation, Payment method & Cost, and Knowledge of Product and Business. We then divided these main seven categories of criteria into a further thirty-seven sub-criteria. The second contribution is using the novel Grey AHP method to arrange these criteria and sub-criteria to select certification bodies. It has been organized in the form of a comparison matrix. Due to the competitive market of certification bodies, Reputation, Payment Method & Cost, and Quality of Auditors are significant indicators with of 0.2120, 0.1704, and 0.1406 weight score of top-ranked criteria in the selection process of CB to achieve SDGs. Simultaneously, the auditor's knowledge about the management standards and review meetings ranked the most critical sub-criterion among the rest of the 35 sub-criteria lists with a weight of 0.5009 and 0.4918.

This study's third contribution is the micro-level identification of ISO and GRI as the top two standards recommend as sustainability standards/tools required to align with and fulfill SDGs. These standards have multiple benefits for organizations wanting to advance CSR and TBL performance. A methodological contribution includes the application of a Grey Absolute Decision Analysis method to prioritize certification bodies. This is the first time this type of prioritization has been performed with the top six most adaptable certification bodies being: Bureau Veritas (BV), *Société Générale de Surveillance* (SGS), *Det Norske Veritas Germanischer Lloyd* (DNVGL), *Lloyds Register* (LR), British Standards Institution (BSI), and the *Technischer Überwachungs-Verein* (TUV). This study assesses and compares these certification bodies based on their sizes, financial position, reputation, team of quality auditors, number of customers, year of services provided, and how many countries they are in. This study concluded that BV and SGS are the most appropriate options in selecting certification bodies by obtaining the highest weight scores of 0.873, and 0.734 to align with and help achieve SDGs. Finally, this study is also the first to apply an integrated approach using three novel grey models to add new insights to the existing body of literature.

Organizations in developing countries often struggle with available resources for certifications aligned with the UN SDGs and improvements in corporate management of environmental, social, and governance performance. Those seeking CBs typically need help in problem solving, decision making, and communication with stakeholders so that limited resources are not wasted. The role of a CB is important as they can help with technical and management options based on prior experience. The quality of these engagements is crucial for advancing industries at a macro-level and individual enterprises at a micro-level. Business decision makers will benefit from our findings in more quickly getting up to speed in choosing a CB to partner with. To this end, we have tried to develop a customizable approach for certification decision-making needs. Organizations, the United Nations, and other authorities can benefit from this study as CBs enable tangible cost-benefit results toward global goals of sustainable development. Local governments, international agencies, businesses, and stakeholders all have opportunities to align with global goals for more sustainable business practices. This study's outcomes help provide a more dynamic value proposition involved with the selection and use of CBs. We hope the results will help enable a more intelligent allocation of scarce resources while also aligning investments with faster progress toward the UN SDGs.

This study provides relevant theoretical and practical implications. This study contributes to the enrichment of SDGs literature, especially regarding the final processes of selecting the proper sustainability certification. Practical contributions include, but are not limited to, this work serving as a decision model in acquiring certification bodies' services. Outcomes will help managers and government agencies to identify the certifying company that best contributes to their needs. We also see our work helping develop strategies for selecting certification bodies to align with SDGs. Scholars can consider emerging criteria from the literature in the future and build on this study. Future research can build on this study, replicate its methods, and combine experts' insights to further developing a more integrated approach to the certification body selection process. The methods used in this study support decision-making in a dynamic and relatively practical way. These methods can and should be applied to further replications by future studies.

Acknowledgments

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References

Ali, J., Yusuf, N., 2019. International Quality Certification and Business Performance of Indian Firms: Evidence from Enterprise Survey Data. Glob. Bus. Rev. https://doi.org/10.1177/0972150919825514

Alvarenga, A.D., Salgado, E.G., Mendes, G.H. de S., 2018. Ranking criteria for selection of certification bodies for ISO 9001 through the Analytic Hierarchy Process (AHP). Int. J. Qual. Reliab. Manag. 35, 1321–1342. https://doi.org/10.1108/IJQRM-12-2016-0217

Azapagic, A., 2015. Sustainable Production and Consumption: A Decision-Support Framework Integrating Environmental, Economic and Social Sustainability, Computer Aided Chemical Engineering. Elsevier. https://doi.org/10.1016/B978-0-444-63578-5.50017-7

Barbosa-Póvoa, A.P., da Silva, C., Carvalho, A., 2018. Opportunities and challenges in sustainable supply chain: An operations research perspective. Eur. J. Oper. Res. 268, 399–431. https://doi.org/10.1016/j.ejor.2017.10.036

Bian, J., Liao, Y., Wang, Y.-Y., Tao, F., 2020. Analysis of Firm CSR Strategies. Eur. J. Oper. Res.

Brundtland, 1987. World Commission on Environment and Development: Our Common Future.

BSI, 2019. The british standards institution annual report and financial statements 2019.

Cai, S., Jun, M., 2018. A qualitative study of the internalization of ISO 9000 standards: The linkages among firms' motivations, internalization processes, and performance. Int. J. Prod. Econ. https://doi.org/10.1016/j.ijpe.2017.12.001

Cândido, C.J.F., Coelho, L.M.S., Peixinho, R.M.T., 2019. Why firms lose their ISO 9001 certification: Evidence from Portugal. Total Qual. Manag. Bus. Excell. https://doi.org/10.1080/14783363.2019.1625266

Castka, P., Prajogo, D., Sohal, A., Yeung, A.C.L., 2015. Understanding firms′ selection of their ISO 9000 third-party certifiers. Int. J. Prod. Econ. 162, 125–133. https://doi.org/10.1016/j.ijpe.2015.01.012

Chan, F.T.S., Chan, H.K., 2010. An AHP model for selection of suppliers in the fast changing fashion market. Int. J. Adv. Manuf. Technol. https://doi.org/10.1007/s00170-010-2683-6

Chan, F.T.S., Kumar, N., 2007. Global supplier development considering risk factors using fuzzy extended AHP-based approach. Omega. https://doi.org/10.1016/j.omega.2005.08.004

Chiarini, A., 2019. Why are manufacturing SMEs cancelling their ISO 9001 certification? Research from Italy. Prod. Plan. Control. https://doi.org/10.1080/09537287.2019.1566840

Chkanikova, O., Sroufe, R., 2020. Third-party sustainability certifications in food retailing: Certification design from

a sustainable supply chain management perspective. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2020.124344
Chowdhury, M.M.H., Paul, S.K., Sianaki, O.A., Quaddus, M.A., 2020. Dynamic sustainability requirements of stakeholders and the supply portfolio. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2020.120148
Diaz-Balteiro, L., González-Pachón, J., Romero, C., 2017. Measuring systems sustainability with multi-criteria methods: A critical review. Eur. J. Oper. Res. https://doi.org/10.1016/j.ejor.2016.08.075
DNVGL, 2019. Sustainability performance Report 2019.
DNVGL, 2015. 2030 forecast: key conclusions The main findings from DNV GL's model-based forecast of the most likely future for our planet [WWW Document]. URL https://www.dnvgl.com/technology-innovation/spaceship-earth/forecast-key-conclusions.html (accessed 11.11.20).
Elkington, J., 1997. Cannibals with Forks: Triple Bottom Line of 21st Century Business, The Triple Bottom Line.
Fonseca, Carvalho, 2019. The Reporting of SDGs by Quality, Environmental, and Occupational Health and Safety-Certified Organizations. Sustainability 11, 5797. https://doi.org/10.3390/su11205797
Gallego-Álvarez, I., Lozano, M.B., Rodríguez-Rosa, M., 2018. An analysis of the environmental information in international companies according to the new GRI standards. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2018.01.240
Gómez Martín, E., Giordano, R., Pagano, A., van der Keur, P., Máñez Costa, M., 2020. Using a system thinking approach to assess the contribution of nature based solutions to sustainable development goals. Sci. Total Environ. 738, 139693. https://doi.org/10.1016/j.scitotenv.2020.139693
Govindan, K., Khodaverdi, R., Jafarian, A., 2013. A fuzzy multi criteria approach for measuring sustainability performance of a supplier based on triple bottom line approach. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2012.04.014
GRI, 2020. Linking the SDGs and the GRI Standards.
Haeri, S.A.S., Rezaei, J., 2019. A grey-based green supplier selection model for uncertain environments. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2019.02.193
Heras-Saizarbitoria, I., Boiral, O., 2013. ISO 9001 and ISO 14001: Towards a Research Agenda on Management System Standards. Int. J. Manag. Rev. https://doi.org/10.1111/j.1468-2370.2012.00334.x
Hernandez-Vivanco, A., Domingues, P., Sampaio, P., Bernardo, M., Cruz-Cázares, C., 2019. Do multiple certifications leverage firm performance? A dynamic approach. Int. J. Prod. Econ. https://doi.org/10.1016/j.ijpe.2019.07.016
Hernandez, H., 2010. Quality audit as a driver for compliance to ISO 9001:2008 standards. TQM J. https://doi.org/10.1108/17542731011053361
Hoy, Z., Foley, A., 2015. A structured approach to integrating audits to create organisational efficiencies: ISO 9001 and ISO 27001 audits. Total Qual. Manag. Bus. Excell. https://doi.org/10.1080/14783363.2013.876181
Huan, Y., Liang, T., Li, H., Zhang, C., 2021. A systematic method for assessing progress of achieving sustainable development goals: A case study of 15 countries. Sci. Total Environ. 752. https://doi.org/10.1016/j.scitotenv.2020.141875
Huang, S.H., Keskar, H., 2007. Comprehensive and configurable metrics for supplier selection. Int. J. Prod. Econ. https://doi.org/10.1016/j.ijpe.2006.04.020
Hu, X., Chong, H.-Y., Wang, X., 2019. Sustainability perceptions of off-site manufacturing stakeholders in Australia. J. Clean. Prod. 227, 346–354. https://doi.org/10.1016/j.jclepro.2019.03.258
Ikram, M., Sroufe, R., Zhang, Q., 2020a. Prioritizing and overcoming barriers to integrated management system (IMS) implementation using AHP and G-TOPSIS. J. Clean. Prod. 254, 120121. https://doi.org/10.1016/j.jclepro.2020.120121
Ikram, M., Zhang, Q., Sroufe, R., 2020b. Future of quality management system (ISO 9001) certification: novel grey forecasting approach. Total Qual. Manag. Bus. Excell. 1–28. https://doi.org/10.1080/14783363.2020.1768062
Ikram, M., Zhang, Q., Sroufe, R., 2020c. Developing integrated management systems using an AHP-Fuzzy VIKOR approach. Bus. Strateg. Environ. 29, 2265–2283. https://doi.org/10.1002/bse.2501
Ikram, M., Zhang, Q., Sroufe, R., Ferasso, M., 2020. The Social Dimensions of Corporate Sustainability: An Integrative Framework Including COVID-19 Insights. Sustainability 12, 8747. https://doi.org/10.3390/su12208747
ISO, 2017. ISO Members. Int. Stand. Stand. URL http://www.iso.org/iso/home/about/iso_members.htm (accessed 1.13.21)
ISO, 2018a. Contributing to the UN Sustainable Development Goals with ISO standards.
ISO, 2018b. HOW ISO STANDARDS HELP MEET THE SDGS [WWW Document]. URL https://www.iso.org/sdgs.html (accessed 10.16.20).

Ivanova, A., Gray, J., Sinha, K., 2014. Towards a unifying theory of management standard implementation. Int. J. Oper. Prod. Manag. https://doi.org/10.1108/ijopm-03-2013-0117
Jamal, K., Sunder, S., 2011. Is mandated independence necessary for audit quality? Accounting, Organ. Soc. https://doi.org/10.1016/j.aos.2011.03.001
Javed, S.A., Liu, S., 2019. Bidirectional Absolute GRA/GIA Model for Uncertain Systems: Application in Project Management. IEEE Access. https://doi.org/10.1109/ACCESS.2019.2904632
Javed, S.A., Mahmoudi, A., Liu, S., 2020. Grey Absolute Decision Analysis (GADA) Method for Multiple Criteria Group Decision-Making Under Uncertainty. Int. J. Fuzzy Syst. https://doi.org/10.1007/s40815-020-00827-8
Javed, S.A., Ikram, M., Tao, L., Liu, S., 2020. Forecasting key indicators of China's inbound and outbound tourism: optimistic–pessimistic method. Grey Syst. Theory Appl. ahead-of-p. https://doi.org/10.1108/GS-12-2019-0064
Javed, S.A., Zhu, B., Liu, S., 2020b. Forecast of biofuel production and consumption in top CO2 emitting countries using a novel grey model. J. Clean. Prod. 276, 123997. https://doi.org/10.1016/j.jclepro.2020.123997
Jespersen, A.H., Hasle, P., 2017. Developing a concept for external audits of psychosocial risks in certified occupational health and safety management systems. Saf. Sci. https://doi.org/10.1016/j.ssci.2016.11.023
Jiang, W., Xie, C., Luo, Y., Tang, Y., 2017. Ranking Z-numbers with an improved ranking method for generalized fuzzy numbers. J. Intell. Fuzzy Syst. https://doi.org/10.3233/JIFS-16139
Kannan, D., De Sousa Jabbour, A.B.L., Jabbour, C.J.C., 2014. Selecting green suppliers based on GSCM practices: Using Fuzzy TOPSIS applied to a Brazilian electronics company. Eur. J. Oper. Res. https://doi.org/10.1016/j.ejor.2013.07.023
Lamichane, S., Egilmez, G., Gedik, R., Bhutta, M.K., Erenay, B., 2020. Benchmarking OECD Countries' Sustainable Development Performance: A Goal-Specific Principal Component Analysis Approach. J. Clean. Prod. 125040.
Lloyd's Register, 2019. Reaching new heights for a safer world.
Ma, Z., Shao, C., Ma, S., Ye, Z., 2011. Constructing road safety performance indicators using fuzzy delphi method and grey delphi method. Expert Syst. Appl. 38, 1509–1514. https://doi.org/10.1016/j.eswa.2010.07.062
Ofosu-Adarkwa, J., Xie, N., Javed, S.A., 2020. Forecasting CO2 emissions of China's cement industry using a hybrid Verhulst-GM(1,N) model and emissions' technical conversion. Renew. Sustain. Energy Rev. 130, 109945. https://doi.org/10.1016/j.rser.2020.109945
Pan, X., Sinha, P., Chen, X., 2020. Corporate social responsibility and eco-innovation: The triple bottom line perspective. Corp. Soc. Responsib. Environ. Manag.
Poksinska, B., Dahlgaard, J.J., Eklund, J.A.E., 2006. From compliance to value-added auditing - Experiences from Swedish ISO 9001:2000 certified organisations. Total Qual. Manag. Bus. Excell. https://doi.org/10.1080/14783360600595294
Power, D., Terziovski, M., 2007. Quality audit roles and skills: Perceptions of non-financial auditors and their clients. J. Oper. Manag. https://doi.org/10.1016/j.jom.2006.02.005
Power, D., Terziovski, M., 2005. The process, practice and outcomes of non-financial auditing: Five Australian case studies. Int. J. Manuf. Technol. Manag. 7, 52–82. https://doi.org/10.1504/IJMTM.2005.006502
Prajogo, D., Castka, P., Searcy, C., 2020a. Paymasters and Assurance Providers: Exploring Firms' Discretion in Selecting Non-financial Auditors. J. Bus. Ethics. https://doi.org/10.1007/s10551-020-04539-9
Prajogo, D., Castka, P., Yiu, D., Yeung, A.C.L., Lai, K.H., 2016. Environmental Audits and Third Party Certification of Management Practices: Firms' Motives, Audit Orientations, and Satisfaction with Certification. Int. J. Audit. https://doi.org/10.1111/ijau.12068
Prajogo, D., Nair, A., Castka, P., 2020b. The Effects of External Auditors and Certification Bodies on the Operational and Market-Oriented Outcomes of ISO 9001 Implementation. IEEE Trans. Eng. Manag. https://doi.org/10.1109/TEM.2020.2982657
Prajogo, D.I., 2011. The roles of firms' motives in affecting the outcomes of ISO 9000 adoption. Int. J. Oper. Prod. Manag. https://doi.org/10.1108/01443571111098753
Ristono, A., -, P., Santoso, P.B., Tama, I.P., 2018. A literature review of criteria selection in supplier. J. Ind. Eng. Manag. https://doi.org/10.3926/jiem.2203
Sethi, S.P., Rovenpor, J.L., Demir, M., 2017. Enhancing the Quality of Reporting in Corporate Social Responsibility Guidance Documents: The Roles of ISO 26000, Global Reporting Initiative and CSR-Sustainability Monitor. Bus. Soc. Rev. 122, 139–163. https://doi.org/10.1111/basr.12113
SGS, 2019. 2019 Corporate Sustainability Report.
Sharma, K.K., Seal, A., 2019. Modeling uncertain data using Monte Carlo integration method for clustering. Expert Syst. Appl. https://doi.org/10.1016/j.eswa.2019.06.050
Shete, P.C., Ansari, Z.N., Kant, R., 2020. A Pythagorean fuzzy AHP approach and its application to evaluate the enablers of sustainable supply chain innovation. Sustain. Prod. Consum. 23, 77–93.

https://doi.org/10.1016/j.spc.2020.05.001
Singh, P.J., Power, D., Chuong, S.C., 2011. A resource dependence theory perspective of ISO 9000 in managing organizational environment. J. Oper. Manag. https://doi.org/10.1016/j.jom.2010.04.002
Skirbekk, G. (1994). Marxism and ecology. Capitalism Nature Socialism, 5(4), 95-104.
Souza, J.P.E., Alves, J.M., 2018. Lean-integrated management system: A model for sustainability improvement. J. Clean. Prod. 172, 2667–2682. https://doi.org/10.1016/j.jclepro.2017.11.144
Stević, Ž., Stjepanović, Ž., Božičković, Z., Das, D.K., Stanujkić, D., 2018. Assessment of conditions for implementing information technology in a warehouse system: A novel fuzzy PIPRECIA method. Symmetry (Basel). https://doi.org/10.3390/sym10110586
Tavares de Aquino, A., Maciel de Melo, R., 2016. Multicriteria model for selecting TQM consultancy and certification services. Benchmarking. https://doi.org/10.1108/BIJ-05-2015-0049
Torkabadi, A.M., Pourjavad, E., Mayorga, R. V., 2018. An integrated fuzzy MCDM approach to improve sustainable consumption and production trends in supply chain. Sustain. Prod. Consum. 16, 99–109. https://doi.org/10.1016/j.spc.2018.05.008
Tsalis, T.A., Malamateniou, K.E., Koulouriotis, D., Nikolaou, I.E., 2020. New challenges for corporate sustainability reporting: United Nations' 2030 Agenda for sustainable development and the sustainable development goals. Corp. Soc. Responsib. Environ. Manag. https://doi.org/10.1002/csr.1910
TÜV SÜD, 2019. Combined management Report 2019.
UN World, 2019. 'Not Keeping Pace' in Implementing Sustainable Development Goals, Secretary-General Stresses, as Development Financing Forum Begins. [WWW Document]. URL https://www.un.org/press/en/2019/ecosoc6972.doc.htm (accessed 11.11.20).
United Nations, 2020. The Sustainable Development Goals 2020.
United Nations Development Programme (UNDP), 2020. Background and Goals. accessed at www.undp.org/undp/en/jhome/sustainable-development-gaols/background
USTUN, O., DEMIRTAS, E., 2008. An integrated multi-objective decision-making process for multi-period lot-sizing with supplier selection☆. Omega 36, 509–521. https://doi.org/10.1016/j.omega.2006.12.004
Veritas, B., 2019. Sustainability report and 2019 UNIVERSAL REGISTRATION DOCUMENT.
Wang, H., Pan, C., Wang, Q., Zhou, P., 2020. Assessing sustainability performance of global supply chains: An input-output modeling approach. Eur. J. Oper. Res. https://doi.org/10.1016/j.ejor.2020.01.057
White, L., Lee, G.J., 2009. Operational research and sustainable development: Tackling the social dimension. Eur. J. Oper. Res. 193, 683–692. https://doi.org/10.1016/j.ejor.2007.06.057
Wiengarten, F., Humphreys, P., Onofrei, G., Fynes, B., 2017. The adoption of multiple certification standards: perceived performance implications of quality, environmental and health & safety certifications. Prod. Plan. Control. https://doi.org/10.1080/09537287.2016.1239847
Zare, A., Feylizadeh, M.R., Mahmoudi, A., Liu, S., 2018. Suitable computerized maintenance management system selection using grey group TOPSIS and fuzzy group VIKOR: A case study. Decis. Sci. Lett. 7, 341–358. https://doi.org/10.5267/j.dsl.2018.3.002
Zareinejad, M., Kaviani, M.A., Esfahani, M.J., Masoule, F.T., 2014. Performance evaluation of services quality in higher education institutions using modified SERVQUAL approach with grey analytic hierarchy process (G-AHP) and multilevel grey evaluation. Decis. Sci. Lett. https://doi.org/10.5267/j.dsl.2013.12.002
Zhao, X., Castka, P., Searcy, C., 2020. ISO Standards: A Platform for Achieving Sustainable Development Goal 2. Sustainability 12, 9332. https://doi.org/10.3390/su12229332

Contribution of Certification Bodies and Sustainability Standards to Sustainable Development Goals: An Integrated Grey Systems Approach

Muhammad Ikram[a], Qingyu Zhang[a*], Robert Sroufe[b], Marcos Ferraso[c]

[a] *College of Management, Research Institute of Business Analytics and Supply Chain Management, Shenzhen University, Shenzhen, 518060, China*
[b] *Donahue Graduate School of Business, Duquesne University, 820 Rockwell Hall, 600 Forbes Avenue, Pittsburg, PA, USA*
[c] *Institute of Scientific Research and graduate school, Universidad de Lima, Lima, Peru Av. Javier Prado Ests, 4600 Santiago de Surco 15023 Lima-Peru*

mikram@szu.edu.cn, q.yu.zhang@gmail.com, sroufer@duq.edu, admmarcosferasso@gmail.com

Abstract

The International Organization for Standardization and the Global Reporting Initiative promote their alignment with the United Nations' Sustainable Development Goals. This convergence of standards and information is causing the selection of a certification body to register these sustainability standards to become more complicated. Therefore, this study fills a gap in the literature by developing a framework to prioritize the most critical attributes for selecting a certification body. Initially, we focus on seven attributes using the Grey Delphi method. Further, Grey Analytical Hierarchy Process is employed to find primary and sub-indicators relative importance. Finally, we used a novel Grey Absolute Decision Analysis model to rank the widely adapted top six certification bodies. Results reveal that Reputation, Payment Method & Cost, and Quality of Auditors are significant indicators with weighted scores of 0.2120, 0.1704, and 0.1406. In contrast, the auditor's knowledge about the sustainability standards, and review of the meetings obtained the highest weight score 0.5009 and 0.4918 and are seen as essential among 35 sub-criteria. Within this study, we identify which ISO and GRI sustainability standards contribute to a given SDG. We also find Bureau Veritas and Société Générale de Surveillance are the best-suited certification bodies help to achieve SDGs as they obtained the highest weights 0.8731, and 0.7340. This study is among the first of its kind to address selecting the right certification body for aligning with the SDGs by using the integration of three novel grey models. Outcomes of this study can help assist scholars, managers, government agencies, and decision-makers in selecting certification bodies to achieve SDGs while simultaneously improving sustainability practices.

Keywords: Sustainable Development Goals; Certification bodies; Sustainability Standards; ISO; GRI; Grey systems approach

Nomenclature	
$[a_j^1, a_j^s]$	Range values of grey classes
a_j^k	Lower bound of *kth* grey classes
b_j^k	Upper bound of *kth* grey classes
$f_j^k(x)$	Weight function of k grey class
η_j^k	Wights of *j* indicator in synthetic clustering

* Corresponding Author (Qingyu Zhang; q.yu.zhang@gmail.com)

$\{\sigma_j^k\} = \{\sigma_j^{k^*}\}$	Evaluated decision vectors
$\otimes G_{ij}$	Grey number
$\otimes \overline{G_{ij}}$	The Upper limit of the grey number
$\otimes \underline{G_{ij}}$	The Lower limit of the grey number
λ	Grey coefficient
(a_j)	Decision Object
$[a_{ij}]$	Response decision measurement
$C(k)$	Decision criteria
$A_{(j)}$	Alternative
ε	Absolute grey relational grade
$[\varepsilon]$	Pairwise comparison matrix
$\alpha_i = \frac{1}{N}(\varepsilon_{i1} + \varepsilon_{i2} .... + \varepsilon_{iN})$	Javed alpha
$\delta(k)$	Simulated weight of criteria
$(\beta_1(k), \beta_2(k), ..., \beta_N(k))$	Geometric mean
ȑ	Grey Absolute Decision Analysis Index
(R_j', R_j'')	Set of relative weights

1. Introduction

The Sustainable Development Goals (SDGs) started at the United Nations Conference on Sustainable Development in Rio de Janeiro in 2012. The objective was to produce a set of universal goals that meet our world's wicked environmental, political, and economic challenges. These challenges are rooted in an understanding that the greater the industrial development in a country, the greater the potential for detrimental impacts to the very resources capital development depends on, the earth, people, and social systems (Skirbekk, 1994). The SDGs replaced the Millennium Development Goals (MDGs), which started a global effort in 2000 with measurable, globally-agreed objectives for addressing poverty, hunger, diseases, expanding education to all children, and other priorities. The SDGs are a call to action for the world to take a more sustainable path into the future. All 17 Goals interconnect, meaning success in one affects success for others. Dealing with the threat of climate change impacts how we manage our fragile natural resources, achieving gender equality or better health helps eradicate poverty, and fostering peace and inclusive societies will reduce inequalities and help economies prosper. The SDGs coincided with another historic agreement reached in 2015 at the COP21 Paris Climate Conference. Together with the Sendai Framework for Disaster Risk Reduction, signed in Japan in March 2015, these agreements provide a set of common standards and achievable targets to reduce carbon emissions, manage the risks of climate change and natural disasters, and rebuild more resiliently after a crisis (UNEP, 2020)

In 2015 the United Nations presented the 2030 Agenda with accompanying 17 Sustainable Development Goals (SDGs). These goals have gotten the attention of academics, business leaders, activists, and policy decision-makers to stimulate a transformation in world affairs (Huan et al., 2021; Zhao et al., 2020). In 2019, United Nations (UN) reported that financial support is a big hurdle to implement the SDGs agenda in developing countries (UN World, 2019). At the same time, the political leaders worldwide are not fully aware of Agenda of 2030. Local governments, international agencies, various businesses, and stakeholders also have no clear distinctions of responsibilities to fulfill these global goals in developed and developing countries (Zhao et al., 2020). It probably comes as no surprise then that Lamichane et al. (2020) found that achieving the SDGs has been happening at a minimal level in developing countries.

Climate change, increasing carbon dioxide emission, continued natural disasters (Ofosu-Adarkwa et al., 2020), and event the COVID-19 pandemic have disrupted the SDGs' plans and have made it more challenging for companies and governments to achieve. Therefore, it is increasingly important for the SDGs and Agenda 2030 to find new ways to progress. Hence, International Organizations for Standardization (ISO), Global Reporting Initiative (GRI), and other international bodies have already provided a tangible mechanism to help achieve SGDs by implementing sustainability tools.

According to Det Norske Veritas Germanischer Lloyd's (DNVGL) forecasting report on SDGs' global progress, the world will not meet even half of SDGs till 2030 (DNVGL, 2015). The report presents the world's division into five regions, i.e., USA, China, OECD, BRISE (Brazil, Russia, India, South Africa and other emerging economies), and ROW (rest of other world) as characterized by their size and economic development status. The details of the regions and progress of SDGs are in Figure 1. The report indicates that the USA and OECD countries (excl. the USA) have likely to be reached (Goal 2, goal 3, goal 4, goal 6, goal 9, and goal 11) a target fulfillment of more than 95%. At the same time, China is also on the right track and has achieved more than 95% (goal2, goal3, goal4, goal7, and goal9). Furthermore, BRISE has not achieved any goal of more than 95% and is at the transition phase. Finally, the ROW has achieved only one goal (goal 12) at more than 95%. The rest of the other goals are not likely to be reached by more than 50%. The report also indicates that collective efforts are required to reach the SDGs. The green indicator shows the goal likely to be reached (i.e., target fulfillment of 95% achieved). In contrast, orange indicates the goal reached more than 50% between today's status, and the goal is likely closed but not fully reached. Meanwhile, the red indicator shows that the goal reached less than 50% reached. Additionally, SDGs 2, 3, 4, 6, and 9 have a positive outcome and move in the right direction. Conversely, the pace is too slow to reach SDG climate action goal 13, affordable and clean energy goal 7 due to energy production through fossil fuels and non-renewable energy consumption, which accumulate predicted carbon emissions of 2,900 Giga tones by the year 2037.

Figure 1. Likelihood of meeting the 17 SDGs in five world regions. (*Source (DNVGL, 2015)*)

ISO is one of the largest standard developers, with more than 22,000 sustainability management standards used in 170 countries to fulfill the SDGs requirements (ISO, 2018a). ISO standards cover all the aspects of social, economic, and environmental concerns and help enable more sustainable practices. ISO standards ensure consistency, transparency, and transfer of sustainability practices across the world. Moreover, ISO standards play a vital role in international trade partnerships. ISO introduced the SDGs action plan regarding Agenda 2030, in which ISO standards provide essential tools to assist governments, industrial sectors, and consumers in achieving each of the SDGs.

Across many of the SDGs, ISO has developed significant standards that contribute to sustainability. For example, the Environmental Management System (EMS) ISO 14001 standard covers SDG 13 on climate action and 15 regarding life on land. Further, ISO 26000, the Social Responsibility standards, provides guidelines and supports SDGs 2 Zero Hunger, 5 Gender Equality, and 10 Reduce Inequalities. Add to this, ISO 24000 addressing Sustainable Procurement, and this standard helps to achieve SDGs 1 No Poverty, 2 Zero Hunger, and 12 Sustainable Production and Consumption. ISO 15001 helps develop an Energy Management System (EnMS) to enable SDG 7 Affordable and Clean Energy. The complete details of how the ISO technical committee developed each standard to help achieve SDGs are within the ISO website (ISO, 2018b).

GRI is the second-largest sustainability reporting non-profit and multi-stakeholder international organization. It was established in 1997 and launched in partnership with the United Nations Environmental Program (UNEP) in 2000 (Sethi et al., 2017). GRI provides assistance to ensure economic, social, and environmental sustainability through sustainability guidelines and standards. GRI is a catalyst for a more sustainable future that provides a consistent and transparent sustainability disclosure and standards to fulfill stakeholder needs (Hu et al., 2019; Gallego-Álvarez et al., 2018). The GRI is a defacto global sustainability reporting standard used in more than 90 countries. It provides services for multinational companies, government agencies, Small Medium Enterprises (SMEs), NGOs, industrial groups, and United Nations Global Compact (Gómez Martín et al., 2020; Tsalis et al., 2020). Since 2016, GRI has been included in the UN SDGs and developed standards and disclosure, contributing to these goals. GRI standard 207: 1-4, 203-2 and 207-2 align with SDG 1; standard 403 contributes to SDG 3 Good Health and well-being; GRI standard 404 enables guidelines for SDG 4 Quality Education; GRI standard 201 covers the SDG 8 Decent Work and Economic Growth; GRI standard 302 helps climate vulnerability SDG 13 Climate Action; GRI 403 promote peaceful societies for sustainable development SDG 16; GRI 303 contributes to the availability of clean water and sanitation SDG 6. The complete details of GRI standards that cover each SDG are available on the GRI Website (GRI, 2020).

When implementing sustainability standards such as certified and assured tools, non-certified tools, international guidelines, management programs, and disclosures, companies need to decide if an external certification body should align with SDGs. To make the business more sustainable, the companies need to become ISO, GRI, and other standards registered companies. These external sustainability certifications can positively affect organizational performance (Ali and Yusuf, 2019). With registration by highly reputable certification bodies, sustainability certification can provide more recognition and competitive advantage to companies (Chkanikova and Sroufe, 2020).

There are various certification bodies (CB) such as DNVGL, Bureau Veritas (BV), SGS, Lloyd's Register (LR), BSI, TUV, Intertek, and others. They provide their services to companies to register ISO, GRI, and other sustainability standards with competitive prices, quality auditors, and assurance to develop more sustainable business practices aligned with SDGs through effective implementation. The selection of an appropriate CB can be a difficult decision for companies. Despite this difficulty, the certification literature has paid little attention to this decision, and few studies have investigated this issue. This lack of attention is evident when comparing this management problem with other management standards, such as integrating management standards (IMS) and barriers to effective IMS implementation.

Further investigation is required to analyze how companies choose their certification bodies to align with and help achieve SDGs. Selecting the inappropriate certification body can have high financial and operational risks. Choosing the appropriate certification bodies to help companies implement sustainability standards can provide various benefits such as cost reduction, competitive advantage, customer complaints reduction, productivity, and customer satisfaction.

Thus, it is important to understand better the range of criteria considered in these decisions and our primary research questions:

RQ1. What sustainability tools are required to align with and help achieve Sustainable Development Goals?

RQ2. What are the main criteria involved in selecting certification bodies, and how to implement sustainability tools?

RQ3. What is the relative importance of these criteria, and how companies prioritize them?

To help answer these research questions, we have identified essential criteria from the literature that include (i) Reputation (ii) Operational Outcomes (iii) Quality of Auditor (iv)Contract Condition (v) Quality of Implementation (vi) Payment Method & Cost (vii) Knowledge of Product and Business. These main certification body criteria are then further classified into sub-criteria. One contribution to this research field is to assess the criteria and sub-criteria while filling a gap in the literature. This study's information can help decision-makers select the right certification body to align practices and help achieve SDGs. The second contribution of this study is to, for the first time, compare the top six certification bodies, providing the outcomes to managers, governments, and policymakers with new insights for how the selection of a certification body can help align practices with the SDGs. Third, we develop a framework for which sustainability tools/standards are required to achieve SDGs. Finally, this study is the first of its kind to use the integration of three novel grey models such as Grey Delphi (to finalize the classification of certification body selection criteria); Grey analytic hierarchy process approach GAHP (compute the weights for criteria and sub-criteria); and we use a novel Grey Absolute Decision Analysis (GADA) method based on GAHP to identify and prioritize the categories of the top six certification bodies.

The paper's structure is as follows: Section 2 presents the conceptual background of SDGs, sustainability standards, study criteria, and sub-criteria for selecting the certification bodies to align with and help achieve SDGs. Section 3 discusses the steps involved in developing Grey Delphi, GAHP, and GADA methodologies. Section 4 presents the results and discussion. The study's conclusion and policy implication, along with options for future research, are in the last section.

2. Literature Review

2.1 The United Nations Sustainable Development Goals (SDGs)

The basis of the SDGs was introduced by United Nations (UN), through the 'Our Common Future' report produced by the Commission on Environment and Development, also known as the 'Brundtland report' (Brundtland, 1987). Within this report, the definition of sustainable development is expressed as meeting the needs of a current generation without compromising the needs of future generations. Sustainable development's primary focus is a better life for human beings while staying within environmental limits. Later, a three-dimensional model of sustainability was proposed by Elkington. Widely known as the Triple Bottom Line (TBL), representing the economic development (profit), environmental progress (planet), and the social equity (people) dimensions (Elkington, 1997). Next, the UN Agenda for Development updated the sustainable development definition by integrating the previous Brundtland definition and the TBL dimensions. Sustainable development is considered in a multidimensional approach promoting a higher quality of life for human beings. It is noted that the interdependent basis of sustainable development is dependent upon economic development, social progress, and environmental protection (Abid et al., 2021; Barbosa-Póvoa et al., 2018).

According to Govindan et al. (2013), although there are significant contributions from the Brundtland Report definition, the challenge for sustainability lies in its operationalization. It also needs to be defined in a way so that it can be measured. Add to this the idea that Corporate Social Responsibility (CSR) is considered essential to organizations' long-term success. It integrates the TBL model when stating that profitability, social responsibility, and environmental preservation are crucial CSR elements (Pan et al., 2020). The literature shows no universal definition, although, for sustainability practices, the available definitions include economic, social, and environmental dimensions. The central idea is to create a balanced development successfully grounded in social progress and preserve the natural environment, with the generation of value for stakeholders (such as partners, customers, employees, and society).

Aligned with the planned Agenda for Sustainable Development 2030, created by the UN and launched in 2015, the 17 SDGs divided into the economy, society, and biosphere (in Figure 2), represent a historical demarcation of the sustainable development agenda. In this approach, the SDGs integrate the three pillars of sustainability: environmental sustainability, social development, and economic development (Rehman et al., 2020; White and Lee, 2009). Underlying this is the importance of sustainability goals for creating global policies and governance. The SDGs include core features of development concerning issues and opportunities involving the COVID-19 pandemic, energy, water, oceans, transportation, urbanization, technology, science, and climate. The SDGs were conceived to be addressed as individual goals and indicators for monitoring progress and reporting in the UN's reports (United Nations, 2020).

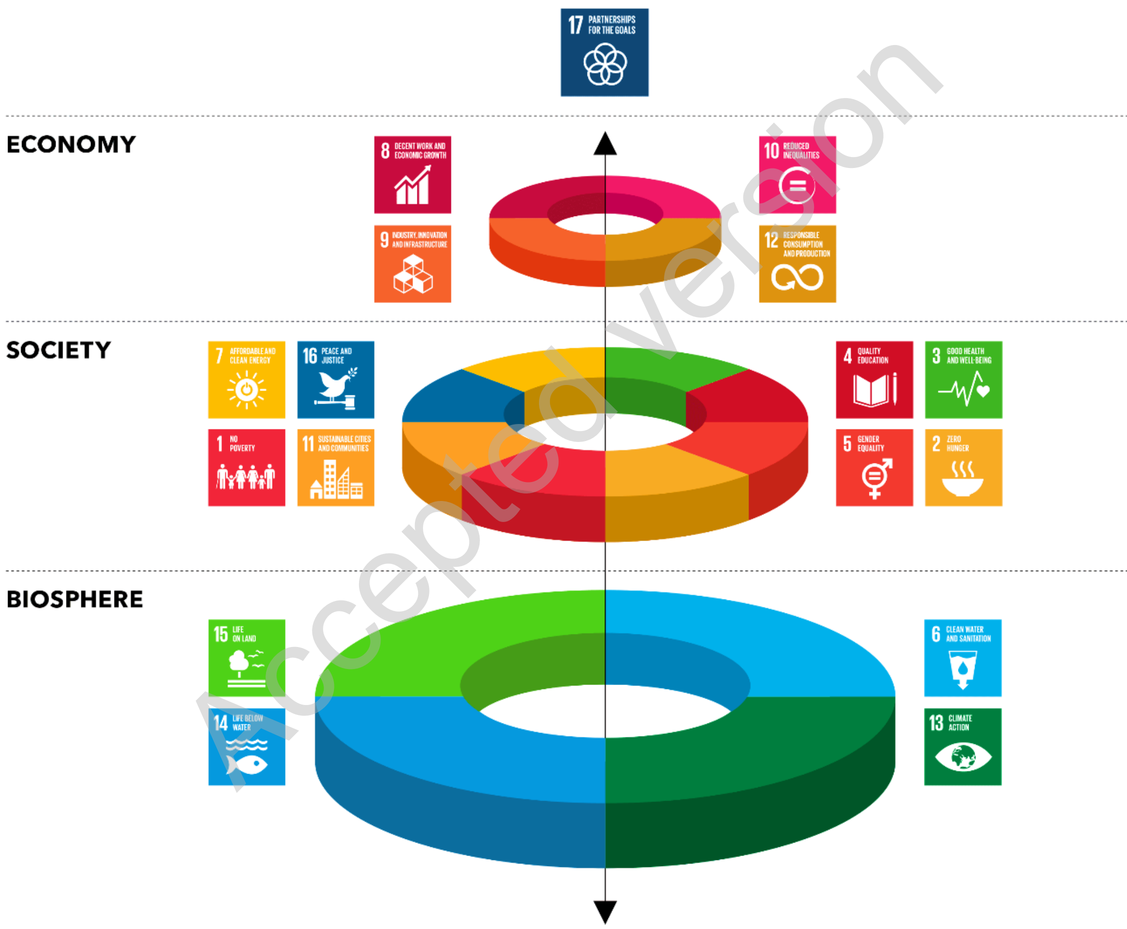


Figure 2. The United Nations Sustainable Development Goals (SDGs); *source (United Nations, 2020)*

2.2 Sustainability Tools to Achieve Sustainable Development Goals

The International Organization of Standardization – ISO is formed by members of more than 160 countries that provide services and a supportive platform for developing standardization to guide entire industries. The standards are presented and discussed by experts' committees appointed by relevant organizations or even state members (ISO, 2017). The standards defined by ISO members form the basis for the definition of global standards and then incorporate this into quality management infrastructure that is the combination of initiatives, institutions, organizations, activities, and people that help ensure products and services meet the requirements of customers (Ikram et al., 2020b). Even though quality management can be operationalized in other measurement or guiding procedures, the ISO standards are considered core patterns and provide relevant mechanisms for codifying and transforming knowledge into organizational knowledge. Table 1 summarizes sustainability tools that foster the achievement of SDGs.

The voluntary sustainability standards joined forces with globalization, turning sustainability into a regulatory mechanism to achieve stakeholders' requirements regarding companies' internationalization and their value chains (Diaz-Balteiro et al., 2017; Chowdhury et al., 2020). The standards that stand out were the management systems established by the ISO related to quality management (ISO 9001:2015), the environmental management system (ISO 14001:2015). The British Standards Institution published the Occupational Health and Safety Standards (OHSAS) (OHSAS 18001; ISO 45001:2018). One primary feature of these management systems is the possibility of being certified by Certification Bodies that perform independent third-party auditing services. They also assess whether the adherence of a given management system according to the determined international standards (e.g., ISOs 9001, 14001, 45001, and the OHSAS 18001) and recommendations for achieving intended results (Fonseca and Carvalho, 2019).

Table 1: Summary of Sustainability Standards Aligned with and Helping to Achieve Sustainable Development Goals (SDGs)

Goal	Goal Description according to SDGs	Required Sustainability Standard	Standards Descriptions
Goal 1: No Poverty	"End poverty in all its forms everywhere." This goal targets the eradication of extreme poverty everywhere and implementing solutions and protection measures. To achieve such a goal, it needs a shared platform containing best practices in economic systems, providing sustainable food production, and balancing resource use and consumption towards SD.	ISO 20400, ISO 37001, GRI 207	ISO 20400:2017: guides organizations on integrating sustainability within procurement (per ISO 26000). ISO 37001: refers to the anti-bribery management system that targets transparency and trust related to organizations' credibility and reputation. GRI 207: relates to economic topic-specific standards. This standard refers to the economic dimension of organizations' sustainability, focusing on their financial conditions.
Goal 2: Zero Hunger	"End hunger achieve food security and improved nutrition and promote sustainable agriculture." This goal is devoted to increasing rural infrastructure investments, agricultural research, technology development, and plant and livestock gene banks to enhance agricultural productive capacity in developing countries.	ISO 22000 family, ISO 26000, ISO 20400, ISO 34101 series, GRI 411	ISO 22000: relates the requirements for food safety management system by mapping the organizations' needs and controlling food safety hazards. ISO 26000: relates the social responsibility, guiding organizations in assessing their commitments towards the environment and society. ISO 20400 (described in goal 1). ISO 34101: refers to sustainable cocoa. This standard provides management systems for cocoa sustainability production and traceability. GRI 411: relates the impacts on indigenous peoples' rights, proposing remediation plans that organizations can follow.
Goal 3: Good Health and Well-Being	"Ensure healthy lives and promote well-being for all at all ages." This goal relates to health and population, sustainable transport, and National Sustainable Development Strategies (NSDS).	ISO 11137 series, ISO 7153, ISO 37101, GRI 403	ISO 11137-1:2006: specifies the sterilization of health care products on radiation and establishes the requirements for development, validation, and routine control of sterilization processes for medical devices. ISO 7153-1:2016: relates to metal materials to manufacture surgical instruments (general surgery, orthopedics, and dentistry). ISO 37101:2016: specifies the SD in communities, guiding cities and communities to establish SD policies based on a holistic approach.

			GRI 403: relates to occupational health and safety.
Goal 4: Quality Education	"Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all." Education for Sustainable Development (ESD) is an integral element of quality education and an enabler of SD.	ISO 21001, ISO 29993, GRI 404	ISO 21001:2018: offers the requirements and guidance for management systems for educational organizations. ISO 29993: relates to learning services outside formal education, including all types of life-long learning. GRI 404: refers to training and education and organizations' approach for training and upgrading employees' skills, performance, and career development.
Goal 5: Gender Equality	"Achieve gender equality and empower all women and girls". This goal is related to gender equality and women's empowerment.	ISO 26000, SA8000, GRI 404	ISO 26000: (described in goal 2). SA8000: guides organizations to develop, maintain and implement socially acceptable practices in workplaces. GRI 404: (described in goal 4).
Goal 6: Clean Water and Sanitation	"Ensure availability and sustainable management of water and sanitation for all." Water resources suffered from an increase in pollution and severe water stress around the world. Water crises are causing implications for public health, environmental sustainability, food, energy security, and economic development.	ISO 24518, ISO 24521, ISO 30500, GRI 303	ISO 24518:2015: specifies the activities relating to drinking water and wastewater services and water utilities crisis management. ISO 24521:2016: guides the management of essential on-site domestic wastewater services. ISO 30500:2018: refers to non-sewer sanitation systems, establishing the general safety and performance requirements for designing and testing prefabricated integrated treatment units. GRI 303: refers to water and effluents and how organizations manage their uses.
Goal 7: Affordable and Clean Energy	"Ensure access to affordable, reliable, sustainable, and modern energy for all." Energy is a core element for achieving SDGs due to climate change. Then, ensuring affordable, reliable, and sustainable sources of energy will allow better opportunities for future generations.	ISO 50001, EU Ecolabel, GRI 302	ISO 50001: relates to energy management, establishes basic guidelines for organizations committed to addressing its impacts, conserving resources, and improving energy management. EU Ecolabel: a label of environmental excellence, Ecolabel promotes the circular economy, less waste, and CO2 generation and manufacturing processes. GRI 302: presents the general requirements about energy use.
Goal 8: Decent Work and Economic Growth	"Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all."	ISO 45001, SA8000, GRI 201	ISO 45001: relates to occupational health and safety, this standard guides organizations in improving employee safety, reducing workplace risks, and creating safer working conditions. SA8000: (described in goal 5).

	This goal is related to the green economy, sustainable tourism, employment, decent work for all, and social protection.		GRI 201: guides the general requirements on economic performance.
Goal 9: Industry, Innovation, and Infrastructure	"Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation." Related to industry and sustainable transportation, this goal intends to reinforce social and industrial development to promote social objectives and employment creation, gender equality, labor standards, and access to health and education.	ISO 56002, ISO 44001, GRI 203	ISO 56002: provides general guidance for an innovation management system to organizations. ISO 44001:2017: establishes the requirements and framework for collaborative business relationship management systems. GRI 203: refers to the requirements on indirect economic impacts of organizations.
Goal 10: Reduce Inequalities	"Reduce inequality within and among countries." This goal aims to advance societies on complex quality and compliance systems, ensure proper market functioning, protect people's health and safety, and preserve the environment.	ISO 26000, SA8000, SMETA, GRI 405	ISO 26000: (described in goal 2). SA8000: (described in goal 5). SMETA: guides organizations on proving they are working in an ethical manner (ethical audit). GRI 405: refers to general guidance for diversity and equal opportunity requirements for organizations.
Goal 11: Sustainable Cities and Communities	"Make cities and human settlements inclusive, safe, resilient, and sustainable." It is related to disaster risk reduction, sustainable transportation, voluntary local reviews, sustainable cities and human settlements, and NSDS.	ISO 37101, GRI 203	ISO 37101: (described in goal 3). GRI 203: (described in goal 9).
Goal 12: Responsible Consumption and Production	"Ensure sustainable consumption and production patterns." It encompasses chemicals and waste, sustainable consumption and production, and sustainable tourism. The way societies are producing and consuming is fundamental when addressing SD.	ISO 20400, ISO 15392, ISO 20245, GRI 302	ISO 20400: (described in goal 1). ISO 15392: provides general guidance for organizations towards sustainability in buildings and civil engineering works. ISO 20245: refers to the cross-border trade of second-hand goods. GRI 302: (described in goal 7).
Goal 13: Climate Action	"Take urgent action to combat climate change and its impacts". This goal comprises the atmosphere, climate change, small island developing States, and NSDS.	ISO 14001, EMAS, EU Ecolabel, GRI 302	ISO 14001:2015: offers requirements and guidance for environmental management systems. EMAS: The Eco-Management and Audit Scheme was developed by European Commission to assess and report their environmental performance. EU Ecolabel: (described in goal 7). GRI 302: (described in goal 7).
Goal 14:			ISO/TC 234: presents the requirements for standardization in the field of fisheries and aquaculture.

Life Below Water	“Conserve and sustainably use the oceans, seas and marine resources for SD”. Within this goal, oceans, seas, small island developing States, and sustainable tourism are addressed.	ISO/TC 234, ISO/TC 8, GRI 305	ISO/TC 8: comprises the ships and marine technology requirements. GRI 305: related to emissions covering greenhouse gases (GHGs).
Goal 15: Life on Land	“Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.” It involves the biodiversity and ecosystems, forests, mountains, NSDS, desertification, land degradation, and drought.	ISO 14055, ISO 14001, EMAS, EU Ecolabel, ISO 38200, GRI 306	ISO 14055-1:2017: presents the good practices framework and guidelines for establishing good practices for combatting land degradation and desertification. ISO 14001: (described in goal 13). EMAS: (described in goal 13). EU Ecolabel: (described in goal 7). ISO 38200: comprises the requirements for chain of custody of wood and wood-based products. GRI 306: refers to effluents and waste management.
Goal 16: Peace, Justice, and strong institutions	Promote peaceful and inclusive societies for SD, provide access to justice for all, and build effective, accountable, and inclusive institutions at all levels. This goal includes the information for integrated decision-making and participation, the institutional frameworks and international cooperation for SD, violence against children, and NSDS.	ISO 19600, ISO/DIS 37000, ISO/TC 309, GRI 403	ISO 19600:2014: offers guidelines for compliance management systems. ISO/DIS 37000: presents guidance for governance and ethics in organizations. ISO/TC 309: specifies the standardization for direction, control, and accountability for organizations' governance. GRI 403: (described in goal 3).
Goal 17: Partnership for the Goals	“Strengthen the means of implementation and revitalize the global partnership for SD.” It involves capacity development, finance, financial inclusion, multi-stakeholder partnerships and voluntary commitments, science, technology, trade, and NSDS.	GRI 207, ISO	GRI 207: (described in goal 1). ISO: independent, a non-governmental international organization for standardization.

Source: International Organization for Standardization, (ISO, 2019); Global Reporting Initiative (GRI, 2018); Social Accountability (SA8000, 2016); UN, 2020); Sedex Members Ethical Trade Audit (SMETA, 2018); Eco-Management and Audit Scheme (EMAS, 2019); European Union Ecolabel (EU Ecolabel, 2018)

2.3 Establishing criteria for selecting certification bodies

Since certifications are essential for several organizations to achieve their trade goals, there are several motivations to obtain certifications from certification bodies. These enterprises and their auditors suffer pressure from these organizations to get more certifications (Alvarenga et al., 2018). They also audit organizations to see whether they are compliant or not. The assurance of these procedures is complex. Certifications require the certification bodies' auditors' professionalism in being impartial, checking whether an organization's compliance is following the predetermined standard. In assessing quality standards, the audits have a broader task in checking if what the company says it is doing is what they are doing in practice (Hoy and Foley, 2015). In ISO 9000, and the principles for continuous improvements, external auditors are expected to look at evidence and facts rather than only look at what is written in reports when comparing financial audit practices. From the side of organizations, they expect auditors to act more as consultants than auditors. It is noteworthy that auditors underestimate the needed balance related to the continuous improvements and compliance with the standard within the auditing process (Ikram et al., 2020c).

Sustainability certification is granted to an organization after an accredited external auditing service (Tavares de Aquino and Maciel de Melo, 2016). Along the certification process, an organization must present the proofs that it meets certification requirements. The organization already selected the certification/registration body that it is targeting to achieve at this stage. The organization's goal related to this external auditing is to confirm whether the organization meets SD requirements (Castka et al., 2015). The external audit can contribute to the company's opportunities to improve its sustainability practices towards the SDGs (Bian et al., 2020). An organization decides which standards/certification process it will be enrolled in by selecting a specific certification body.

In this study, we selected the top six reputable certification bodies such as BV, SGS, DNVGL, LR, BSI, and TÜV based on their sizes, support of SDGs, competitive prices, a team of quality auditors, financial position, and factors presented in Table 2. We posit the organization's decision is based on many criteria, such as getting industry knowledge, recognition concern, costs reduction, gaining experience, improving quality of communication, among others. These criteria come from both the literature and support from elements of the certification standards. The knowledge of the organization's product and business is grounded in ensuring the auditor understands the client's goals while possessing knowledge of the industry. A CB's reputation is vital as this third parties' involvement helps build confidence in the certification process and its outcomes. There is always a concern with payments and operational outcomes, as this reflects the importance of providing a cost/benefit for the certification. As was noted in the introduction of this study, organizations in developing countries often struggle with available resources for certifications as a lack of resources prohibits certification and therefore progress on meeting the UN SDGs and improvements in corporate social responsibility (CSR) Lamichane et al. (2020). We would like to note the importance of high-quality auditors. Organizations seeking CBs need help solving ambiguous problems, judgment based on prior experience, and good communication. To this end, selecting the correct CB can quickly help get an organization up to speed with technical and management decisions. Additionally, the vast range of auditing services prices and quality levels led are also considered critical factors for an organization to decide which service to hire (USTUN and DEMIRTAS, 2008).

Due to the variety of certification bodies, an organization may face difficulties choosing the best certification body according to its goal. If the organization makes a poor decision in selecting its suppliers and certification body, direct and indirect consequences arise (Chan and Kumar, 2007). In this way, Prajogo et al. (2020) reported the appearance of inconsistencies related to the certification body's quality of service, especially those offered in developing countries. In summary, selecting the wrong supplier can lead the organization to incur operational and financial difficulties.

The establishment of successful partnerships depends on selecting suitable suppliers and strategic choices when managing suppliers (Wang et al., 2020). Selecting the right certification body operationalizes strategic choices, and these decisions involve tangible and intangible criteria. Previous researchers have stated that a general model for decision-making comprising preestablished criteria sets is almost impracticable (Huang and Keskar, 2007). We see a practical approach to identifying criteria based on our review and summary of the literature. Each of these selection criteria has additional sub-criteria. The sub-criteria help to provide a deeper level of insight. We add to this additional expert feedback. The resulting criteria and sub-criteria are in Table 2 and Table 3.

Table 2: Comparative Analysis of Top Certification Bodies

Certification Body	Location	Support to Sustainable Development Goals (SDGs)	Number of employees	Providing Services to Countries	Years of Business	Customers	Assessment, Rating and Awards	Revenue in 2019 (USD)	Source
DNVGL	Norway	√	12000	+100	156	100,000	NA	2386.89 million USD	(DNVGL, 2019)
BV	France	√	+78000	+140	192	400,000	MSCI (Rated A), Dow Jones Sustainability Index (B Rated), Sustainanalytics (Low Risk, Ranked 3)	6041.77 million USD	(Bureau Veritas, 2019)
SGS	Switzerland	√	+94000	+140	140	200,000	Member of Dow Jones Sustainability Index, ASM sustainability award Gold class, Received the new platinum rating from Ecovadis	7243.99 million USD	(SGS, 2019)
LR	United Kingdom	√	8000	80	260	60,000	NA	945.1 million USD	(Lloyd's Register, 2019)
BSI	United Kingdom	√	5089	195	118	84,000	NA	725.53 million USD	(BSI, 2019)
TÜV	Germany	√	25000	50	150	90000	NA	2469.86 million USD	(TÜV SÜD, 2019)

DNVGL=*Det Norske Veritas Germanischer Lloyd*, **BV**= *Bureau Veritas*, **SGS**= *Société Générale de Surveillance*

LR= *Lloyds Register*, **BSI**= *British Standards Institution*, *TUV*= *Technischer Überwachungs-Verein*

Table 3: Decision Sub-criteria of Certification Bodies Selection and Supporting Information

Criteria	Sub-Criteria	Supporting Information	Authors
Reputation (REP)	The financial position of certification bodies gives confidence to customers. (RC1) The level of acceptance of the certification bodies in the market. (RC2) The global recognition of certification body. (RC3)	At ISO 17021, topic 4.5.1, the CB provides public access, or disclosure of financial situation, CB's hierarchical structure, the license status, i.e. (the granting, maintenance of certification). These procedures are adopted to gain the confidence and credibility of the CB.	(Chan & Chan, 2010; Haeri & Rezaei, 2019)

	The total number of auditors working in the certification body. (RC4) The total number of customers the certification body has. (RC5)		
Operational outcomes (OP)	Improve effectiveness and consistency of the operating procedures (OP1) Improved conformance to specification (OP2) Reduction in customer complaints (OP3) The effective tracking of documents and records (OP4) The reduction in managerial losses (OP5) Creating a transparent organizational culture (OP6)	Collective action theory: Companies may find incentives in private benefits for engaging a sustainability certification. Among the benefits are customer loyalty, corporate reputation, regulatory relief, improved product prices, and increased market shares. Moreover, the benefits from sustainability certification include improvements in economic efficiency by integrating sustainability into products' supply. Another benefit is reducing the coordination costs that arise in inter-firm relations, harmonizing sourcing requirements, and reaching production scales.	(D. I. Prajogo, 2011; Cai & Jun, 2018; Kannan et al., 2014)
Quality of Auditor (QA)	The auditors have much knowledge about the management standards (QA1) The auditors are responsive to the needs of the organization (QA2) The auditors have a valid auditing license (QA3) The auditors conduct the auditing work in an appropriate way (QA4) The auditors have high ethical standards (QA5) The auditors have practical communication skills (QA6)	At ISO 10019, topic 4.2.6, it is finding the auditor's requirements related to work experience (such as technical, management, and professional features). Some of the experience's features recommended for the auditor are related to solving problems, judgment ability, and communication skills with parties. The auditor's profile may consider one or more specified requirements.	(Poksinska et al., 2006; Power & Terziovski, 2007)
Contract Condition (CC)	The certification body's requirement on the maturity of the standard (CC1) The clarification of term and conditions (CC2) To comply with certification body rules and policies (CC3) The cancellation/termination of contract (CC4) Forbidden to disclose confidential information without prior written approval (CC5) Limitation of Liability and Indemnity of certification body (CC6)	In CB25, recommendations about consultancy selection procedures are found by analyzing the strategic adaptation concerning an organization's target goals, organizational culture, and contract conditions.	(Heras-Saizarbitoria & Boiral, 2013; Hernandez-Vivanco et al., 2019)

Quality of Implementation (QI)	Documented the quality policy and standard operating procedures (QI1) Review and evaluation of management system documents (QI2) Conducting audits on a semiannual basis to check the performance (QI3) Conducting the awareness session about the importance of management standards (QI4) Properly communicate the audit result to the organization's top management team (QI5) To check management review meeting minutes (QI6) Follow up the corrective actions on non-conformances (QI7) Corruption free and transparency in implementation (QI8)	The CB25 states that auditors need to prove their technical competency and professional profiles implemented in previous projects. These proofs ensure that the auditor can manage the project under his/her responsibility and ensure the management standard.	(Ivanova et al., 2014; Ikram, Sroufe, et al., 2020)
Payment Method & Cost (PC)	The price for the overall three-year program of certification audit and surveillance (PC1) Payment methods and conditions that fit the internal planning of an organization (PC2) Location influences on cost, for example, auditors transport costs (PC3)	CB25 stated the requirements for selecting a consultant, such as defining projects' deadlines and determining prices based on cost/benefits, previously formalizing the contract. At ISO 10019 (annex A, topic A.1.2.e), it is expressed that the costs for supporting all activities must be written in the contract.	(Castka, et al., 2020; Jamal & Sunder, 2011)
Knowledge about the client's products and business (KN)	Ability to determine the specific cultural and social customs related to the company's production process (KN1) Knowledge of the sector of activity (by type of process) and the associated risks concerning the place of the company processes (KN2) Technical knowledge is needed to address constraints that may emerge from the partnership (KN3)	In ISO 10019 (topic 4.2.5), recommendations about the CB should possess reasonable knowledge on an organization's processes and products and know customers' expectations previously of offering the consultancy service. Moreover, consultants must understand and capture core points from the industry in which the company operates.	(Power & Terziovski, 2005; Prajogo et al., 2016; Azapagic, 2015)

Note: CB= Certification body

3. Research Methods

Within this study, we have applied Grey Delphi, Grey AHP, and GADA, respectively, to sort out the most relevant indicators from the literature and assess the influence of shortlisting the main criteria and sub-criteria. The schematic of the methodology is in Figure 3. The main advantage of grey system theory is that it ensures accuracy with a small set of data and partial information (Javed et al., 2020). The initial step is to assess the main criteria involved in selecting certification bodies that help align with and achieve SDGs through a thorough literature review. To shortlist only the relevant and most important indicators, Grey Delphi is employed. In fuzzy Delphi, experts were asked to score each indicator for screening out fewer essential barriers. The next step is to apply GAHP for obtaining the relative importance of finalized CB indicators and sub-indicators. Finally, the overall ranking of sub-criteria is achieved by summing each alternative's respective weights under individual criteria involved in selecting the right certification body. Then, GADA obtains comparative weights of the top six most adoptable CB's.

To the best of the author's knowledge, there is no standard available for how many experts should be involved in this analysis. Prior studies used a different number of experts for data analysis to obtain reliable results. For example, Shete et al. (2020) selected six experts for data analysis for sustainable supply chain innovation to achieve sustainability in supply chain management. Whereas, Torkabadi et al. (2018) selected five experts using the FAHP to identify the supply chain barriers to improve sustainable production and consumption.

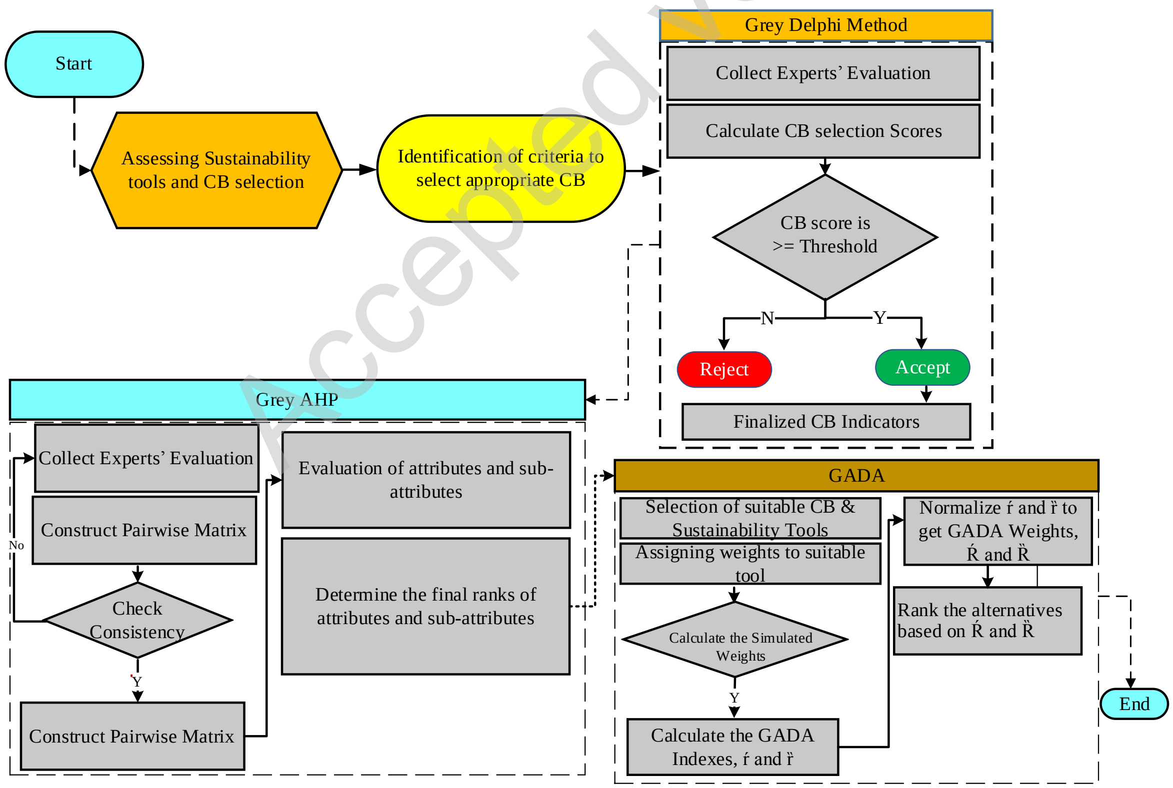


Figure 3: Grey Integrated Research Methodology

Note: CB=Certification body, GAHP= Grey Analytical Hierarchy Process, GADA=Grey Absolute Decision Analysis, R' =Relative Weight

Therefore, in this study, eight academic and industrial experts were involved. We asked participants through an online survey: academic professors, research analysts, policymakers, stakeholders, and managers to give a score based on TFN and grey linguistic variables to prioritize the certification body's selection indicators. The experts have more than 15 years of work experience, and they are very familiar with the industry, sustainability standards, and certification research context. Their details are in the supplementary information (S1), whereas the questionnaire used for experts' feedback is in the supplementary information (S2).

3.1 Grey Delphi Method

We used the Grey Delphi method by combining Deng's Grey system Theory and the Delphi method (Ma et al., 2011) to select the evaluation indicator using the grey weight function based on Delphi questionnaires. This study employs grey numbers, shown in Table 4, to construct the membership degree function. The development process of the Grey Delphi Method comprises four steps, which are as follows:

Step 1: This step involves the development and defining the range values of *s* grey classes for evaluation indicator *j* $\left[a_j^1, a_j^s\right]$ and split into *k* grey classes.

Step 2: As per the evaluation criterion, half trapezoidal whitening weight function is used for *k* and *s* grey classes and *k*=1. The formulation is as follows:

$$f_j^1(x)=\begin{cases} 1, & x \le a_j^1 \\ \frac{b_j^1 - x}{b_j^1 - b_j^1}, & a_j^1 < x \le b_j^1, \\ 0 & x > b_j^1, \end{cases} \quad (1)$$

$$f_j^s(x)=\begin{cases} 0, & x < a_{j,}^s \\ \frac{x - b_j^s}{b_j^s - b_j^s}, & a_j^s \le x < b_j^s, \\ 1 & x \ge b_j^s, \end{cases} \quad (2)$$

The triangular whitening weight function is used for k=*l* (*l*=2, 3…*s*-1). The formulation is as follows:

$$f_j^k(x)=\begin{cases} 0, & x \notin \left[a_{j,}^k\ b_j^k\right], \\ \frac{x - a_j^k}{\lambda_j^k - a_j^k}, & x \in \left[a_{j,}^k\ \lambda_j^k\right], \\ \frac{b_j^k - x}{b_j^k - \lambda_j^k} & x \in \left[\ \lambda_j^k, a_j^k\right], \end{cases} \quad (3)$$

Where, $\lambda_j^k=\left[a_{j+}^k\ b_j^k\right]/2$. a_j^k represent the initial point is called the lower bound of *kth* grey classes, whereas the upper bound of *kth* grey classes is denoted by b_j^k respectively.

Step 3: The calculation of synthetic coefficient denoted by σ_j^k is done in this step. The formulation equation is:

$$\sigma_j^k=\sum_{p=1}^{l} f_j^k(x).\ \eta_j^p \quad (4)$$

Where $f_j^k(x)$ represents the whitening weight function of k grey class for indicator *j*, whereas p presents the experts' feedback. Here, η_j^k indicates the wight of *j* indicator in synthetic, clustering, the name and number of experts' feedback in category *p*.

Step 4: The final step involves the identification of evaluated decision vectors to check whether the indicator as per set judge criterion $max_{1\leq k\leq s}\left\{\sigma_j^k\right\}=\left\{\sigma_j^{k^*}\right\}$ belongs to class k^*.

Table 4: Grey Delphi Scale

Grey class	Linguistics	Range
1	Not Very Important	(1, 2.5)
2	Not Important	(0.5, 3.5)
3	Undecided	(1.5, 4.5)
4	Important	(2.5, 5.5)
5	Very Important	(3.5, 5)

3.2 Grey Analytical Hierarchy Process

We employed Grey Analytical Hierarchy Process (G-AHP) by integrating Grey System theory with Saaty's AHP to determine the weights. In Saaty's AHP, a 1-9 scale is used to construct pairwise judgment matrices. Whereas, in GAHP five grade-level scales such as (1, 3, 5, 7, 9) are used to evaluate the experts' feedback. The details of GAHP linguistic values are in Table 5. The main advantage of using GAHP, over AHP is that GAHP provides a more accurate result with a partial and uncertain set of expert feedback (Zareinejad et al., 2014). Moreover, we select GAHP over Fuzzy AHP because the former already includes fuzzy conditions and, therefore, expertly handles fuzziness in the decision-making.

The development process of G-AHP comprises four following steps:

Step 1: Draw the hierarchal structure by defining the study goal, main criteria, sub-criteria, and Alternatives for the problem to be solved

Step 2: Construct matrices of pairwise comparisons with the help of the grey scale presented in Table 5. Let E indicates the G-AHP pairwise comparison matrix as follows:

$$E=\begin{bmatrix}\otimes G_{11} & \cdots & \otimes G_{1n}\\ \vdots & \ddots & \vdots\\ \otimes G_{m1} & \cdots & \otimes G_{mn}\end{bmatrix}=\begin{bmatrix}\left(\underline{G_{11}},\overline{G_{11}}\right) & \cdots & \left(\underline{G_{1n}},\overline{G_{1n}}\right)\\ \vdots & \ddots & \vdots\\ \left(\underline{G_{m1}},\overline{G_{m1}}\right) & \cdots & \left(\underline{G_{mn}},\overline{G_{mn}}\right)\end{bmatrix} \tag{5}$$

Where grey number is denoted by $\otimes G_{ij}$, $\otimes G_{ij}$ =1 if i=j, and $\otimes \overline{G_{ij}}$ and $\otimes \underline{G_{ij}}$ are the upper and lower limit of $\otimes G_{ij}$ respectively.

Step 3: The normalization of the G-AHP pairwise matrix as in presented as follows:

$$E^* =\begin{bmatrix}\otimes G_{11}^* & \cdots & \otimes G_{1n}^*\\ \vdots & \ddots & \vdots\\ \otimes G_{m1}^* & \cdots & \otimes G_{mn}^*\end{bmatrix}=\begin{bmatrix}\left(\underline{G_{11}}^*,\overline{G_{11}}^*\right) & \cdots & \left(\underline{G_{1n}}^*,\overline{G_{1n}}^*\right)\\ \vdots & \ddots & \vdots\\ \left(\underline{G_{m1}}^*,\overline{G_{m1}}^*\right) & \cdots & \left(\underline{G_{mn}}^*,\overline{G_{mn}}^*\right)\end{bmatrix} \tag{6}$$

The upper and lower limit $\otimes \overline{G_{11}}^*$ and $\otimes \underline{G_{11}}^*$ are the normalized limits obtained as follows:

$$\underline{G_{11}}^* = \left[\frac{2\underline{G}_{ij}}{\sum_{i=1}^{m}\underline{G}_{ij}+\sum_{i=1}^{m}\overline{G_{11}}}\right] \tag{7}$$

$$\overline{G_{11}}^* = \left[\frac{2\overline{G_{ij}}}{\sum_{i=1}^{m}\underline{G}_{ij}+\sum_{i=1}^{m}\overline{G_{11}}}\right] \tag{8}$$

Step 4: Finally, the calculation of relative grey weights for the normalized lower and upper limit as presented as:

$$\otimes \underline{W_i} = \frac{\Sigma\, \underline{G_{ij}^*}}{n} \quad (9)$$

$$\otimes \overline{W_i} = \frac{\Sigma\, \overline{G_{ij}^*}}{n} \quad (10)$$

By converting the grey weights $\otimes W_i$ to definite weights W_i as:

$$W_i = \left[(1-\lambda) * \otimes \underline{W_i}\right] + \left[\lambda * \otimes \overline{W_i}\right] \quad (11)$$

λ is the grey coefficient whose value is set as 0.5. The value of the grey coefficient can be varied based on the decision-makers.

The consistency ratio (CR) is an important indicator to check the reliability of each GAHP pairwise matrices' results was computed. The acceptability of result by using GAHP, when the value of CR records less than 0.1, otherwise the obtained results consider unreliable and process of obtaining experts' feedback must be revised (Ikram et al., 2020)

Table 5. Grey Analytical Hierarchy Process Scale

Scale	Linguistics	$\otimes G$
1	Not Very Important	(1, 2)
3	Not Important	(2, 4)
5	Undecided	(4, 6)
7	Important	(6, 8)
9	Very Important	(8, 10)

3.3 Grey Absolute Decision Analysis (GADA)

Deng introduced the concept of grey system theory in 1981. It has been widely used in these applications in various areas, e.g., healthcare, environmental sustainability, supply chain, publications growth analysis, project management engineering, and energy. In the grey systems paradigm, perfect data does not exist (Javed et al., 2020b). Data that may seem perfectly useful may appear partially useful tomorrow when new information is made available. There are many sources of uncertainty in primary data (Sharma and Seal, 2019). No quantitative information can ideally represent a qualitative concept. Thus, the data's uncertainty or impreciseness corresponds to subjectivity or qualitative concepts, and this can represent the practices looked at in this study.

The Grey Absolute Decision Analysis (GADA) model, proposed by Javed et al. (2020), is a new development in the theory of Multi-Attribute Decision Making (MADM). It is a breakthrough as, unlike conventional methods, it does not require the normalization of data. Thus, the model is practically dimensionless and convenient to use. The key advantage of this method lies in its simplicity and independence over the criteria dimensions. The construction of GADA comprises six steps:

Step 1. Preparing data reporting

This step involves the recording of responses from the expert and constructing a matrix of response decision measurement represented by $[a(k)]$ for "higher the better" criteria, C(k) for each alternative, $A_{(j)}$ for an optimal solution.

$$A_{(1)} \quad \cdots \quad A_{(S)}$$

$$a(k) = \begin{array}{c} E_1 \\ \dots \\ E_N \end{array} \begin{bmatrix} a_{1(1)} & \cdots & a_{1(S)} \\ \vdots & \ddots & \vdots \\ a_{N(1)} & \cdots & a_{N(S)} \end{bmatrix} \tag{12}$$

The reciprocal operation is needed to separate better from lower criteria. The above matrix can be written as:

$$\begin{array}{ccc} & A_{(1)} \quad \dots \quad A_{(S)} \\ r(k) = \begin{array}{c} E_1 \\ \dots \\ E_N \end{array} & \begin{bmatrix} r_{1(1)} & \cdots & r_{S(1)} \\ \vdots & \ddots & \vdots \\ r_{1(N)} & \cdots & r_{S(N)} \end{bmatrix} \end{array} \tag{13}$$

While it should be observed that in the case of lower the better criteria, we need to Reciprocate the Operation (RO) for data analysis, thus, we can say $r_j = RO(a_i)$. Whereas, when in the 'higher the better case, the feedback from experts can be used for data analysis thus $[r_{ij}] = [a_{ij}]$. The GADA approach's main advantage is that there is no need for data normalization due to its dimensionless property.

Step 2. **Calculating alpha values and AGRG Pairwise Comparison Matrix**

Absolute Grey Relational Grade (ε) is calculated among E_i, and can be written as: $E_s; i = 1, 2, 3....N$, which is the AGR pairwise comparison matrix denoted by $[\varepsilon]$ and used to obtain the pairwise comparison matrix. Further, Javed alpha (α) is measured by the average of each row yields (Javed and Liu, 2019). When the value of ε is equal to 1, which shows a strong association, and the ε value at 0.5, presents a poor and weak relationship, the overall value of ε within E_i would always be equal to 1. If the value of ε is recorded less than or near to 0.5, which lies in the unacceptable region, which turns into recollecting the expert's feedback. For one criterion, the equation can be developed as:

$$\begin{array}{cc} & \begin{array}{ccc} E_1 & \dots & E_N \end{array} \\ \begin{array}{c} E_1 \\ \dots \\ E_N \end{array} & \begin{bmatrix} \varepsilon_{11} & \cdots & \varepsilon_{1N} \\ \vdots & \ddots & \vdots \\ \varepsilon_{N1} & \cdots & \varepsilon_{NN} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \dots \\ \alpha_N \end{bmatrix} \begin{bmatrix} \sqrt{\alpha_1} \\ \dots \\ \sqrt{\alpha_N} \end{bmatrix} \end{array} \tag{14}$$

Where $\alpha_i = \frac{1}{N}(\varepsilon_{i1} + \varepsilon_{i2} .... + \varepsilon_{iN})$ is called *Javed's alpha*. The equation can be developed for M criteria $C(k); K = 1, 2, ...M$.

$$\begin{array}{cc} & \begin{array}{ccc} C(1) & \dots & C(M) \end{array} \\ \begin{array}{c} E_1 \\ \dots \\ E_N \end{array} & \begin{bmatrix} \sqrt{\alpha_1(1)} & \cdots & \sqrt{\alpha_1(M)} \\ \vdots & \ddots & \vdots \\ \sqrt{\alpha_N}(1) & \cdots & \sqrt{\alpha_N(M)} \end{bmatrix} \end{array} \tag{15}$$

Step 3. Calculating perceived criteria weights

Now collect perceived weights of criteria from the experts, whereas, k=1,2,3,…, M (M) is total number of criteria

$$\begin{array}{cc} & \begin{array}{ccc} E_1 & \dots & E_N \end{array} \\ \begin{array}{c} C(1) \\ \dots \\ C(M) \end{array} & \begin{bmatrix} \beta_1(1) & \cdots & \beta_N(1) \\ \vdots & \ddots & \vdots \\ \beta_1(M) & \cdots & \beta_N(M) \end{bmatrix} \end{array} \tag{16}$$

$$\begin{array}{c} C(1) \\ \dots \\ C(M) \end{array}\begin{bmatrix} \beta(1) \\ \dots \\ \beta(M) \end{bmatrix}$$

Where, $\beta(k)$ =geometric mean $\beta_1(k), \beta_2(k), \beta_3(k), \dots..\beta_{N(k)}$ and $\sum \beta(k) = 1$.

Step 4. Calculating simulated criteria's weights

Calculating Simulated weight criteria's $\delta(k)$ by multiplying the $\theta(k)\,\beta_i(k)$ and $\sqrt{\alpha_i(k)}$ to get $\Theta(\mathrm{k})$. This approach of obtaining $\Theta(\mathrm{k})$ was proposed by Javed et al. (2020).

$$\begin{array}{c|c} & \begin{array}{ccc} E_1 & \dots & E_N \end{array} \\ \begin{array}{c} C(1) \\ \dots \\ C(M) \end{array} & \begin{bmatrix} \Theta_1(1) & \cdots & \Theta_N(1) \\ \vdots & \ddots & \vdots \\ \Theta_1(M) & \cdots & \Theta_N(M) \end{bmatrix} \end{array} \tag{17}$$

$$\begin{array}{c} C(1) \\ \dots \\ C(M) \end{array}\begin{bmatrix} \delta(1) \\ \dots \\ \delta(M) \end{bmatrix}$$

Where, $\delta(k)$ = geometric mean $\Theta_1(k), \Theta_2(k) \dots \Theta_N(k)$ $\theta_i(k) = \omega_i(k) \times \sqrt{\alpha_i(k)}\,.\,\beta_i(k)$

Step 5. Calculating the overall weight of criterion against each alternative

In the MCDA approach, each criterion's weight is obtained by transforming the individual's weight to group weight with the help of a geometric mean (Stević et al., 2018). Hence, in GADA, the formula of overall weight calculation is as follows:

$$\begin{array}{c|c} & \begin{array}{ccc} A_1 & \dots & A_S \end{array} \\ \begin{array}{c} C(1) \\ \dots \\ C(M) \end{array} & \begin{bmatrix} \grave{\mathrm{r}}_1(1) & \cdots & \grave{\mathrm{r}}_N(1) \\ \vdots & \ddots & \vdots \\ \grave{\mathrm{r}}_1(M) & \cdots & \grave{\mathrm{r}}_N(M) \end{bmatrix} \end{array} \tag{18}$$

Where for each criterion $C(k)$ and for $j = 1, 2, 3, 4...., S$.

$$\ddot{\mathrm{r}}_j = \left(\prod_{j=1}^{S} \grave{\mathrm{r}}_j^{\delta_k}\right)^{1/\sum_{k=1}^{M}\delta_k} \tag{19}$$

The relative weights $\ddot{\mathrm{R}}_j$ would be yielded by normalization of weights of alternatives within each criterion, can be written as:

$$\ddot{\mathrm{R}}_j = \frac{\ddot{\mathrm{r}}_j}{\sum_{j=1}^{S} \ddot{\mathrm{r}}_j} \tag{20}$$

The weights mentioned above can be arranged in a definite sequence to obtain the local rankings of alternatives.

Step 6. Calculating the relative weight of each criterion against each alternative

In the final step of the GADA construction method, the calculation of global weights is developed by aggregating the simulated weights of criteria such as:

$$\begin{array}{c} \\ \begin{bmatrix} \check{\mathrm{r}} \\ \check{\mathrm{R}} \\ Rank \end{bmatrix} \end{array} = \begin{array}{c} \begin{array}{ccc} A_1 & \dots & A_S \end{array} \\ \begin{bmatrix} \check{\mathrm{r}}_j & \cdots & \check{\mathrm{r}}_j \\ \check{\mathrm{R}}_j & \ddots & \check{\mathrm{R}}_j \\ - & \cdots & - \end{bmatrix} \end{array} \tag{21}$$

Where,

$$r_j'' = \left(\prod_{j=1}^{S} r_j'^{\delta(k)} \right)^{1/\sum_{K=1}^{M} \delta(k)} \tag{22}$$

With the normalized vector,

$$\check{\mathrm{R}}_j = \frac{\check{\mathrm{r}}_j}{\sum_{j=1}^{S} \check{\mathrm{r}}_j} \tag{23}$$

Later, $\check{\mathrm{R}}_j$ can be utilized to prioritize suitable *S* alternatives in order to provide the optimal solution.

4. Results and Discussion

The results of this study comprised three phases. In the first phase, the result of Grey Fuzzy is developed in which the main criteria and sub-criteria of certification bodies are finalized. The second phase presents the GAHP results in which the main and sub-indicators of certification bodies are ranked as per the experts' relative importance. Finally, the ranking of alternatives to show which certification body is the best option for managers to select to achieve the SGDs is in the last phase of the study.

4.1 Grey Delphi results

We used the five grey classes from 1 to 5 with range [1, 2.5], [0.5, 3.5], [1.5, 4.5], [2.5, 5.5] and [3.5,5] which represents not very important, no important, undecided, important, and very important, respectively. Further, this study calculated the synthetic clustering coefficient σ_j^k to obtain the values of decision vectors of certification bodies indicators. To check the validity of Grey Delphi result, there are two selection parameters: (a) If the important or very important (4 and 5) classes have maximum value in the vector decision that belongs to class *k**, this certification body indicator is accepted. Whereas (b) If the class's value, namely important and very important, which is the ratio of class k* is more than 1.0 or over 50% degree except for undecided class, then this certification body indicator is accepted. Table 6 presents the eleven certification bodies' indicators. In contrast, seven indicators are selected, and the rest of the four certification bodies' indicators are not included as they do not meet the acceptability criteria. According to the expert judgmental and evaluation approach, the four indicators, such as political instability, location, ownership, institutional corruption, and nepotism, are rejected because they do not fit within the study's scope.

Table 6: Result of the Grey Delphi Method

Indicators	Grey decision vector	Selected √/ Rejected ×	Indicator code

Reputation	(0.67, 4.33, 8.67, 5.67, 2)	√	REP
Operational outcomes	(1.67, 3.67, 6.33, 6.67, 2.67)	√	OP
Location	(2.33, 6, 7.33, 4.67, 1)	×	
Quality of Auditor	(3, 8, 7.33, 2.67, 0.33)	√	QA
Contract Condition	(0, 0.67, 3.67, 7.67, 7.67)	√	CC
Political instability	(0, 3.33, 10.67, 5.67, 1.67)	×	
Quality of Implementation	(6.67, 4.33, 8.67, 8.67, 2)	√	QI
Ownership	(5.8, 5.33, 2, 0.33)	×	
Payment Method & Cost	(3, 4, 3.67, 6, 3.67)	√	PC
Institutional corruption and nepotism	(0, 2, 7.33, 7, 4.33	×	
Knowledge of products and business	(1.33, 2.33, 5.67, 8.33, 3.33)	√	KN

4.2 Grey analytical hierarchy process results

In this study, GAHP is used to compute the weights of main and sub-indicators to select certification bodies to align with SDGs. Firstly, we draw the hierarchical structure of the problem presented in Figure 4. There are three steps involved in GAHP. The first step involves the study's goal to be achieved; second, a ranking of main and sub-criteria of top five certification bodies worldwide, which ensure the achievement of SD; finally, the overall ranking by using GAHP is presented in the third section respectively.

4.2.1 Ranking of main indicators

The relative weights of criteria and sub-criteria are obtained by solving the pairwise matrix. The results are in Table 7. It can be seen that the Reputation (REP) received the highest weight of 21.20 % followed by Payment Method & Cost (PC) 17.04%, Operational outcomes (OP) 16.10%, Quality of Auditor (QA) 14.06%, Quality of Implementation (QI) 13.87%, Contract Condition (CC) 10.20% and Knowledge of products and business (KN) 7.53%. It must be noted that the difference between the weights of all the main criteria is not more than 3%. Hence, we can say that all of these barriers are important. Nonetheless, REP is relatively more significant, and KN is comparatively less significant.

We can see why reputation is important as certification bodies help with transparency, disclosure of information, and build confidence in the CB, certification process, and its outcomes. This visibility is helpful to all parties and promotes confidence in the certification process of reputable CBs. There is good face validity with payments and operational outcomes also ranking high. These support the importance of providing a cost/benefit for the certification process and strong value proposition. As was noted in the introduction of this study, organizations in developing countries often struggle with available resources for certifications as a lack of resources prohibits certification and therefore progress on meeting the UN SDGs and improvements in corporate social responsibility (CSR) Lamichane et al. (2020). We would like to note the importance of high-quality auditors. Organizations seeking CB need help quickly solving problems, good judgment, and communication with these third parties, so resources are not wasted. The CB can quickly help with technical and management options based on prior experience. The quality of implementation, contract, and knowledge of the organization’s product and business is grounded in ensuring the auditor can manage the project, understand the goals and organizational culture of the client while possessing practical knowledge of the industry at a macro-level and the client's products and processes at a micro-level.

Table 7: Ranking of selecting Certification Body criteria

Main Criteria	Code	Weight	Rank
Reputation	REP	0.2120	1st
Paymen Method & Cost	PC	0.1704	2nd
Operational Outcomes	OP	0.1610	3rd
Quality of Auditor	QA	0.1406	4th
Quality of Implementation	QI	0.1387	5th
Contract Condition	CC	0.1020	6th
Knowledg of Products & Business	KN	0.0753	7th

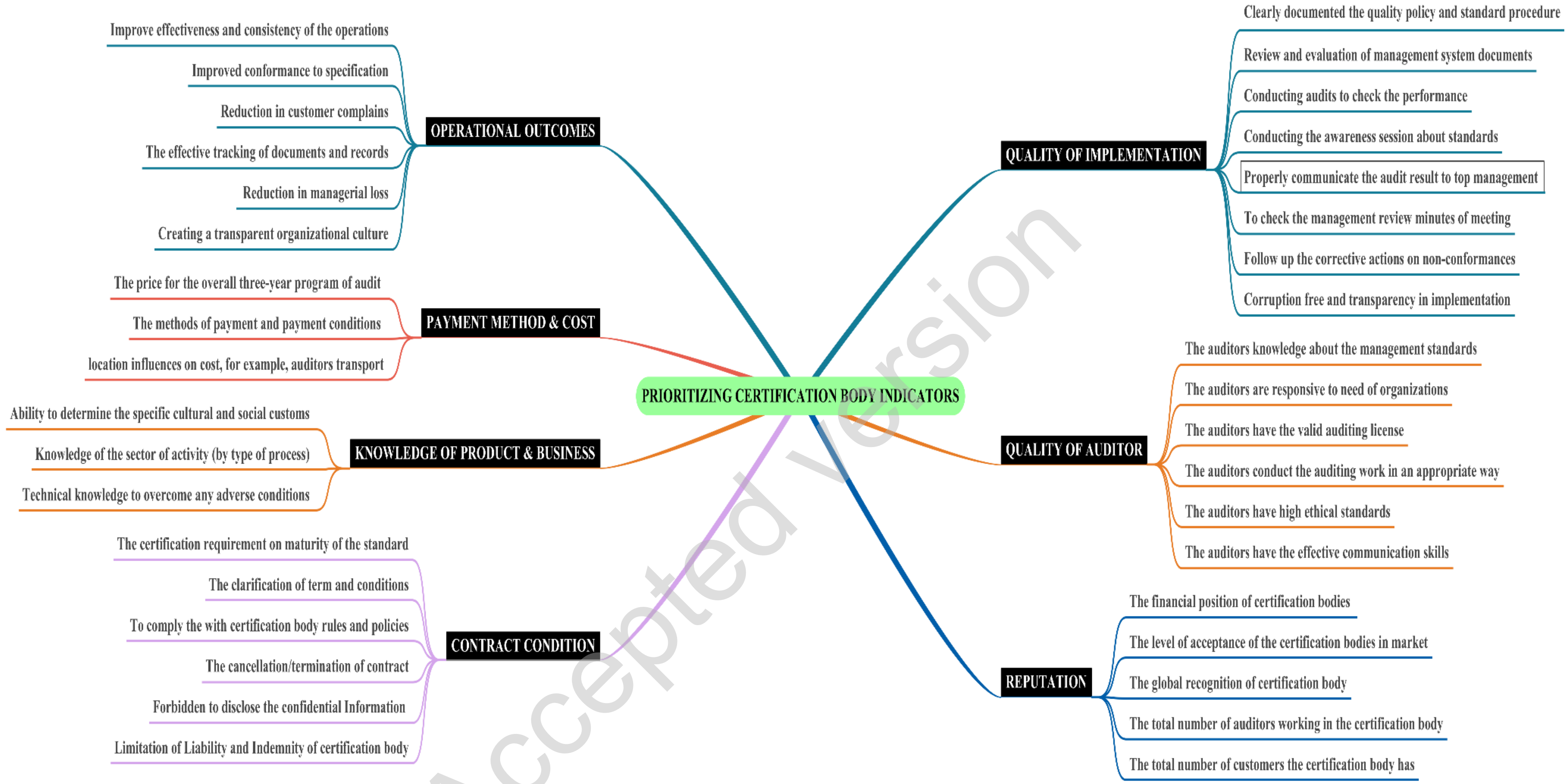


Figure 4: Hierarchal Structure Operationalized in this Study; *source (self-developed)*

4.2.2 Ranking of sub-Indicators

Once we obtained the main criteria weights, we now quantify the degree to which each indicator affects each certification body alternative. To do so, we have constructed 37 pairwise comparison matrices, one for each category of indicator. Next, experts were asked to compare sub-criteria under each category. The weights of sub-criteria under each category are calculated by solving these matrices, as shown in Fig 5-10. It should be noted that any sub-criteria receiving higher weight shall be considered more affected for the selection of certification body to align with and help achieve SD. Hence, the lower the weights, the least effective the rankings of the sub-indicators.

Under REP, weights received by sub-criteria (from higher to lower) are as follows: The level of acceptance of the certification bodies in the market (REP2) 30.84%> the global recognition of certification body (REP3) 23.13%> can be seen in Figure 5. The total number of auditors working in the certification body (REP4) 16.20% > The total number of customers the certification body has (REP5) 15.10% > The financial position of certification bodies to give more confidence to customers (REP1) 14.73%. Other studies that investigated the service sector (Ristono et al., 2018) observed that a third party's reputation plays a vital role in supplier selection. Since the process of certifying the management standard is a bit complex and expensive, organizations rely on reputable certification firms and qualified auditors to effectively implement sustainability certifications. Moreover, certification providers transfer value and credibility to accreditation companies. Therefore, selecting reputable certification bodies ensures internal and external stakeholders that the accreditation body effectively follows management standards (Alvarenga et al., 2018).

Under OP, the sequence is: Reduction in customer complaints (OP3) 25.02%> Improve effectiveness and consistency of the operating procedures (OP1) 23.94%> The reduction in managerial losses (OP5) 18.02%> The effective tracking of documents and records (OP4) 12.51%> Improved conformance to specification (OP2) 10.96%> Creating a transparent organizational culture (OP6) 9.52% presented in Figure 6. According to other studies, Ikram et al. (2020), Chkanikova & Sroufe (2020), our result indicates that the third-party sustainability certification enables dynamic capabilities, improves the production process, reduction in customer complaints, customer satisfaction, and opportunity for the organization to create competitive advantage.

The following sequence was obtained for a pairwise comparison of the Quality of Auditor category (QA): Figure 7 represents the ranking of Quality of Auditor sub-criteria category in which the auditors know the management standards (QA1) 32.30%> The auditors conduct the auditing work in an appropriate way (QA4) 20.21%> The auditors have high ethical standards (QA5) 14.64%> The auditors are responsive to need of organizations (QA2) 13.24%> The auditors have the valid auditing license (QA3) 11.01%> The auditors have practical communication skills (QA6) 8.61%. Previous studies Wiengarten et al. (2017), Singh et al. (2011) endorse our result in which effective implementation of ISO 9001 management standard needs to be handled by the quality auditor, which has an immediate positive effect on organization production outcome. External auditors' expertise, such as practical communication skills, ethical behavior, and enough knowledge about management standards, may boost the company’s confidence and willingness to adopt sustainability standards (Hernandez, 2010).

The pairwise comparison of Contract of Condition (CC) presents the following sequence in Figure 8: The clarification of term and conditions (CC2) 28.06%> The cancellation/termination of the contract (CC4) 25.93%> Forbidden to disclose this confidential information to any person or entity without the prior written approval (CC5)

14.90> Limitation of Liability and Indemnity of certification body (CC6) 10.59%> To comply the with certification body rules and policies (CC3) 11%> The certification body's requirement on the maturity of the standard (CC1) 9.940%. Cândido et al. (2019) conducted a study to investigate if the certified companies may lose/withdraw their ISO certifications due to compliance issue with CB rule and regulation, cancellation of the contract because of less economic benefits, and facing difficulties to manage on-time cost and expenditures of the audit fee.

Figure 9 shows the following sequences of sub-criteria regarding the Under Quality of Implementation (QI): Clearly documented the quality policy and standard operating procedures (QI1) come first in the row and obtained the higher weights of 21.88% within the category followed by> Conducting audits at semiannual basis to check the performance (QI3) 16.38%> Conducting the awareness session about the importance of management standards (QI4) 13.66%> Follow up the corrective actions on non-conformances (QI7) 12.02%> Review and evaluation of management system documents (QI2) 10.44%, corruption-free and transparency in implementation (QI8) 9.671%> Properly communicate the audit result to organization top management team (QI5) 9.360%> and To check management's review of the minutes from meeting (Q16) received the lowest weight 6.6000 % within the category respectively. The certification firms adopt a systematic management approach through identification and evaluate the indicators that assist the effective implementation of management standards. Therefore, the sufficient quality of management standards implementation needs to overcome the repetition/duplication of documents through integration techniques to create synergy between systems (Ikram et al., 2020).

Under the Payment Method and Cost (PC) category, the weights received by sub-criteria (from higher to lower) are in Figure 10: The methods of payment and payment conditions that fit the internal planning of the organization (PC2) received the highest weight of 39.10% followed by> The price for the overall three-year program of certification audit and surveillance (PC1) 36.23%> location influences on cost, for example, auditors transport costs (PC3) 24.67%. Our result is similar to other studies in which payment method and cost are essential criteria when selecting a CB (Jespersen & Hasle, 2017; Poksinska et al., 2006). The cost of certification is considered more as an investment than an expense when quality or technical departments maintain strong connections with their customer through timely fulfilling orders (Chiarini, 2019).

Pairwise comparison of Knowledge about the client's products and business (KN) presents the following sequence: Knowledge of the sector of activity (by type of process) and the associated risks concerning the place of the company processes (KN2) 36.04%> Ability to determine the specific cultural and social customs related to the

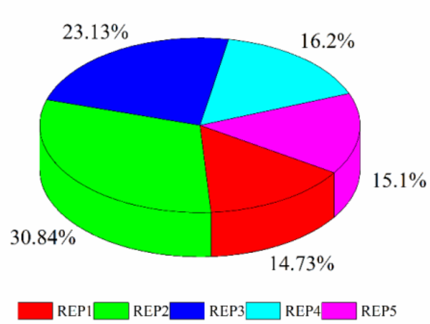


Fig. 5: Reputation Sub-criteria

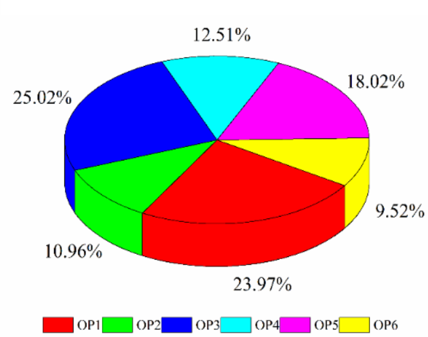
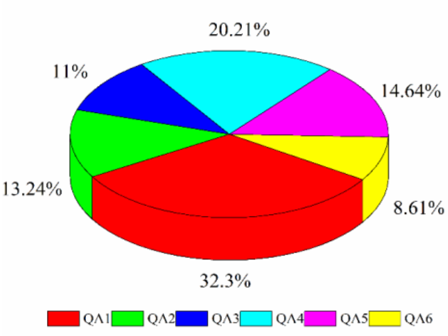


Fig. 6: Operational Outcome Sub-criteria



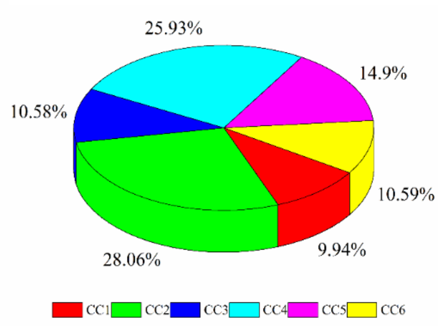
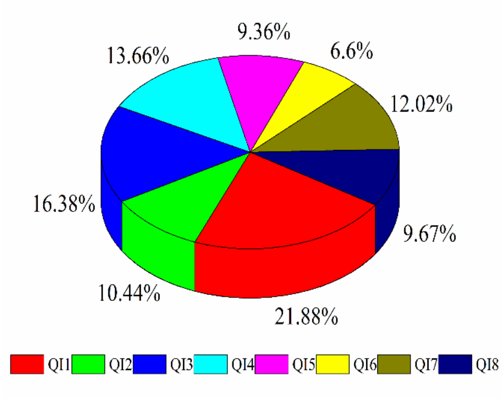


Fig. 7: Quality of Auditor Sub-criteria **Fig. 8:** Contract Conditions Sub-criteria



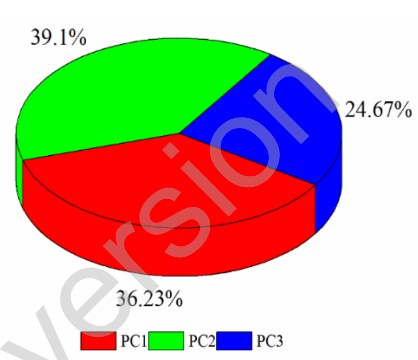


Fig. 9: Quality of Implementation Sub-criteria **Fig. 10:** Payment Method & Cost Sub-criteria

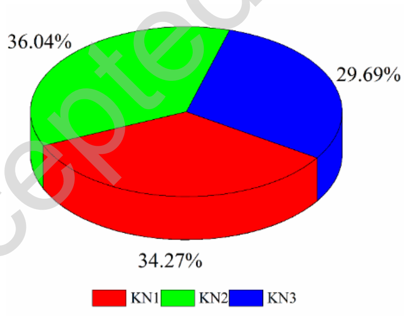


Fig. 10: Knowledge of Product and Business Sub-criteria

Note= The complete form of sub-criteria codes is in table 3.

company's production process (KN1) 34.27%> Technical knowledge to overcome any adverse conditions that are part of the partnership scope (KN3) 29.69%.

4.3 The final ranking of sub-indicators

After calculating the weights of sub-indicators, it is important to determine the overall ranking of sustainability tools' sub-indicators. Weights of sub-criteria under each category of main criteria are multiplied with that criteria weight to obtain the final ranking.

To check the reliability of each GAHP pairwise matrices, the CR was computed. When the value of CR records less than 0.1, the results using GAHP are acceptable; otherwise the obtained results are considered unreliable and the process of obtaining experts' feedback must be revised (Ikram et al., 2020). The CR of all the attributes in this study was recorded less than 0.1 and is presented in Table 8. Finally, each sub-criteria's weights are summed to obtain cumulative weight, which is the base of the final ranking provided in Table 8. The final ranking of sub-attributes for the selection of appropriate certification bodies to develop SD is as follows:

QA1>QI6>QI1>KN2>QA5>KN1>PC1>QI4>>QA3>OP6>REP4>CC5>QI3>REP1>REP3>KN3>OP5>PC2>QI2>CC4>QI7>CC6>CC1>OP2>OP3>QI8>QI5>QA6>QA2>REP5>REP2>QA4>OP4> PC3>CC3>OP1>CC2.

To select the appropriate CB which ensure to achieve SD in the organization, the auditors have much knowledge about the management standards (QA1) receive the highest weight of 0.5009, followed by, To check the management review minutes of the meeting (Q16) 0.4918 and Clearly documented the quality policy and standard operating procedures (QI1) 0.4789 are ranked at the second and third position of the essential sub-criteria respectively (see Figure 11).

Table 8: Final Ranking

Main Attributes	Main criteria codes	Main Attribute Weights	Sub-attributes codes	Consistency Ratio (CR)	Normalized weight
Reputation	REP	0.2120	REP1	0.0083	0.3496
			REP2		0.1488
			REP3		0.3359
			REP4		0.3719
			REP5		0.1546
Operational outcomes	OP	0.1610	OP1	0.0086	0.0980
			OP2		0.2117
			OP3		0.1958
			OP4		0.1408
			OP5		0.3139
			OP6		0.3889
Quality of Auditor	QA	0.1406	QA1	0.0129	0.5009
			QA2		0.1589
			QA3		0.3889
			QA4		0.1471
			QA5		0.4439
			QA6		0.1705
Contract Condition	CC	0.1020	CC1	0.0003	0.2145
			CC2		0.0857
			CC3		0.1190
			CC4		0.2967
			CC5		0.3692
			CC6		0.2318
Quality of Implementation	QI	0.1387	QI1	0.0178	0.4789
			QI2		0.3044
			QI3		0.3543
			QI4		0.4189
			QI5		0.1891
			QI6		0.4918
			QI7		0.2917
			QI8		0.1932
Payment Method & Cost	PC	0.1704	PC1	0.0007	0.4339
			PC2		0.3069
			PC3		0.1192
	KN		KN1		0.4367

Knowledge of products and business	0.0753	KN2	0.0112	0.4717
		KN3		0.3223

Note= The complete form of sub-criteria codes is presented in table 3.

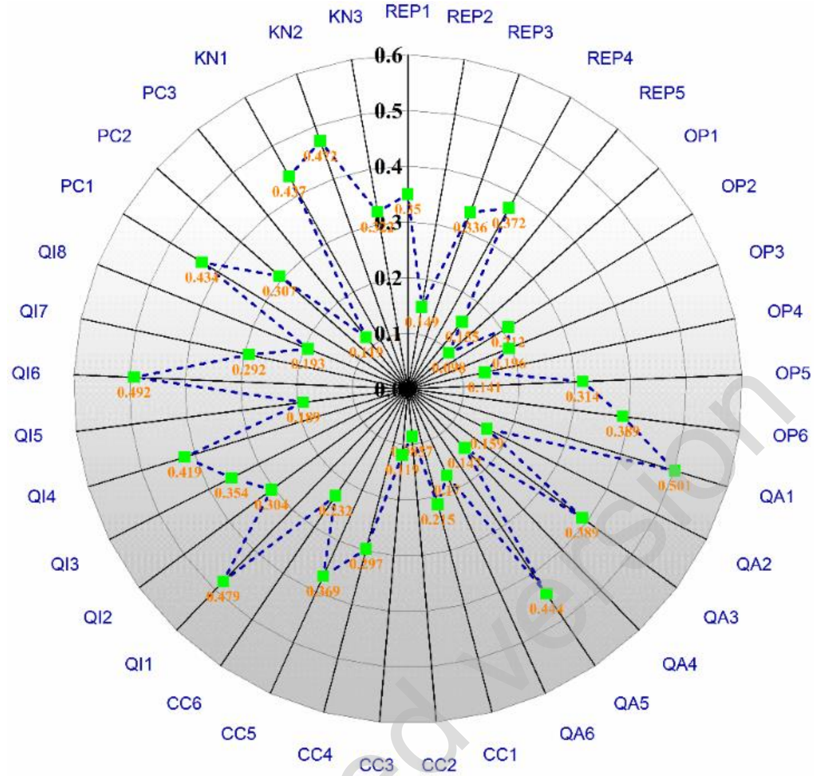


Fig. 11: Relative Weights of Sub-criteria**;** *source (self-developed)*

Note= The complete form of sub-criteria codes is presented in table 3.

4.4 Grey absolute decision analysis results

Aiming to expand insights from analyses, we want to know which sustainability tools effectively align with and manage sustainability practices. The ranking of six main categories of certification bodies, namely BVC (CB1), LR (CB2), DNVGL (CB-3), BSI (CB-4), SGS (CB-5), TÜV (CB-6), and systems, were analyzed by a novel GADA approach. The authors benefited from skilled experts who provided valuable inputs and were considered in a 'group decision-making approach' to determine results. The GADA method is an MCDA technique by which it is possible to obtain optimum relative weights of the criteria. This is achieved through the 'simulated weights of the criteria' δ(k), normalized to get more precise relative weights, which, in turn, is used for improving the ranking of other methods. We have considered all criteria, including, in this manuscript, as a benefit for developing corporate sustainability towards six types of tools involving systems, standards reporting, and programs.

In implementing the GADA method, we have six alternatives/possibilities and eight independent decision-makers experts, who evaluated them against two unequally weighted criteria C(1): benefit criteria (Support to SDGs, assessment rating and awards, providing services to countries); C(2) cost-type criteria. (number of employees, year of business, revenue, number of customers). These experts attributed weights to the set of two criteria, as follows (0.2, 0.8), (0.3, 0.7), (0.4, 0.6), (0.2, 0.8), (0.4, 0.6), (0.3, 0.7), (0.4, 0.6) and (0.2, 0.8). Table 9 scores were recorded on 5-

point Likert-scale, where 1 represents the lowest score, and 5 represents the highest score. For benefit, 5 implies the highest benefit, and cost at 1 implies the least expensive. Here, criterion 1 is the worst concerning the benefit, and concerning the cost criterion, 5 is the worst. Tables 10 and 11 presents the execution of the GADA method.

The certification bodies selection overall ranking for the development of SDGs are based on final scores presented in Table 11 and is as follows:

CB-1>CB-5>CB-3>CB-6>CB-4>CB-2

Based on global weight recording, our results recommend the BVC (CB-1) and SGS (CB-5) the best suited CB's help to achieve SDGs by obtaining the highest weight 0.8731, and 0.7340 respectively, followed by DNVGL (CB-3) with 0.6782 weight, TUV (CB-6) scored at 0.5344, BSI (CB-4) with 0.5212 weight, and LR (CB-2) with a weight of 0.4213. The ordered ranking of different certification bodies and packages is significant due it provided the overall values for SDGs. While the weight comparison helps to prioritize certification bodies, all the identified CB's can achieve SDGs. Additionally, these certification bodies are commonly adopted, ensuring business improvements related to quality and sustainable business practices (Souza and Alves, 2018).

Companies can benefit from our findings in looking at certification bodies in a new light, including our study's outcomes in decision-making and utilizing the approach taken in this study to develop a customized approach for their organization and certification decision-making needs. Organizations, the United Nations, and other authorities can benefit from this study because certification bodies enable tangible cost-benefit results toward global goals. Local governments, international agencies, businesses, and stakeholders all have opportunities to help fulfill these global goals. In 2019, United Nations (UN) reported that financial support is a big hurdle to implement the SDGs agenda in developing countries (UN World, 2019). This study's results and additional benefits come from understanding a more complex value proposition involved in certification bodies. We hope the results will help with a more intelligent allocation of scarce resources to help align investments with faster progress toward global goals for sustainable development.

Table 9: Experts' Evaluation Input Data of Certification Bodies

Experts	**Benefit *C(1)***						**Cost *C(2)***					
	a_1	a_2	a_3	a_4	a_5	a_6	a_1	a_2	a_3	a_4	a_5	a_6
E_1	4	3	3	5	4	3	2	1	2	3	1	4
E_2	5	3	4	3	4	5	3	2	1	2	4	2
E_3	4	5	3	5	4	3	3	2	2	2	3	1
E_4	4	3	4	5	4	5	2	3	4	2	3	5
E_5	5	5	4	4	3	4	1	2	4	2	3	1
E_6	4	3	4	4	4	5	2	5	2	1	5	1
E_7	3	5	3	4	3	4	2	3	4	4	1	3
E_8	5	5	3	4	4	3	4	2	3	2	3	1

Note: E=Expert, a=alternative

Table 10. The Cost and Benefits Result Based on Expert Feedback

Experts			**Benefit *C(1)***							**Cost *C(2)***		
	a_1	***a_2***	***a_3***	***a_4***	***a_5***	***a_6***	***a_1***	***a_2***	***a_3***	***a_4***	***a_5***	***a_6***
E_1	4	3	3	5	4	3	0.5	1	0.5	0.3333	1	0.25
E_2	5	3	4	3	4	5	0.3333	0.5	1	0.5	0.25	0.5
E_3	4	5	3	5	4	3	0.3333	0.5	0.5	0.5	0.3333	1
E_4	4	3	4	5	4	5	0.5	0.3333	0.25	0.5	0.3333	0.2
E_5	5	5	4	4	3	4	1	0.5	0.25	0.5	0.3333	1
E_6	4	3	4	4	4	5	0.5	0.2	0.5	1	0.2	1
E_7	3	5	3	4	3	4	0.5	0.3333	0.25	0.25	1	0.3333
E_8	5	5	3	4	4	3	0.25	0.5	0.3333	0.5	0.3333	1
GM	4.6953	3.8730	3.4642	4.1952	3.7221	3.9040	0.4524	0.4395	0.4001	0.4753	0.3976	0.5491
Weighted GM using α (ȑ)	4.8311	3.9005	3.1248	4.2143	3.9215	3.9110	0.4902	0.4777	0.3974	0.4502	0.4051	0.5880

Note: E=Expert, a=alternative, GM=Geometric mean, :ȑ=GADA

Table 11: Final Evaluation and Ranking of Certification Bodies

	(ȑ)	***a_1***	***a_2***	***a_3***	***a_4***	***a_5***	***a_6***
Inter-criteria ranking	*C(1)*	4.8317	3.9001	3.1244	4.2146	3.9210	3.9112
	Ranking	1	5	6	2	3	4
	C(2)	0.4902	0.4777	0.3974	0.4502	0.4051	0.5880
	Ranking	2	3	6	5	5	1
Global ranking	ȑ	0.8733	0.5342	0.7344	0.6781	0.4210	0.5210
	Ȑ	0.3544	0.2901	0.2505	0.3112	0.2311	0.2090
	Ranking	1	4	3	2	6	5

Note:ȑ=GADA index, Ȑ =GADA weight

5. Conclusions

While we hope companies, governments and decision makers can benefit from this study, yet also need to acknowledge some limitations. First, we enlisted the help of eight experts from different companies and sectors. Although this study aligns with other studies using MCDA models in quality management problems, it would be interesting, for future work, to increase the number of experts participating in the study to increase the validation of results. We also recommend the inclusion of experts from other economies, i.e., from the well-established to emergent, to gather a diversified set of options related to different experts' profiles. In doing so, we recommend the Delphi method panels to compare and enrich the criteria and sub-criteria considered in this research. Finally, the selection criteria can change according to specific conditions and vary according to external disturbances (such as the current COVID-19 pandemic). Future work should focus on companies from different countries. We recommend building a comparative scorecard to identify global insights related to certification body selection processes worldwide.

In meeting the objectives and primary research questions of this study, we have multiple contributions to the field and new insights regarding relationships and the SDGs. First, this study contributes to academic research that attempts to explain the right certification bodies' selection to align with and achieve SDGs. The outcomes of his study helps to establish seven important criteria and thirty-seven sub-criteria. These criteria are developed by starting with an extensive review of the literature, and then we build on this with the help of experts. Simultaneously, we used a novel Grey Delphi method to meet this objective. Initially, we identified eleven main criteria from the literature. We classified them into seven of the most influential and vital criteria, such as Reputation, Operational Outcomes, Quality of Auditor, Contract Condition, Quality of Implementation, Payment method & Cost, and Knowledge of Product and Business. We then divided these main seven categories of criteria into a further thirty-seven sub-criteria. The second contribution is using the novel Grey AHP method to arrange these criteria and sub-criteria to select certification bodies. It has been organized in the form of a comparison matrix. Due to the competitive market of certification bodies, Reputation, Payment Method & Cost, and Quality of Auditors are significant indicators with of 0.2120, 0.1704, and 0.1406 weight score of top-ranked criteria in the selection process of CB to achieve SDGs. Simultaneously, the auditor's knowledge about the management standards and review meetings ranked the most critical sub-criterion among the rest of the 35 sub-criteria lists with a weight of 0.5009 and 0.4918.

This study's third contribution is the micro-level identification of ISO and GRI as the top two standards recommend as sustainability standards/tools required to align with and fulfill SDGs. These standards have multiple benefits for organizations wanting to advance CSR and TBL performance. A methodological contribution includes the application of a Grey Absolute Decision Analysis method to prioritize certification bodies. This is the first time this type of prioritization has been performed with the top six most adaptable certification bodies being: Bureau Veritas (BV), *Société Générale de Surveillance* (SGS), *Det Norske Veritas Germanischer Lloyd* (DNVGL), *Lloyds Register* (LR), British Standards Institution (BSI), and the *Technischer Überwachungs-Verein* (TUV). This study assesses and compares these certification bodies based on their sizes, financial position, reputation, team of quality auditors, number of customers, year of services provided, and how many countries they are in. This study concluded that BV and SGS are the most appropriate options in selecting certification bodies by obtaining the highest weight scores of 0.873, and 0.734 to align with and help achieve SDGs. Finally, this study is also the first to apply an integrated approach using three novel grey models to add new insights to the existing body of literature.

Organizations in developing countries often struggle with available resources for certifications aligned with the UN SDGs and improvements in corporate management of environmental, social, and governance performance. Those seeking CBs typically need help in problem solving, decision making, and communication with stakeholders so that limited resources are not wasted. The role of a CB is important as they can help with technical and management options based on prior experience. The quality of these engagements is crucial for advancing industries at a macro-level and individual enterprises at a micro-level. Business decision makers will benefit from our findings in more quickly getting up to speed in choosing a CB to partner with. To this end, we have tried to develop a customizable approach for certification decision-making needs. Organizations, the United Nations, and other authorities can benefit from this study as CBs enable tangible cost-benefit results toward global goals of sustainable development. Local governments, international agencies, businesses, and stakeholders all have opportunities to align with global goals for more sustainable business practices. This study's outcomes help provide a more dynamic value proposition involved with the selection and use of CBs. We hope the results will help enable a more intelligent allocation of scarce resources while also aligning investments with faster progress toward the UN SDGs.

This study provides relevant theoretical and practical implications. This study contributes to the enrichment of SDGs literature, especially regarding the final processes of selecting the proper sustainability certification. Practical contributions include, but are not limited to, this work serving as a decision model in acquiring certification bodies' services. Outcomes will help managers and government agencies to identify the certifying company that best contributes to their needs. We also see our work helping develop strategies for selecting certification bodies to align with SDGs. Scholars can consider emerging criteria from the literature in the future and build on this study. Future research can build on this study, replicate its methods, and combine experts' insights to further developing a more integrated approach to the certification body selection process. The methods used in this study support decision-making in a dynamic and relatively practical way. These methods can and should be applied to further replications by future studies.

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References

Ali, J., Yusuf, N., 2019. International Quality Certification and Business Performance of Indian Firms: Evidence from Enterprise Survey Data. Glob. Bus. Rev. https://doi.org/10.1177/0972150919825514

Alvarenga, A.D., Salgado, E.G., Mendes, G.H. de S., 2018. Ranking criteria for selection of certification bodies for ISO 9001 through the Analytic Hierarchy Process (AHP). Int. J. Qual. Reliab. Manag. 35, 1321–1342. https://doi.org/10.1108/IJQRM-12-2016-0217

Azapagic, A., 2015. Sustainable Production and Consumption: A Decision-Support Framework Integrating Environmental, Economic and Social Sustainability, Computer Aided Chemical Engineering. Elsevier. https://doi.org/10.1016/B978-0-444-63578-5.50017-7

Barbosa-Póvoa, A.P., da Silva, C., Carvalho, A., 2018. Opportunities and challenges in sustainable supply chain: An operations research perspective. Eur. J. Oper. Res. 268, 399–431. https://doi.org/10.1016/j.ejor.2017.10.036

Bian, J., Liao, Y., Wang, Y.-Y., Tao, F., 2020. Analysis of Firm CSR Strategies. Eur. J. Oper. Res.

Brundtland, 1987. World Commission on Environment and Development: Our Common Future.

BSI, 2019. The british standards institution annual report and financial statements 2019.

Cai, S., Jun, M., 2018. A qualitative study of the internalization of ISO 9000 standards: The linkages among firms' motivations, internalization processes, and performance. Int. J. Prod. Econ. https://doi.org/10.1016/j.ijpe.2017.12.001

Cândido, C.J.F., Coelho, L.M.S., Peixinho, R.M.T., 2019. Why firms lose their ISO 9001 certification: Evidence from Portugal. Total Qual. Manag. Bus. Excell. https://doi.org/10.1080/14783363.2019.1625266

Castka, P., Prajogo, D., Sohal, A., Yeung, A.C.L., 2015. Understanding firms′ selection of their ISO 9000 third-party certifiers. Int. J. Prod. Econ. 162, 125–133. https://doi.org/10.1016/j.ijpe.2015.01.012

Chan, F.T.S., Chan, H.K., 2010. An AHP model for selection of suppliers in the fast changing fashion market. Int. J. Adv. Manuf. Technol. https://doi.org/10.1007/s00170-010-2683-6

Chan, F.T.S., Kumar, N., 2007. Global supplier development considering risk factors using fuzzy extended AHP-based approach. Omega. https://doi.org/10.1016/j.omega.2005.08.004

Chiarini, A., 2019. Why are manufacturing SMEs cancelling their ISO 9001 certification? Research from Italy. Prod. Plan. Control. https://doi.org/10.1080/09537287.2019.1566840

Chkanikova, O., Sroufe, R., 2020. Third-party sustainability certifications in food retailing: Certification design from

a sustainable supply chain management perspective. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2020.124344
Chowdhury, M.M.H., Paul, S.K., Sianaki, O.A., Quaddus, M.A., 2020. Dynamic sustainability requirements of stakeholders and the supply portfolio. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2020.120148
Diaz-Balteiro, L., González-Pachón, J., Romero, C., 2017. Measuring systems sustainability with multi-criteria methods: A critical review. Eur. J. Oper. Res. https://doi.org/10.1016/j.ejor.2016.08.075
DNVGL, 2019. Sustainability performance Report 2019.
DNVGL, 2015. 2030 forecast: key conclusions The main findings from DNV GL's model-based forecast of the most likely future for our planet [WWW Document]. URL https://www.dnvgl.com/technology-innovation/spaceship-earth/forecast-key-conclusions.html (accessed 11.11.20).
Elkington, J., 1997. Cannibals with Forks: Triple Bottom Line of 21st Century Business, The Triple Bottom Line.
Fonseca, Carvalho, 2019. The Reporting of SDGs by Quality, Environmental, and Occupational Health and Safety-Certified Organizations. Sustainability 11, 5797. https://doi.org/10.3390/su11205797
Gallego-Álvarez, I., Lozano, M.B., Rodríguez-Rosa, M., 2018. An analysis of the environmental information in international companies according to the new GRI standards. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2018.01.240
Gómez Martín, E., Giordano, R., Pagano, A., van der Keur, P., Máñez Costa, M., 2020. Using a system thinking approach to assess the contribution of nature based solutions to sustainable development goals. Sci. Total Environ. 738, 139693. https://doi.org/10.1016/j.scitotenv.2020.139693
Govindan, K., Khodaverdi, R., Jafarian, A., 2013. A fuzzy multi criteria approach for measuring sustainability performance of a supplier based on triple bottom line approach. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2012.04.014
GRI, 2020. Linking the SDGs and the GRI Standards.
Haeri, S.A.S., Rezaei, J., 2019. A grey-based green supplier selection model for uncertain environments. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2019.02.193
Heras-Saizarbitoria, I., Boiral, O., 2013. ISO 9001 and ISO 14001: Towards a Research Agenda on Management System Standards. Int. J. Manag. Rev. https://doi.org/10.1111/j.1468-2370.2012.00334.x
Hernandez-Vivanco, A., Domingues, P., Sampaio, P., Bernardo, M., Cruz-Cázares, C., 2019. Do multiple certifications leverage firm performance? A dynamic approach. Int. J. Prod. Econ. https://doi.org/10.1016/j.ijpe.2019.07.016
Hernandez, H., 2010. Quality audit as a driver for compliance to ISO 9001:2008 standards. TQM J. https://doi.org/10.1108/17542731011053361
Hoy, Z., Foley, A., 2015. A structured approach to integrating audits to create organisational efficiencies: ISO 9001 and ISO 27001 audits. Total Qual. Manag. Bus. Excell. https://doi.org/10.1080/14783363.2013.876181
Huan, Y., Liang, T., Li, H., Zhang, C., 2021. A systematic method for assessing progress of achieving sustainable development goals: A case study of 15 countries. Sci. Total Environ. 752. https://doi.org/10.1016/j.scitotenv.2020.141875
Huang, S.H., Keskar, H., 2007. Comprehensive and configurable metrics for supplier selection. Int. J. Prod. Econ. https://doi.org/10.1016/j.ijpe.2006.04.020
Hu, X., Chong, H.-Y., Wang, X., 2019. Sustainability perceptions of off-site manufacturing stakeholders in Australia. J. Clean. Prod. 227, 346–354. https://doi.org/10.1016/j.jclepro.2019.03.258
Ikram, M., Sroufe, R., Zhang, Q., 2020a. Prioritizing and overcoming barriers to integrated management system (IMS) implementation using AHP and G-TOPSIS. J. Clean. Prod. 254, 120121. https://doi.org/10.1016/j.jclepro.2020.120121
Ikram, M., Zhang, Q., Sroufe, R., 2020b. Future of quality management system (ISO 9001) certification: novel grey forecasting approach. Total Qual. Manag. Bus. Excell. 1–28. https://doi.org/10.1080/14783363.2020.1768062
Ikram, M., Zhang, Q., Sroufe, R., 2020c. Developing integrated management systems using an AHP-Fuzzy VIKOR approach. Bus. Strateg. Environ. 29, 2265–2283. https://doi.org/10.1002/bse.2501
Ikram, M., Zhang, Q., Sroufe, R., Ferasso, M., 2020. The Social Dimensions of Corporate Sustainability: An Integrative Framework Including COVID-19 Insights. Sustainability 12, 8747. https://doi.org/10.3390/su12208747
ISO, 2017. ISO Members. Int. Stand. Stand. URL http://www.iso.org/iso/home/about/iso_members.htm (accessed 1.13.21)
ISO, 2018a. Contributing to the UN Sustainable Development Goals with ISO standards.
ISO, 2018b. HOW ISO STANDARDS HELP MEET THE SDGS [WWW Document]. URL https://www.iso.org/sdgs.html (accessed 10.16.20).

Ivanova, A., Gray, J., Sinha, K., 2014. Towards a unifying theory of management standard implementation. Int. J. Oper. Prod. Manag. https://doi.org/10.1108/ijopm-03-2013-0117
Jamal, K., Sunder, S., 2011. Is mandated independence necessary for audit quality? Accounting, Organ. Soc. https://doi.org/10.1016/j.aos.2011.03.001
Javed, S.A., Liu, S., 2019. Bidirectional Absolute GRA/GIA Model for Uncertain Systems: Application in Project Management. IEEE Access. https://doi.org/10.1109/ACCESS.2019.2904632
Javed, S.A., Mahmoudi, A., Liu, S., 2020. Grey Absolute Decision Analysis (GADA) Method for Multiple Criteria Group Decision-Making Under Uncertainty. Int. J. Fuzzy Syst. https://doi.org/10.1007/s40815-020-00827-8
Javed, S.A., Ikram, M., Tao, L., Liu, S., 2020. Forecasting key indicators of China's inbound and outbound tourism: optimistic–pessimistic method. Grey Syst. Theory Appl. ahead-of-p. https://doi.org/10.1108/GS-12-2019-0064
Javed, S.A., Zhu, B., Liu, S., 2020b. Forecast of biofuel production and consumption in top CO2 emitting countries using a novel grey model. J. Clean. Prod. 276, 123997. https://doi.org/10.1016/j.jclepro.2020.123997
Jespersen, A.H., Hasle, P., 2017. Developing a concept for external audits of psychosocial risks in certified occupational health and safety management systems. Saf. Sci. https://doi.org/10.1016/j.ssci.2016.11.023
Jiang, W., Xie, C., Luo, Y., Tang, Y., 2017. Ranking Z-numbers with an improved ranking method for generalized fuzzy numbers. J. Intell. Fuzzy Syst. https://doi.org/10.3233/JIFS-16139
Kannan, D., De Sousa Jabbour, A.B.L., Jabbour, C.J.C., 2014. Selecting green suppliers based on GSCM practices: Using Fuzzy TOPSIS applied to a Brazilian electronics company. Eur. J. Oper. Res. https://doi.org/10.1016/j.ejor.2013.07.023
Lamichane, S., Egilmez, G., Gedik, R., Bhutta, M.K., Erenay, B., 2020. Benchmarking OECD Countries' Sustainable Development Performance: A Goal-Specific Principal Component Analysis Approach. J. Clean. Prod. 125040.
Lloyd's Register, 2019. Reaching new heights for a safer world.
Ma, Z., Shao, C., Ma, S., Ye, Z., 2011. Constructing road safety performance indicators using fuzzy delphi method and grey delphi method. Expert Syst. Appl. 38, 1509–1514. https://doi.org/10.1016/j.eswa.2010.07.062
Ofosu-Adarkwa, J., Xie, N., Javed, S.A., 2020. Forecasting CO2 emissions of China's cement industry using a hybrid Verhulst-GM(1,N) model and emissions' technical conversion. Renew. Sustain. Energy Rev. 130, 109945. https://doi.org/10.1016/j.rser.2020.109945
Pan, X., Sinha, P., Chen, X., 2020. Corporate social responsibility and eco-innovation: The triple bottom line perspective. Corp. Soc. Responsib. Environ. Manag.
Poksinska, B., Dahlgaard, J.J., Eklund, J.A.E., 2006. From compliance to value-added auditing - Experiences from Swedish ISO 9001:2000 certified organisations. Total Qual. Manag. Bus. Excell. https://doi.org/10.1080/14783360600595294
Power, D., Terziovski, M., 2007. Quality audit roles and skills: Perceptions of non-financial auditors and their clients. J. Oper. Manag. https://doi.org/10.1016/j.jom.2006.02.005
Power, D., Terziovski, M., 2005. The process, practice and outcomes of non-financial auditing: Five Australian case studies. Int. J. Manuf. Technol. Manag. 7, 52–82. https://doi.org/10.1504/IJMTM.2005.006502
Prajogo, D., Castka, P., Searcy, C., 2020a. Paymasters and Assurance Providers: Exploring Firms' Discretion in Selecting Non-financial Auditors. J. Bus. Ethics. https://doi.org/10.1007/s10551-020-04539-9
Prajogo, D., Castka, P., Yiu, D., Yeung, A.C.L., Lai, K.H., 2016. Environmental Audits and Third Party Certification of Management Practices: Firms' Motives, Audit Orientations, and Satisfaction with Certification. Int. J. Audit. https://doi.org/10.1111/ijau.12068
Prajogo, D., Nair, A., Castka, P., 2020b. The Effects of External Auditors and Certification Bodies on the Operational and Market-Oriented Outcomes of ISO 9001 Implementation. IEEE Trans. Eng. Manag. https://doi.org/10.1109/TEM.2020.2982657
Prajogo, D.I., 2011. The roles of firms' motives in affecting the outcomes of ISO 9000 adoption. Int. J. Oper. Prod. Manag. https://doi.org/10.1108/01443571111098753
Ristono, A., -, P., Santoso, P.B., Tama, I.P., 2018. A literature review of criteria selection in supplier. J. Ind. Eng. Manag. https://doi.org/10.3926/jiem.2203
Sethi, S.P., Rovenpor, J.L., Demir, M., 2017. Enhancing the Quality of Reporting in Corporate Social Responsibility Guidance Documents: The Roles of ISO 26000, Global Reporting Initiative and CSR-Sustainability Monitor. Bus. Soc. Rev. 122, 139–163. https://doi.org/10.1111/basr.12113
SGS, 2019. 2019 Corporate Sustainability Report.
Sharma, K.K., Seal, A., 2019. Modeling uncertain data using Monte Carlo integration method for clustering. Expert Syst. Appl. https://doi.org/10.1016/j.eswa.2019.06.050
Shete, P.C., Ansari, Z.N., Kant, R., 2020. A Pythagorean fuzzy AHP approach and its application to evaluate the enablers of sustainable supply chain innovation. Sustain. Prod. Consum. 23, 77–93.

https://doi.org/10.1016/j.spc.2020.05.001
Singh, P.J., Power, D., Chuong, S.C., 2011. A resource dependence theory perspective of ISO 9000 in managing organizational environment. J. Oper. Manag. https://doi.org/10.1016/j.jom.2010.04.002
Skirbekk, G. (1994). Marxism and ecology. Capitalism Nature Socialism, 5(4), 95-104.
Souza, J.P.E., Alves, J.M., 2018. Lean-integrated management system: A model for sustainability improvement. J. Clean. Prod. 172, 2667–2682. https://doi.org/10.1016/j.jclepro.2017.11.144
Stević, Ž., Stjepanović, Ž., Božičković, Z., Das, D.K., Stanujkić, D., 2018. Assessment of conditions for implementing information technology in a warehouse system: A novel fuzzy PIPRECIA method. Symmetry (Basel). https://doi.org/10.3390/sym10110586
Tavares de Aquino, A., Maciel de Melo, R., 2016. Multicriteria model for selecting TQM consultancy and certification services. Benchmarking. https://doi.org/10.1108/BIJ-05-2015-0049
Torkabadi, A.M., Pourjavad, E., Mayorga, R. V., 2018. An integrated fuzzy MCDM approach to improve sustainable consumption and production trends in supply chain. Sustain. Prod. Consum. 16, 99–109. https://doi.org/10.1016/j.spc.2018.05.008
Tsalis, T.A., Malamateniou, K.E., Koulouriotis, D., Nikolaou, I.E., 2020. New challenges for corporate sustainability reporting: United Nations' 2030 Agenda for sustainable development and the sustainable development goals. Corp. Soc. Responsib. Environ. Manag. https://doi.org/10.1002/csr.1910
TÜV SÜD, 2019. Combined management Report 2019.
UN World, 2019. 'Not Keeping Pace' in Implementing Sustainable Development Goals, Secretary-General Stresses, as Development Financing Forum Begins. [WWW Document]. URL https://www.un.org/press/en/2019/ecosoc6972.doc.htm (accessed 11.11.20).
United Nations, 2020. The Sustainable Development Goals 2020.
United Nations Development Programme (UNDP), 2020. Background and Goals. accessed at www.undp.org/undp/en/jhome/sustainable-development-gaols/background
USTUN, O., DEMIRTAS, E., 2008. An integrated multi-objective decision-making process for multi-period lot-sizing with supplier selection☆. Omega 36, 509–521. https://doi.org/10.1016/j.omega.2006.12.004
Veritas, B., 2019. Sustainability report and 2019 UNIVERSAL REGISTRATION DOCUMENT.
Wang, H., Pan, C., Wang, Q., Zhou, P., 2020. Assessing sustainability performance of global supply chains: An input-output modeling approach. Eur. J. Oper. Res. https://doi.org/10.1016/j.ejor.2020.01.057
White, L., Lee, G.J., 2009. Operational research and sustainable development: Tackling the social dimension. Eur. J. Oper. Res. 193, 683–692. https://doi.org/10.1016/j.ejor.2007.06.057
Wiengarten, F., Humphreys, P., Onofrei, G., Fynes, B., 2017. The adoption of multiple certification standards: perceived performance implications of quality, environmental and health & safety certifications. Prod. Plan. Control. https://doi.org/10.1080/09537287.2016.1239847
Zare, A., Feylizadeh, M.R., Mahmoudi, A., Liu, S., 2018. Suitable computerized maintenance management system selection using grey group TOPSIS and fuzzy group VIKOR: A case study. Decis. Sci. Lett. 7, 341–358. https://doi.org/10.5267/j.dsl.2018.3.002
Zareinejad, M., Kaviani, M.A., Esfahani, M.J., Masoule, F.T., 2014. Performance evaluation of services quality in higher education institutions using modified SERVQUAL approach with grey analytic hierarchy process (G-AHP) and multilevel grey evaluation. Decis. Sci. Lett. https://doi.org/10.5267/j.dsl.2013.12.002
Zhao, X., Castka, P., Searcy, C., 2020. ISO Standards: A Platform for Achieving Sustainable Development Goal 2. Sustainability 12, 9332. https://doi.org/10.3390/su12229332

Supplementary file